MaiAgent Technical Documentation

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Generative AI Quick Start

Large Language Models (LLM)

Key Selection Considerations

When selecting a large language model, consider the following key factors:

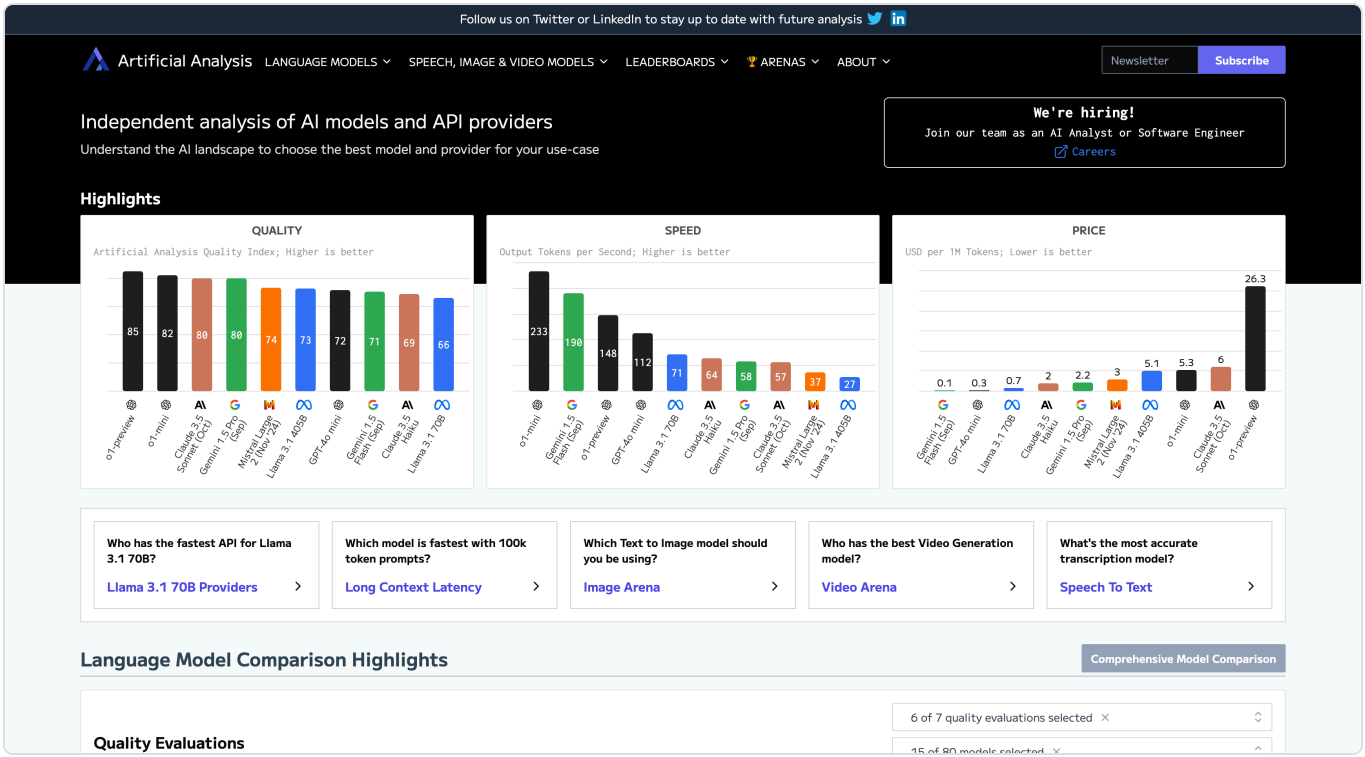
Environment: Determine whether to use cloud-based or on-premise models based on internet connectivity.

Quality: The model's ability to generate responses and adherence to instructions.

Speed: Text generation speed and latency requirements to ensure timely model responses.

Pricing: Consider the model's usage cost and select an appropriate model based on usage requirements. (No need to consider model pricing on MaiAgent)

Other: Whether it supports multimodality and function calling.



Large Language Model Analysis - Artificial Analysis

Large Language Models Supported on MaiAgent

Closed-Source Models

Model Name	Description	Is Reasoning Model	Use Cases
o4-mini	Faster than o3-mini-high, slightly lower quality than o3-mini-high	Yes	High-quality, fast option
o3-mini-high	High quality, medium speed, uses chain-of-thought reasoning for multi-layer computation before answering to provide more complete and accurate responses	Yes	High-difficulty tasks requiring deep reasoning and creativity
o3-mini-medium	Fast speed, medium quality	Yes	Most business applications, simple creative tasks, or regular Q&A
o3-mini-low	**Fastest speed**, basic quality, lacks deep reasoning	Yes	Simple tasks prioritizing speed over depth
o1-mini 2024-09-12	o1 series large language models are trained with reinforcement learning to perform complex reasoning. o1 models think before answering, generating a long internal chain of thought before responding to users. Slowest speed, **good quality**.	Yes	Extremely difficult problems when other LLMs fail
GPT-4o 2024-08-06	Above-average quality and speed	No	Instruction following and logic slightly inferior to Claude 3.5 Sonnet, but faster than Claude 3.5 Sonnet. A commonly used choice👍
GPT-4o mini 2024-07-18	Fast speed, medium quality. Quality slightly lower than Gemini 2.0 Flash	No	Simple tasks, alternative when Gemini 2.0 Flash is unavailable
Claude 4 Sonnet	Medium-to-slow speed, strong structured data generation/extraction capabilities, particularly excels at Tool Calling; logical reasoning and coding performance surpass Claude 3.7 Sonnet, with further reduced hallucination rate.	Hybrid reasoning model	First choice for Agent mode 👍 Suitable for highly complex tasks, professional domain applications, and extended conversations.
Claude 3.7 Sonnet	Medium-to-slow speed, **excels at generating structured** data, has stronger logical reasoning than Claude 3.5 Sonnet. **Low hallucination rate**	Hybrid reasoning model	First choice for most situations👍 Suitable for high-complexity tasks, professional domains, and long conversations

Model Name	Description	Is Reasoning Model	Use Cases
Claude 3.5 Sonnet	Follows role instructions well, weaker logical reasoning compared to Claude 3.7 Sonnet, but **faster speed**. **Low hallucination rate**	No	Can switch to Gemini 2.0 Flash when speed feels too slow
Gemini 2.5 Pro	Better quality than Claude 3.7 Sonnet for longer conversations and code generation, but slightly inferior in Agent mode and tool calling	No	Can be used interchangeably with Claude 3.7 Sonnet
Gemini 2.0 Pro	Similar quality to Claude 3.5 Sonnet, but slower speed	No	Alternative to Claude 3.5 Sonnet
Gemini 2.5 Flash	**Fast speed**, good multimodal capabilities	No	
Gemini 2.0 Flash	**Fast speed**, medium quality	No	First choice for simple tasks 👍
DeepSeek V3	Fast speed, high quality	Yes	Suitable for document retrieval and large-scale database query tasks
DeepSeek R1 Distill Llama 70B	High response quality, medium speed (slower than DeepSeek V3)	Yes	Suitable for tasks requiring multi-step reasoning and background knowledge
DeepSeek R1	Slower response speed, but **strong Chinese language understanding**, **high-quality response content**. Deep thinking and truly adjusts based on role instruction content	Yes	Complex multi-turn Chinese conversations. Handling complex role instructions 👍

Open-Source Models

Below is a comparison table of mainstream open-source models. For hardware requirements, refer to the GPU section.

Model Name	Description	Agent Support	Use Cases
GPT-OSS-120B	Excels at organizing information in table format	Yes	Data analysis, content creation
Gemma3 27B	Good image OCR performance	No	Invoice recognition, image analysis

Model Name	Description	Agent Support	Use Cases
Meta Llama3.3 70B	Highly cost-effective general-purpose model with strong instruction following	No	Voice customer service, RAG Q&A assistant
Meta Llama3.2 90B	Vision capabilities, can process both images and text	No	Professional domain Q&A, high-precision tasks
Meta Llama3.1 405B	Extremely broad knowledge and high reasoning ability	Yes	Customer service Q&A, knowledge Q&A
Llama3 Taiwan 70B	Fine-tuned for Traditional Chinese and Taiwanese culture, accurate localized terminology	No	Customer service Q&A, RAG Q&A assistant
DeepSeek R1	Reasoning-enhanced model, excels at solving complex problems through chain of thought	No	Scientific logical reasoning
DeepSeek V3.2	MoE architecture, fast inference speed and extremely low cost	No	Large-scale text summarization
Qwen3 235B	Code and math specialized, deep logical understanding, suitable for hardcore tasks	Yes	Academic paper assistance
Qwen3 32B	Dynamic balance between high-quality answers and efficient processing	Yes	Multilingual dialogue and instruction following
Qwen3 8B	Medium quality, balanced performance and cost	Yes	Chat assistant, customer service FAQ
Qwen2.5 VL 72B instruct	Vision-language model, accurate image detail recognition	Yes	Multimodal chat assistant
Mistral Large (24.07)	Medium quality, lacks deep reasoning ability, fast speed	No	Customer service Q&A, simple text generation

Is Fine-tuning Always Necessary?

With the rapid development of artificial intelligence technology, language models have acquired powerful language understanding and generation capabilities and are widely applied in many fields. For example, pre-trained language models can easily handle daily conversations, article generation, and simple Q&A tasks.

However, when models face more challenging professional domain tasks, such as healthcare, legal, or technical support, relying solely on pre-trained models may not provide optimal performance. Some

developers choose to perform **Fine-tuning**, which involves additional training for specific domains to enhance the model's professional knowledge.

However, Fine-tuning is not the only solution. There are two other effective methods to achieve this goal: **Prompt Engineering** and **RAG (Retrieval-Augmented Generation)**.

1. Prompt Engineering: Optimizing Model Performance Through Precise Prompts

Prompt Engineering involves designing precise prompt statements to guide the model in generating desired results. The core of this approach is to design detailed expressions based on task requirements, helping the model narrow the scope of answers and understand the context and required output format.

Suppose there is a language model whose purpose is to recommend suitable products based on user needs. In this case, if expressed as "I want to buy a high-performance phone," it may lead the model to generate insufficiently precise answers because "high-performance" can have many different interpretations, possibly referring to processing speed, camera performance, battery life, etc.

To improve recommendation accuracy, we can perform **Prompt Engineering** and design more specific questions or provide additional context to guide the model to better understand user needs. For example, change to the following prompt statement:

"I need a phone with **long battery life** and a **powerful processor**, priced between **$500 and $800**. Please recommend several phones that meet these criteria."

2. RAG: Combining External Knowledge to Enhance Generation Capabilities

RAG enhances model performance by combining external knowledge retrieval with the generation process.

In traditional generation tasks, models rely only on knowledge learned during pre-training. RAG uses a retrieval system to obtain external data in real-time and combines this data with the generation model to answer questions or generate text more accurately.

For example, in healthcare, when a model is asked about a rare disease, RAG can first retrieve relevant information from professional medical databases, then generate more accurate answers based on this information. The advantage of this method is that even if the model hasn't encountered certain data during training, it can still make high-quality responses by retrieving existing knowledge. This is particularly suitable for scenarios requiring real-time knowledge updates and can greatly expand the model's knowledge scope.

> For a more detailed introduction to RAG, see the next chapter "RAG Knowledge Base Retrieval System Explanation".

In summary, Fine-tuning, Prompt Engineering, and RAG each have their advantages and applicable scopes. You can choose the most suitable strategy based on application scenarios and needs, without relying solely on Fine-tuning.

Prompt Engineering provides a low-cost and flexible solution by designing precise prompts to guide the model to produce high-quality results.

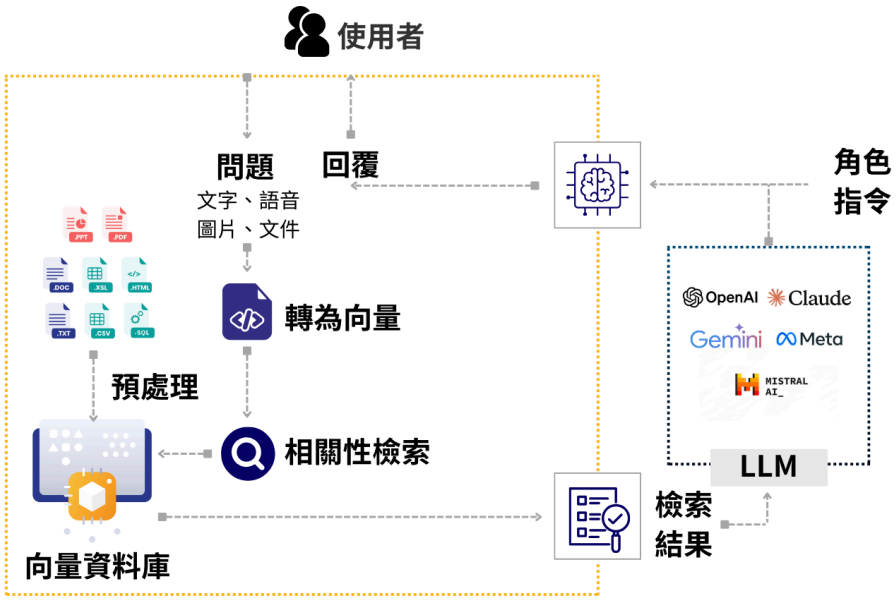
RAG provides a method combining external knowledge with generation capabilities, offering more accurate answers when dynamic knowledge retrieval is needed.

Fine-tuning can significantly improve model performance in specific domains but requires substantial specialized data and consumes additional resources. It should be considered a last resort when both **Prompt Engineering** and **RAG** fail to achieve success.

RAG Knowledge Base Retrieval System

Utilizing external databases or knowledge bases for retrieval, and combining retrieval results with large language models to generate more precise and contextually relevant responses.

The core of RAG technology is combining the language capabilities of generative AI with knowledge retrieval capabilities, enabling models to not only rely on internal training data when answering questions, but also dynamically obtain the latest and more specialized information from external databases and incorporate this information into generated responses.



RAG Flow

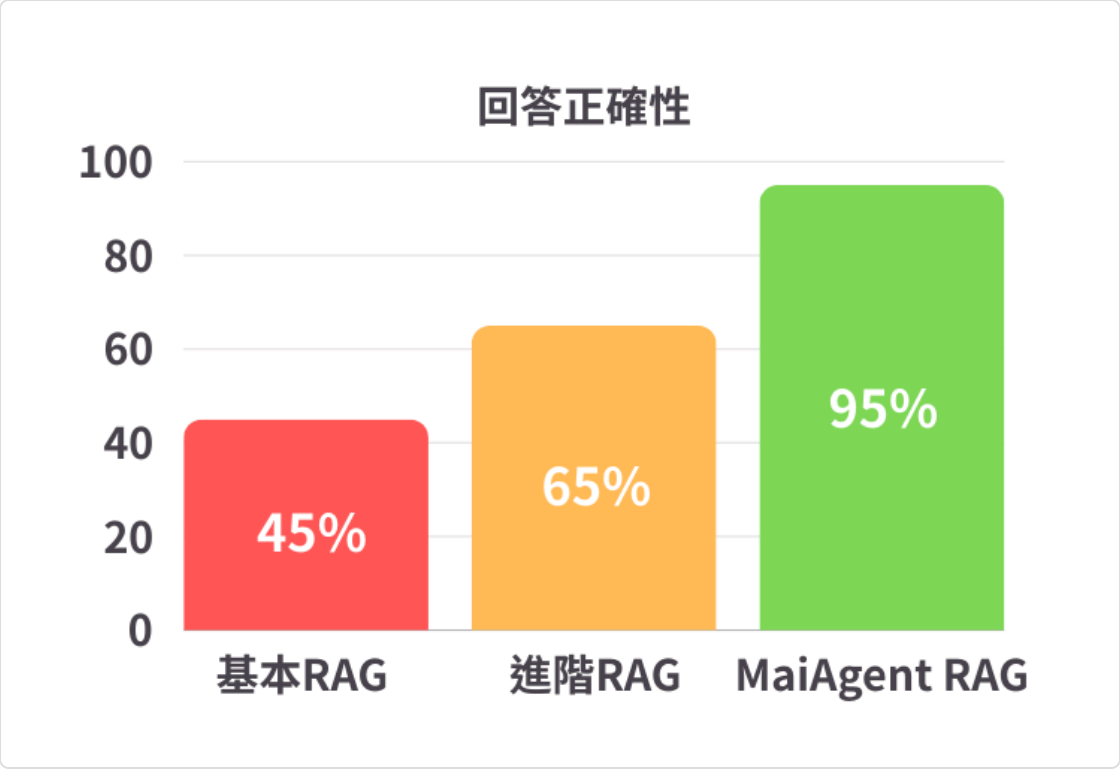
High-Precision RAG System

> While RAG knowledge base retrieval systems can be quickly implemented using Vector Search and released as a basic version, improving their response accuracy further presents challenges. Response accuracy is crucial for user experience as it directly affects users' trust and satisfaction with system responses. If response accuracy is insufficient, users may doubt the system's answers, reducing their willingness to use it.

According to data from the 2023 OpenAI Developer Conference, RAG systems using only simple vector similarity search (Vector Search) and selecting the correct embedding model can achieve 45% accuracy.

Adding HyDE Retrieval, FT Embeddings, and Chunk/Embedding Experiments can achieve 65% response accuracy.

MaiAgent RAG not only includes the RAG technologies mentioned at the OpenAI Developer Conference but also combines various classic NLP algorithms and proprietary retrieval technologies. Through internal datasets compared with OpenAI RAG's response accuracy, both can achieve 95% response accuracy.



MaiAgent RAG Response Accuracy

The MaiAgent platform provides two types of RAG: MaiAgent RAG and OpenAI RAG. Here's a comparison table of various aspects:

	MaiAgent RAG	**OpenAI RAG**
Model Support	Supports all models👍	Only supports OpenAI models
Environment Support	Supports cloud and on-premises👍	Cloud only, data must be sent to OpenAI
Response Accuracy	Very high👍	Very high👍
Supported File Formats	Supports all common formats👍 doc, docx, xls, xlsx, csv, ppt, pptx, pdf, txt, json, jsonl, md	No support for xlsx, csv No support for jsonl No support for legacy Office files (doc, xlsx, ppt)

	MaiAgent RAG	OpenAI RAG
Image Support in Documents	Yes (currently experimental)👍	No
Table Support in Documents	Yes (currently experimental)👍	No
Attachment Upload Support	Supported👍	Supported👍
Data Slice Transparency	Visualized👍	Black box
Debug Difficulty	Normal👍	Black box, cannot debug
Top K Adjustment	Enterprise version customization👍	None
Embedding Model Switch	Enterprise version customization👍	None

Embedding Model

What is Embedding?

In the RAG (Retrieval-Augmented Generation) process, the core task of **Embedding** is to convert large amounts of text data processed by the Parser, such as documents and knowledge base content, into numerical forms that computers can understand and compare - vectors.

This conversion process enables the RAG system to:

1. Understand Semantics

Embedding models capture the deep semantic meaning of text, not just surface vocabulary. This means that even if the wording in queries and documents differs, the system can identify their relevance as long as the semantics are similar.

2. Efficient Retrieval

After converting text to vectors, the system can use efficient vector similarity search algorithms to quickly find the most relevant text segments from massive databases.

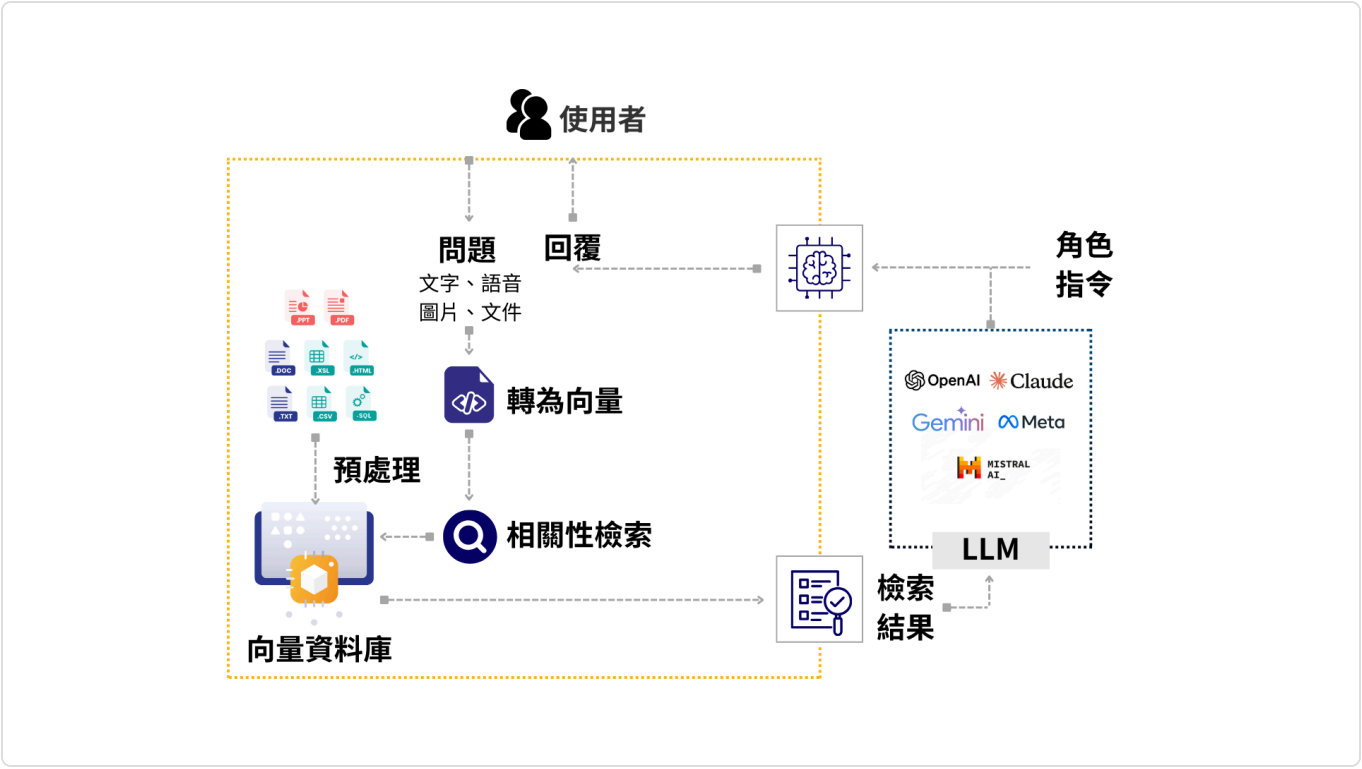
3. Improve Generation Quality

Retrieved relevant text segments are provided to Large Language Models (LLMs) as contextual references, helping LLMs generate more accurate responses.

Embedding Plays Two Key Roles in MaiAgent's RAG Technology

1. Convert knowledge base content into vectors and store them in vector databases

2. Vectorize user questions to compare similarity with vectors in the database to find the most relevant content



RAG Process

Simply put, Embedding is the cornerstone of RAG systems, transforming unstructured text data into computable and comparable vectors, which is a key prerequisite for achieving precise information retrieval and high-quality content generation.

Impact of Embedding on RAG Systems

1. Retrieval Quality

Semantic Understanding Depth: High-quality embedding models can more accurately capture the semantic content of text, improving retrieval relevance

Context Awareness: Excellent embeddings can understand textual context relationships, ensuring retrieval result coherence

Multilingual Support: Powerful multilingual embedding models can handle cross-language knowledge retrieval needs

2. System Performance

Retrieval Speed: The vector dimensions and computational efficiency of embedding models directly affect retrieval response time

Resource Consumption: Different embedding models have varying computational resource requirements, affecting system scalability

Parallel Processing: Efficient embedding models can support large-scale parallel retrieval

Embedding Models Supported by MaiAgent

Model Name	Model Developer	Origin	Features	Open Source	Deployment Method	MTEB Average Score
Cohere Embed v4.0	Cohere	Canada	Multilingual support, highest performance	No	Requires cloud API inference service	Not yet public Reference v3.0: 64.47
Cohere Embed Multilingual v3.0 (Bedrock)	Cohere	Canada	Multilingual support, high performance	No	Requires cloud API inference service	64.47
OpenAI text-embedding-3-Large	OpenAI	USA	Multilingual (especially strong in English context), medium performance	No	Requires cloud API inference service	64.68
EmbeddingGemma	Google	USA	Open source, multilingual support, lightweight	Yes	Can be deployed on cloud or local GPU	61.15
Mxbai-embed-large	Mixedbread AI	USA	Open source, good balance of performance and resources; strong in long context	Yes	Can be deployed on cloud or local GPU	64.68
BGE-Large	BAAI	China	Open source, multilingual support, lightweight	Yes	Can be deployed on cloud or local GPU	64.23
Nomic-embed-text	Nomic AI	USA	Open source, lightweight	Yes	Can be deployed on cloud or local GPU	62.39

Model Name	Model Developer	Origin	Features	Open Source	Deployment Method	MTEB Average Score
Qwen3-Embedding 0.6B	Alibaba	China	Open source, lightweight	Yes	Can be deployed on cloud or local GPU	61.82
Granite-embedding-278m-multilingual	IBM	USA	Open source, multilingual, lightweight	Yes	Can be deployed on cloud or local GPU	56.1

> To ensure semantic representation accuracy and retrieval precision, MaiAgent's embedding models are selected and evaluated based on MTEB (Massive Text Embedding Benchmark) standards. MTEB is currently the mainstream benchmark for semantic vectorization models, covering various task types including: Retrieval Classification Clustering Reranking STS (Semantic Textual Similarity)
> Summarization
> QA
> Pair Classification, etc.

MaiAgent's Embedding Technology Advantages

1. Flexible Model Selection:

Offers multiple embedding model choices to meet different needs

2. Diverse Deployment Options:

Supports both cloud and local deployment to ensure data security

3. Performance Optimization:

Specially optimized for RAG scenarios to provide optimal retrieval results

4. Cost Effectiveness:

Choose appropriate models based on actual needs, balancing performance and cost

Reranker Model

What is a Reranker?

A **Reranker** is a model or system used for **ranking tasks**, commonly employed in information retrieval, recommendation systems, or Natural Language Processing (NLP). Its main purpose is to reorder a set of candidate results to return the most relevant or suitable options.

How Reranker Works

Typically, when a search engine or recommendation system returns a series of candidates (such as search results, recommended products, or articles), they are already ranked according to certain preliminary ranking criteria. These initial rankings may be based on simple matching or basic relevancy indicators. However, such rankings may not fully consider all details or deeper semantic relationships, potentially leading to results that don't completely meet user needs.

For example, if you search online for "best pizza restaurants," the search engine will return many results based on basic criteria like restaurant names, ratings, and addresses. These results might already be ranked based on keywords or simple matching, but they may not fully align with your needs.

This is where the Reranker helps with "**optimizing the ranking**" by examining these results based on more details, potentially considering your past search habits, other people's reviews of the restaurant, or the restaurant's current status, thereby reordering the results to highlight the most relevant restaurants, helping you find the best option more quickly. Using this example, here's how the Reranker works:

Initial Ranking: Use simple ranking models (such as keyword matching, TF-IDF, or other features) to perform preliminary ranking

Search: "best pizza restaurants," the search engine returns a list of restaurants

Candidate Result Filtering: Refine ranking based on more features (such as context understanding, semantic relationships, user behavior, etc.) to re-score candidates

Take a closer look at these restaurants, considering if any are in areas you frequently visit, if they have high ratings from other users, or what promotions they offer

Final Ranking: Return the most relevant or suitable results based on their scores

The Reranker will place the restaurants that best match your needs at the top, making it easier to find the most suitable option

Reranker Model Provided by MaiAgent

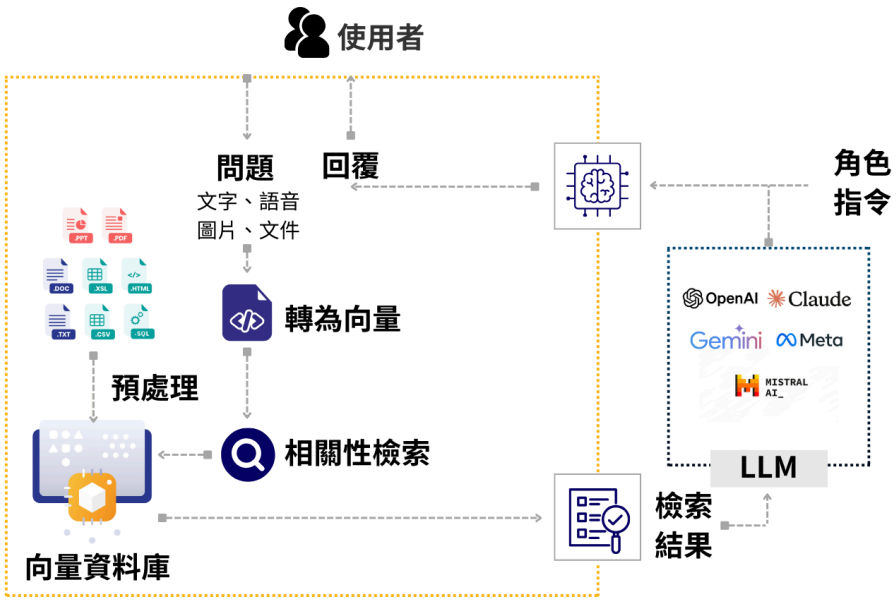
Features of Cohere Rerank v3.5

Cohere Rerank v3.5 is a **reranking** model provided by Cohere, specifically designed to optimize search results or candidate answer rankings. It can perform fine-grained ranking among candidate results, selecting the most relevant results based on deeper semantic understanding and context, thereby improving the final quality.

Parser Tools

What is RAG Parser?

RAG Parser is a critical component in Retrieval-Augmented Generation (RAG) systems, responsible for parsing and breaking down raw data as a preprocessing step for embedding vectorization. It provides the foundation for subsequent vectorization and semantic retrieval, having a decisive impact on overall data quality and retrieval effectiveness.



RAG Workflow

1. Document Parser Types

MaiAgent provides four document parsers, suitable for various document formats including PDF, Word, Excel, images, and more:

Feature	MaiAgent Parser (Default)	MaiAgent Parser (Online)	MaiAgent Parser (Offline)	Vision Parser
Cost	Low	High	Medium	High

Feature	MaiAgent Parser (Default)	MaiAgent Parser (Online)	MaiAgent Parser (Offline)	Vision Parser
Image Content Parsing	Cannot parse text in images	OCR only, can parse text in images	OCR + AI understands image semantics	AI visual understanding, best quality
Uses LLM	No	Yes	Yes	Yes
Text Parsing Quality	Standard	Good	Good (excellent structure preservation)	Good
Table Parsing	Native extraction	AI intelligently extracts text from images	Native structure preservation (best), includes static resources in images besides visual understanding	AI visual recognition generates text content from visual understanding of images
Parsing Speed	Fast	Medium	Slow	Slow
On-Premises (Offline)	Yes	No	Yes, but requires VLM deployment	No
Supported Formats	22 types	20 types	20 types (PDF only supports OCR)	7 types

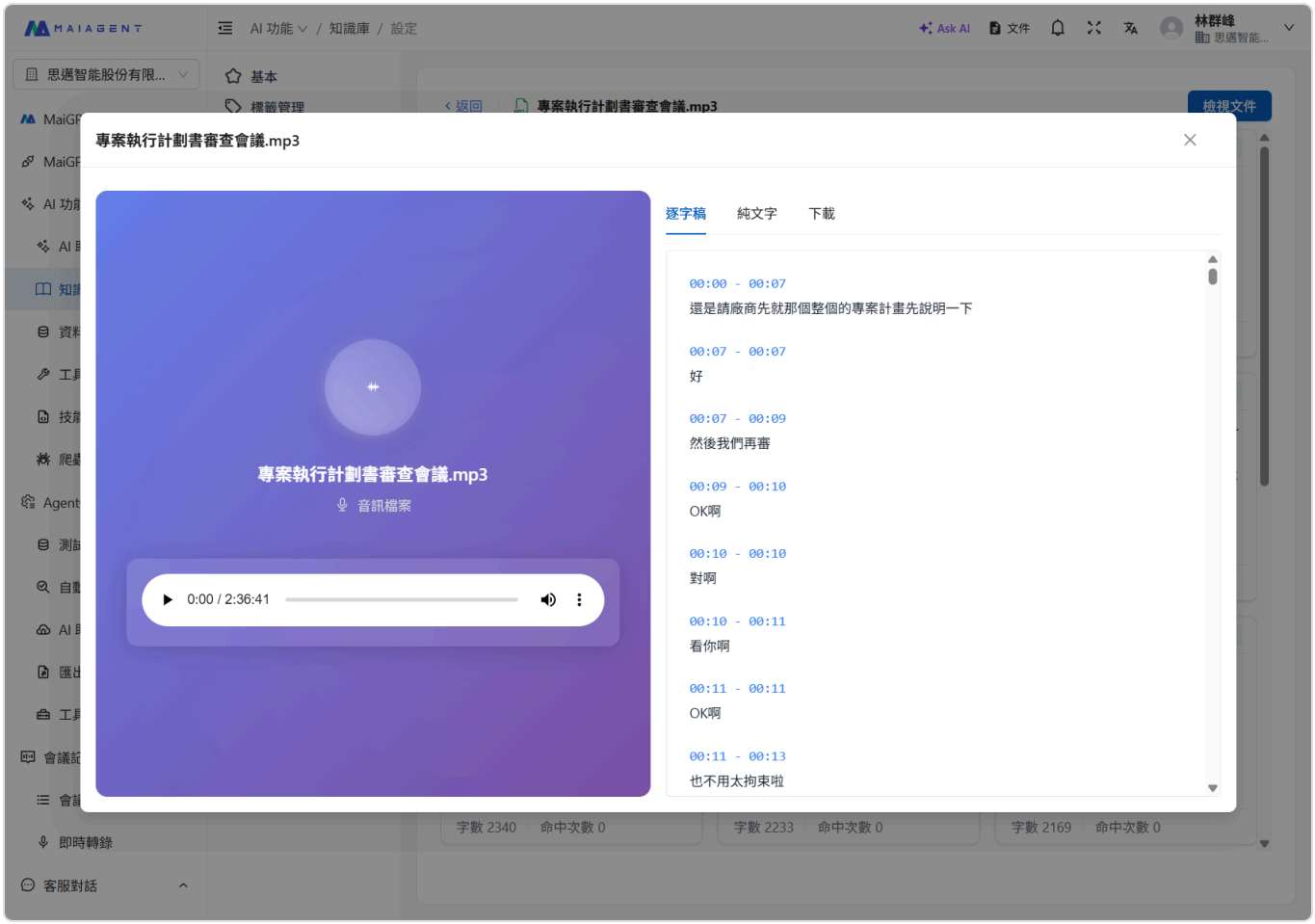
2. Speech-to-Text Parser Types

MaiAgent provides four speech-to-text parsers that can transcribe audio files into text for knowledge base integration:

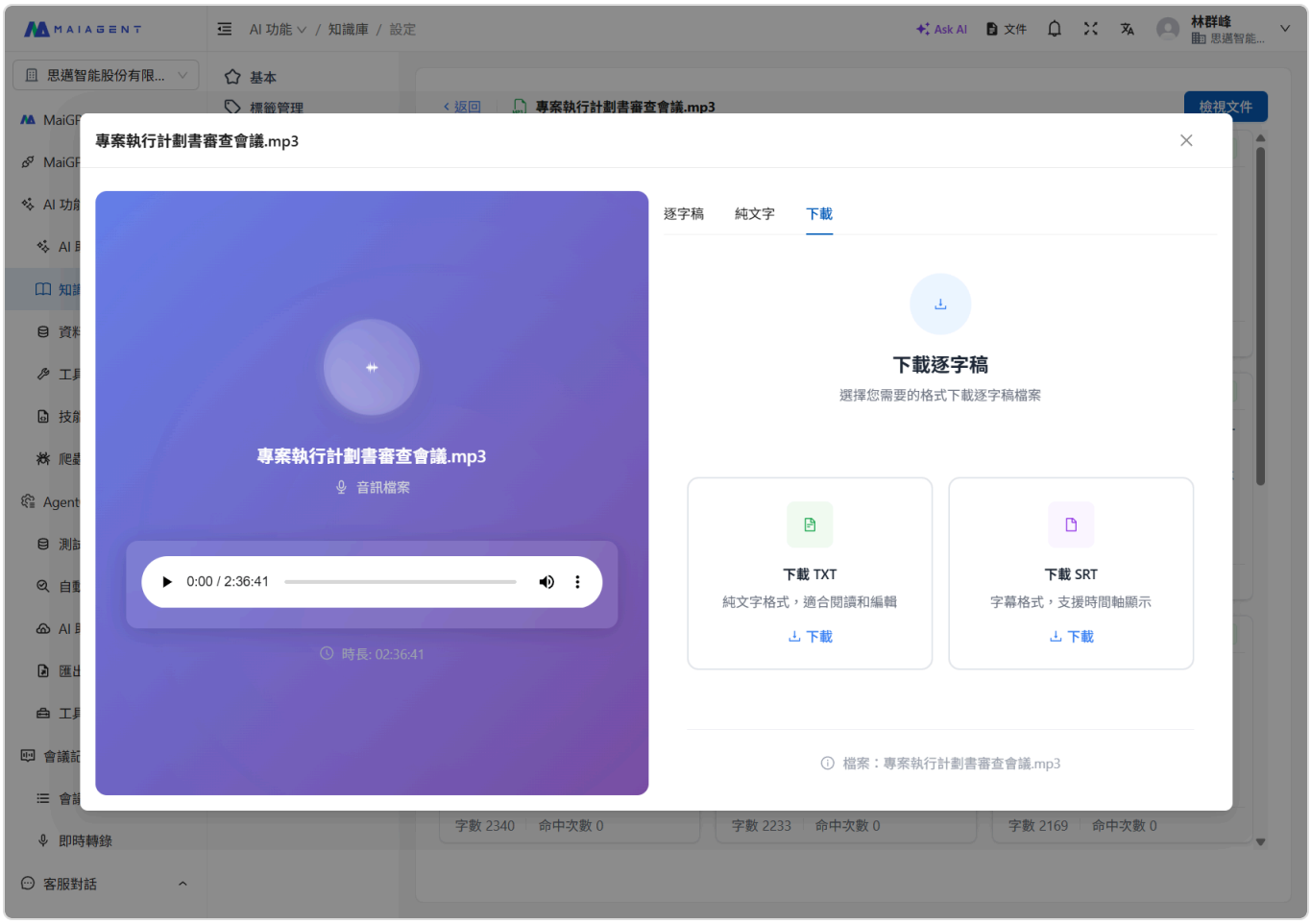
Feature	Azure Speech	Whisper (Groq)	Whisper (OpenAI)	Whisper (Offline)
Cost	High	Low	Medium	Free
Transcription Accuracy	High	High (large-v3)	High (whisper-1)	High (depends on local model)
Uses LLM	No	No	No	No
Parsing Speed	Real-time	Fastest	Medium	Depends on hardware
On-Premises (Offline)	No	No	No	Yes
Multi-language Support	Yes	Yes (auto-detect)	Yes (auto-detect)	Yes (auto-detect)

Feature	Azure Speech	Whisper (Groq)	Whisper (OpenAI)	Whisper (Offline)
Custom Prompts	No	Yes	Yes	Yes
VAD Voice Detection	—	Yes	Yes	Yes
Data Privacy	Cloud (Azure)	Cloud (Groq)	Cloud (OpenAI)	Fully local

After audio files are uploaded to the knowledge base, the system automatically performs speech-to-text parsing and provides transcript viewing and download functionality:



Transcript View: Displays timestamps and corresponding text content



Download Transcript: Supports two formats - TXT (plain text) and SRT (subtitle format)

Image Recognition Support

What is VLM (Vision Language Model)?

Vision Language Model (VLM) is an advanced artificial intelligence model that can understand image content and combine image information with textual information. Unlike traditional image processing technologies, VLM doesn't just "see" images, but can "understand" objects, scenes, relationships within images, and generate text, answer questions, or execute related commands based on image content.

The core capability of VLM lies in its multimodal processing ability - simultaneously processing and understanding information from different sources (visual and textual). This enables VLM to perform more complex and context-aware tasks.

Comparison between VLM and Traditional OCR

Traditional Optical Character Recognition (OCR) technology focuses on detecting and extracting text from images but has limitations in understanding the overall semantics and non-textual content of images.

Characteristic	Traditional OCR	VLM (Vision Language Model)
Main Function	Extract text from images	Understand image content and combine with textual information
Information Processing	Single modal (only processes text pixels)	Multimodal (processes both visual information and text semantics)
Understanding Level	Character-level recognition	Scene understanding, object recognition, relationship inference, context awareness
Main Tasks	Document scanning, text extraction	Image description generation, visual QA (VQA), image retrieval, object detection etc.
Context Awareness	Limited, mainly relies on language models in post-processing	Strong, can understand overall context and details of images
Non-text Content Processing	Usually ignored or unable to process	Can identify and understand objects, scenes, actions etc. in images

VLM's advantages include:
* **Deeper Understanding**: VLM not only reads text but understands semantic content of images, such as recognizing objects, analyzing scenes, understanding relationships between elements in images.

* **Interactivity**: VLM can perform Visual Question Answering (VQA), answering user questions based on image content.
* **Content Generation**: VLM can generate descriptive text for images (Image Captioning).
* **Versatility**: Beyond text-related tasks, VLM can be applied to broader visual understanding tasks.

Real VLM Application Cases

VLM's powerful capabilities enable wide applications across multiple fields:

Visual Question Answering (VQA)

Application: Users upload an image and ask questions about its content, like "What color clothes is the person wearing?" or "Was this photo taken indoors or outdoors?"

Scenarios: Smart assistants, education, visual impairment assistance.

Image Captioning

Application: Automatically generate concise, accurate text descriptions for images.

Scenarios: Automated image annotation, content management systems, social media content generation, assisting visually impaired people understand images.

Content-Oriented Image Retrieval

Application: Allow users to search images using natural language descriptions, e.g., "Find images of people having meetings in offices"

Scenarios: Large image library management, e-commerce product search.

Multimodal Data Analysis

Application: Combine medical images with patient records to assist diagnosis; analyze product images and user reviews for market trend prediction.

Scenarios: Healthcare, retail, finance.

Human-Machine Interaction

Application: Enable robots or virtual assistants to understand their visual environment and interact more naturally with humans.

Scenarios: Smart robots, autonomous vehicles (understanding traffic signs and road conditions).

MaiAgent Image Recognition Advantages

MaiAgent RAG integrates VLM technology to provide technical personnel and developers with convenient and efficient image recognition and understanding solutions, with the following advantages:

Attachment Image VLM Recognition (Image Understanding and Q&A)

MaiAgent RAG can perform VLM recognition on user-uploaded attachment images, providing precise image content understanding.

VLM Recognition of Embedded Images in Attachment Documents

MaiAgent RAG is actively developing VLM recognition capabilities for embedded images in attachment documents (such as PDF, Word documents). This means the system can not only understand document text but also analyze image content, achieving true multimodal document understanding. This is an advanced feature many standard RAG systems lack.

Content Q&A for Document Images

Whether for attachment images or embedded document images, MaiAgent RAG supports content-based Q&A, allowing users to ask specific questions about image details and receive accurate answers.

Content Q&A for Knowledge Base Document Images

MaiAgent RAG supports image Q&A in knowledge base documents (requires Prompt under role instructions, displaying images in Markdown format).

Post-VLM Recognition Q&A for Knowledge Base Document Images

MaiAgent introduces an experimental feature that first performs VLM recognition on images in knowledge base documents, then combines recognition results for in-depth Q&A. This further bridges image information with knowledge base connections, providing more comprehensive knowledge services.

Through MaiAgent's VLM technology, you can more deeply mine the value of image information, achieving more intelligent human-machine interaction and automated processes.

Platform Development

Overview

In addition to providing SaaS services, the MaiAgent platform also offers self-hosted solutions (private cloud, on-premises). The MaiAgent platform itself only requires general computing resources without GPU services. The model services used by MaiAgent (LLM, Embedding Model, Reranker Model) require computing power, which can utilize either cloud API inference services or on-premises GPU infrastructure.

A hybrid cloud architecture is also possible, with MaiAgent on-premises (private cloud, on-premises) and model services (LLM, Embedding Model, Reranker Model) in the cloud. MaiAgent supports LLM, Embedding model, and Reranker model provided by all cloud service providers (CSPs). If data security concerns arise or on-premises costs decrease in the future, switching to on-premises computing power can be done immediately.

MaiAgent Platform Overview

MaiAgent is a comprehensive generative AI platform providing end-to-end services from system backend to user frontend. The platform adopts a scalable microservices architecture and supports mainstream cloud environments (AWS, GCP, Azure, Oracle) as well as on-premises environments (Docker, K8s), allowing flexible deployment based on enterprise needs.

The platform core is built on **Docker**, combining diverse service modules covering API, task scheduling, data storage, cache management, and frontend/backend applications. Its overall design ensures **high availability, scalability, and cross-cloud integration capabilities** while maintaining security and operability.

Service Item	Purpose	AWS	GCP	Azure	VMs
MaiAgent Server	MaiAgent core, API, system management backend	EKS(Fargate)	GKE	AKS	Django on Docker
MaiAgent Worker Server	MaiAgent queue processing, asynchronous service worker, especially for message stream output	EKS(Fargate)	GKE	AKS	Django on Docker
MaiAgent Worker Server (Low-Priority)	MaiAgent queue processing, asynchronous service worker, especially for	EKS(Fargate)	GKE	AKS	Django on Docker

Service Item	Purpose	AWS	GCP	Azure	VMs
	document vectorization				
MaiAgent Admin Frontend	MaiAgent management platform	S3 + CloudFront	GCS+Google Cloud CDN	Azure Blob Storage + Azure CDN	Nginx + Static Files on Docker
MaiAgent Web Chat Frontend	MaiAgent web chat frontend	S3 + CloudFront	GCS+Google Cloud CDN	Azure Blob Storage + Azure CDN	Nginx + Static Files on Docker
Relational Database (RDB) - PostgreSQL	Store MaiAgent data	RDS	Cloud SQL	Azure Database	PostgreSQL on Docker
Vector Database (Vector DB) - Elasticsearch	Store vectors needed for RAG and memory functions	Elasticsearch	Elasticsearch	Elasticsearch	Elasticsearch on Docker
Static Storage	Store static files and web pages	S3	GCS	Azure Blob Storage	MinIO on Docker
Memory Cache - Redis	API cache, queue scheduling service queue	ElastiCache	Memorystore	Azure Cache	Redis on Docker

Model Services

The **MaiAgent** platform is designed to be compatible with both **cloud API inference services** and **self-hosted GPU environments**.

With **cloud API inference services**, MaiAgent can directly integrate with various LLM, Embedding, and Reranker APIs, enabling rapid scaling and support for dynamic traffic demands, facilitating experimentation and quick deployment.

In **self-hosted GPU mode**, MaiAgent can integrate with models deployed locally or in private data centers, fully utilizing GPU resources and optimizing inference while ensuring data privacy and compliance.

Users can freely switch between flexibility and cost based on their needs, or even use a hybrid approach, making **MaiAgent** a unified inference and service management layer.

	Cloud API Inference Services	Self-hosted GPU
MaiAgent Platform Compatibility	AWS Bedrock Google Vertex AI Azure AI Oracle OCI	HPE Advantech Cisco Dell
Model Capabilities	Closed-source models: High Open-source models: Same as self-hosted GPU	Depends on open-source model releases
Speed	Faster Claude 4 Sonnet: 80 token/s Gemini 2.5 Pro: 156 token/s	Medium Llama3.3 70B: 25.01 token/s (H100 example)
Investment Cost	Token API fees (pay-as-you-go)	Hardware costs Data center costs Hardware and model maintenance personnel costs Hardware depreciation
Concurrent Users	Based on cloud service provider support	Based on GPU quantity (Currently about 25 users per H100 GPU)
Data Security	Using cloud service providers (AWS, GCP, Azure, Oracle) that promise not to use data for training	Highest confidentiality, most secure
Personal Data Issues	Use DLP Server or services to remove personal data	None

Platform Deployment Environments

To meet different customer needs during development, testing, and deployment processes, our software platform provides multiple layered environments. These environment setups ensure proper validation and control of every process from development to production.

Environment Architecture

Our platform setup follows industry best practices and flexibly provides the following environments based on customer needs:

Environment Name	Notes	Main Purpose	Characteristics
PROD	Internal	Production Environment	Official external service, uses real data, requires high stability and security

Environment Name	Notes	Main Purpose	Characteristics
UAT	Internal	User Acceptance Testing	For customer and business unit acceptance, verifying system functionality meets requirements, environment similar to production
SIT	Internal, may need to connect to internal systems	System Integration Testing	Validates integration and compatibility between different modules and services, uses near-real test data
DEV	External development for added features to speed up	Development Testing Environment	For developer program development and unit testing, uses simulated data, frequent updates, tolerates errors

Environment Combinations

Different customers can choose suitable environment combinations based on project requirements, such as:

PROD Only: Suitable for small projects or simple deployment needs, directly deployed to production environment.

PROD + UAT: Suitable for projects requiring acceptance testing, ensuring functionality meets requirements before going live.

Complete Environment Combination (DEV + SIT + UAT + PROD): Suitable for large or complex projects requiring complete development, integration testing, and acceptance processes.

We flexibly configure based on customer needs, ensuring optimal balance between cost and quality.

Platform Environments (SaaS/Private Cloud/On-Premises)

MaiAgent offers multiple deployment environments to meet diverse customer needs. With the public cloud SaaS model, customers can quickly go live and enjoy maximum scalability at a lower initial cost. For customers who need to handle sensitive data, they can choose to deploy on a dedicated private cloud, offering high privacy and customizable environments. For those with higher security requirements, customers can opt for on-premises deployment which, despite higher deployment complexity and costs, provides maximum privacy and regulatory compliance solutions.

	Public Cloud (SaaS)	Private Cloud	On-Premises
Platform Deployment Location	MaiAgent Cloud	Customer Cloud (AWS, Azure, GCP, Oracle)	Customer Data Center
Large Language Models	**All Models** GPT Series, Claude, Gemini, Llama3, etc.	**All Models** GPT Series, Claude, Gemini, Llama3, etc.	Open Source Models Llama3, Gemma, Mistral, etc. Can also connect to GPT Series, Claude, Gemini
Response Quality	**Optimal**	**Optimal**	Good (Optimal, depends on model used)
Deployment Complexity	Standard	Medium	High
Initial Cost	**Low**	Medium	High
Usage Cost	Usage-based	Usage-based	**Fixed**
Time to Launch	**Quick**	Standard	High
Scalability	**Highest**	High	Low
Maintenance Difficulty	Standard	High	Highest
Privacy	Standard	High	**Highest**
Suitable Scenarios	General Use	Sensitive Data, Specific Industries	Highly Confidential, Regulatory Restrictions
Pricing Model	Fixed Price	Fixed Price	Project-based Quotation

Deployment Architecture

MaiAgent is a scalable generative AI platform that supports diverse application scenarios. To accommodate different business requirements and resource allocation methods, the platform offers two main deployment modes: **Monolithic Deployment** and **Distributed Deployment**.

This section will explain the differences between these two architectures, their applicable scenarios, their respective advantages and disadvantages, and provide practical deployment references.

Monolithic Deployment

Architecture Overview

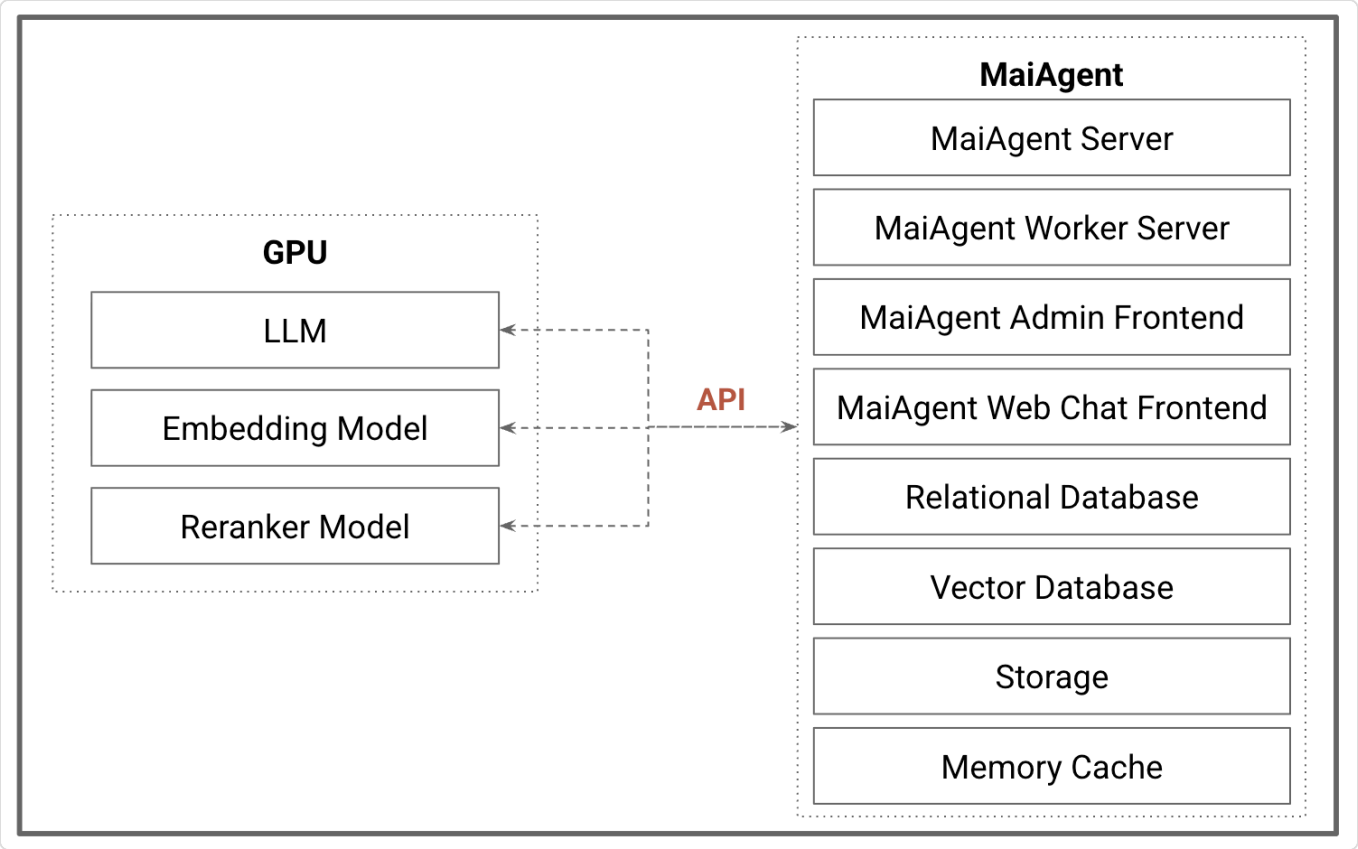
In the monolithic deployment mode, all core components of MaiAgent (such as main service, task scheduling service, data storage, database, frontend service, etc.) are installed and run on the same server. It features **centralized management**, simple deployment, and is suitable for rapid launch and testing environments.

The MaiAgent platform can operate without GPU, allowing smooth deployment and execution in standard CPU environments. However, when deployed on machines with GPU resources, the platform can also be deployed alongside models in the same environment to fully utilize hardware acceleration capabilities. Below are two common architecture diagrams for reference.

Architecture Diagrams

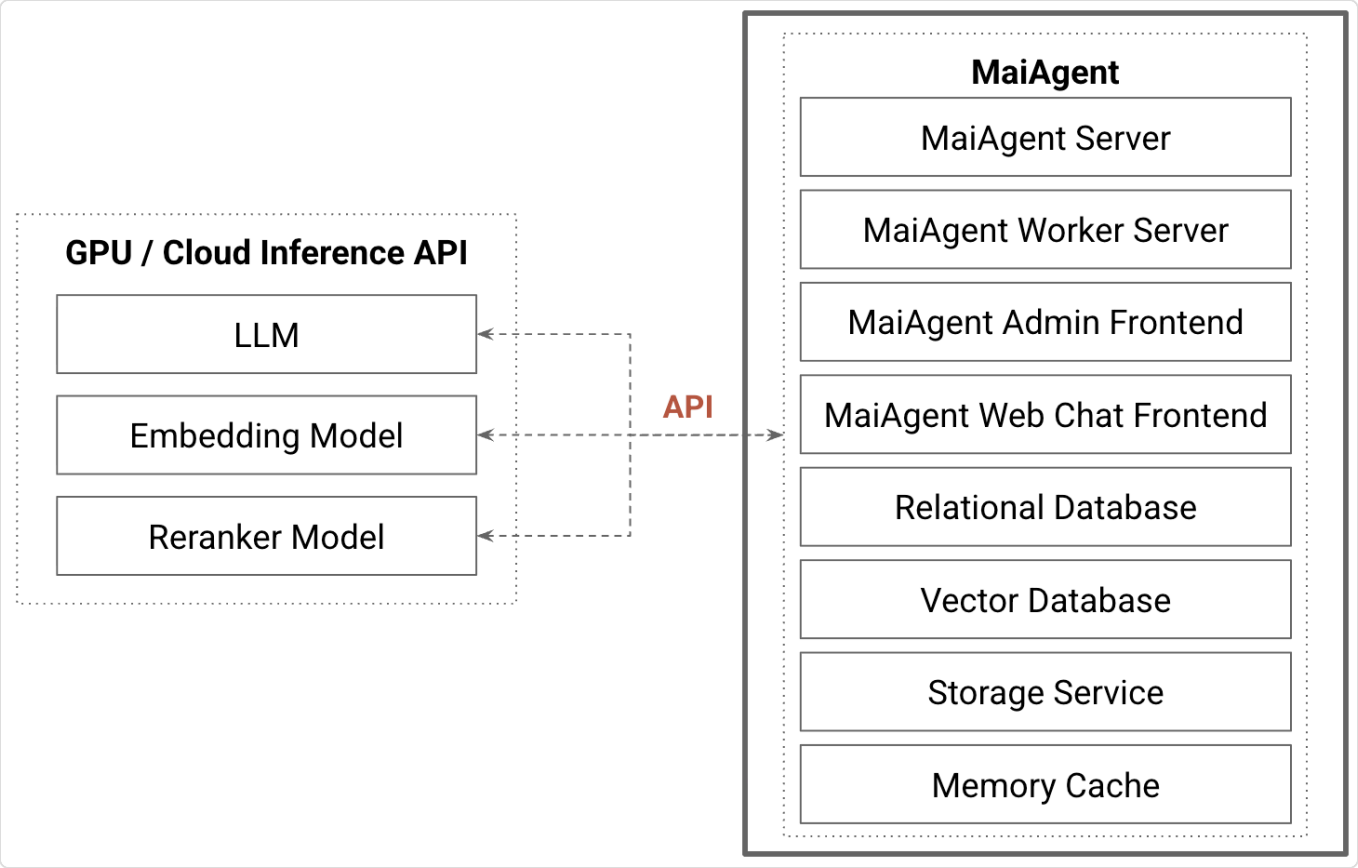
Deployment on GPU-enabled server:

MaiAgent platform and model services are installed on the same machine, with the platform handling request coordination and traffic control through internal APIs, while the model utilizes GPU for efficient inference capabilities. When the platform and model services are placed together, there's no need to purchase additional servers solely for running the platform, reducing overall hardware costs.



Deployment on non-GPU server:

When MaiAgent is deployed on a server without GPU, since model services are still required, it needs to integrate with GPU servers or cloud API inference services through API connections. When the platform and model services are separated, they can be scaled independently, allowing flexible increase or decrease of computing power based on needs, making the architecture more flexible and maintainable.



Distributed Deployment

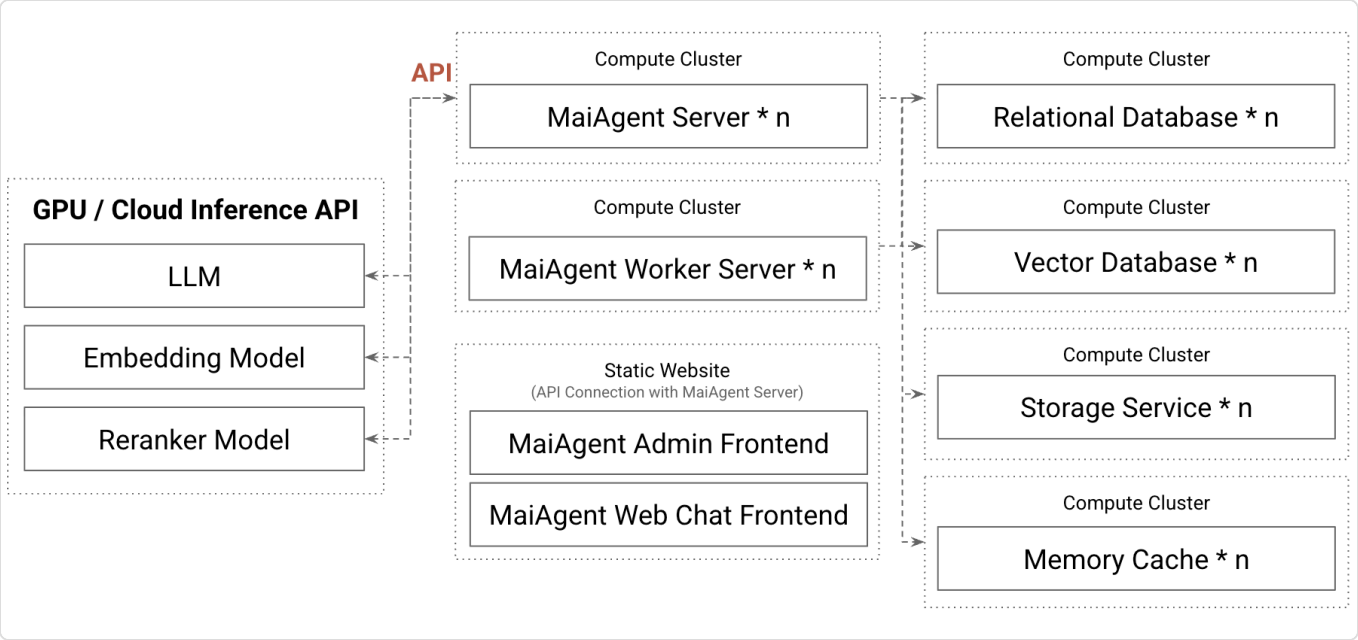
Architecture Overview

In the distributed deployment mode, MaiAgent's core modules are split into independent services and distributed across multiple servers. Different modules can be horizontally scaled according to needs, achieving high availability and large-scale processing capabilities.

Cloud Platform (Cloud PaaS) Environment
In public or private cloud environments, you can directly utilize Platform as a Service (PaaS) capabilities, such as Kubernetes, AWS ECS/EKS, GCP Cloud Run, Azure App Service, etc. These services provide container orchestration, load balancing, auto-scaling, and monitoring mechanisms, enabling quick deployment and dynamic resource adjustment of distributed modules while reducing infrastructure maintenance burden.

On-Premise VM Environment
Even in on-premise VM scenarios, you can set up container platforms or application service frameworks through virtual machines or bare metal servers to achieve distributed management and scalability similar to cloud environments. Although cluster resources, monitoring, and redundancy mechanisms need to be planned independently, high availability and elastic scaling can still be achieved.

Architecture Diagram



Deployment Mode Comparison Table

Characteristics	Monolithic Deployment	Distributed Deployment
Architecture Design	All components centralized on a single server/container	Components split into independent services, distributed across multiple nodes
Infrastructure Cost	Low, single server sufficient	High, requires multiple servers or cloud resources
Deployment Cost	Low	High, complex deployment, requires DevOps team
Maintenance Cost	Low, centralized management	High, requires cross-server and cross-service maintenance and monitoring
Scalability	None, limited by single machine resources	Yes, can independently scale bottleneck modules
High Availability	None, single point of failure leads to system-wide outage	Yes, single service failure doesn't affect overall system
Suitable Scenarios	PoC, development testing, small-scale applications	Production deployment, large-scale, multi-department

Cloud Model Inference API Service

Large Language Models (LLM)

	AWS Bedrock	Google Vertex AI	Azure AI
Claude Model	✅	✅	
GPT Model			✅
Gemini Model		✅	

Embedding Models

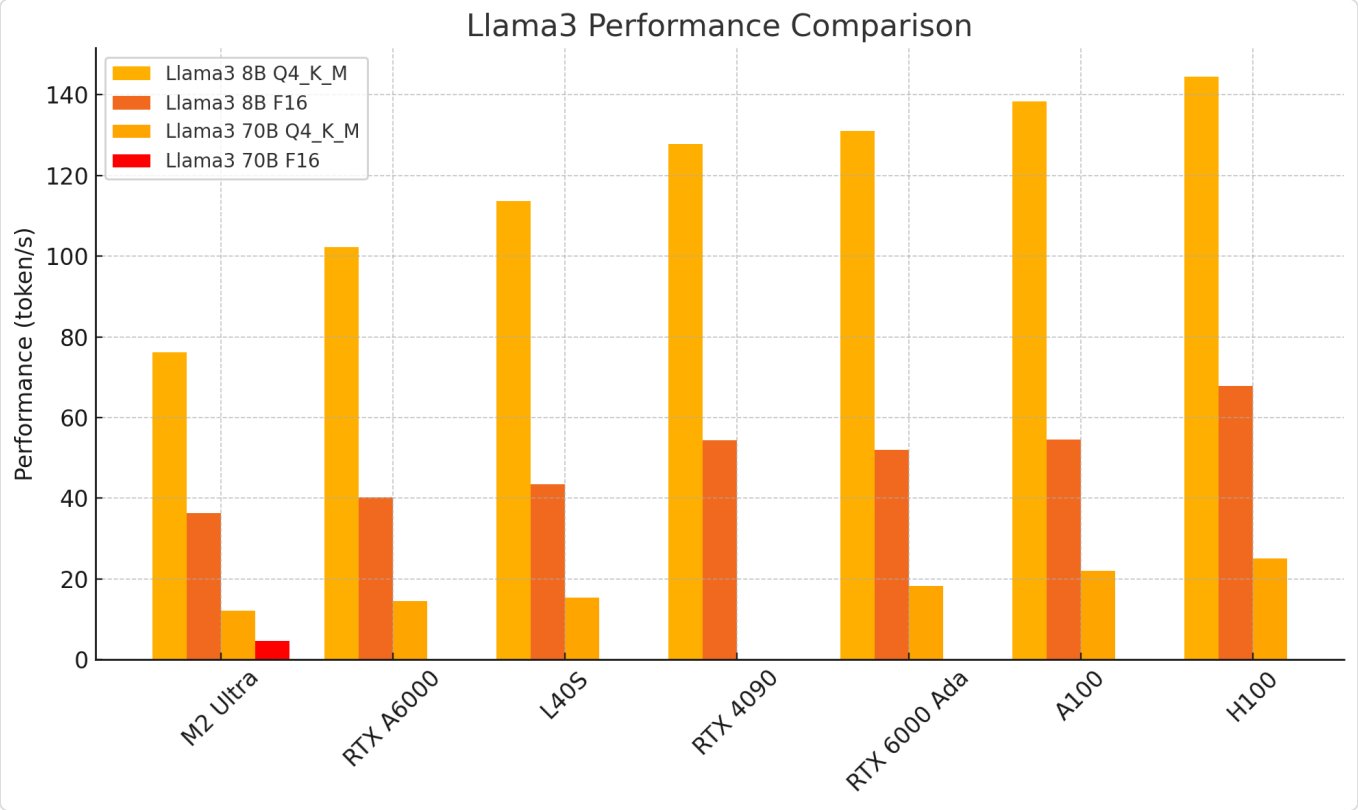
	AWS Bedrock	Google Vertex AI	Azure AI
Cohere Embedding	✅		✅
OpenAI Embedding			✅
Gemini Embedding		✅	

Reranker Models

	AWS Bedrock	Google Vertex AI	Azure AI
Cohere Reranker	✅		✅
Gemini Reranker		✅	

GPU Computing Hardware Planning

Llama3 Inference Speed on GPUs (tokens/second)



Performance Comparison of Mainstream GPUs on Llama3 8B
70B

GPU	Memory(VRAM)	8B Q4_K_M	8B F16	70B Q4_K_M	70B F16
RTX 4090	24GB	127.74	54.34	Out of Memory	Out of Memory
RTX A6000	48GB	102.22	40.25	14.58	Out of Memory
L40S	48GB	113.60	43.42	15.31	Out of Memory
RTX 6000 Ada	48GB	130.99	51.97	18.36	Out of Memory
A100	80GB	138.31	54.56	22.11	Out of Memory
H100	80GB	144.49	67.79	25.01	Out of Memory
M2 Ultra	192GB	76.28	36.25	12.13	4.71

VRAM Requirements for Llama3 Models

Model	Q4_K_M (Quantized)	F16 (Original)
Llama3 8B	4.58 GB	14.96 GB
Llama3 70B	39.59 GB	131.42 GB

Source

> 📺 **預覽連結：**https://github.com/XiongjieDai/GPU-Benchmarks-on-LLM-Inference

Hardware Configuration Recommendations

MaiAgent recommends two combinations suitable for different groups:

Two H100(80GB): Higher budget, prioritizing quality and performance
L40S(48GB) and RTX 6000 Ada(48GB): Standard budget, focusing on cost-effectiveness

For more detailed information, please contact MaiAgent's professional consultants at sales@maiagent.ai

Celery Periodic Tasks Configuration and OAuth Token Refresh

This document explains how the MaiAgent platform uses the Celery Beat scheduler to execute periodic tasks, with a particular focus on the implementation of automatic OAuth Token refresh.

1. The Role of Celery in MaiAgent

Celery is a distributed task queue system that plays a critical role in the MaiAgent platform:

Task Type	Description	Execution Method
Asynchronous Tasks	Time-consuming background processing such as document parsing and vectorization	Celery Workers receive and execute tasks
Periodic Tasks	Scheduled maintenance jobs such as Token refresh and data cleanup	Celery Beat triggers tasks on schedule
Delayed Tasks	Tasks that need to run at a specific time, such as sending scheduled notifications	Executed with countdown or eta parameters
Task Chains	Complex workflows requiring sequential execution of multiple steps	Multiple tasks combined using Celery Chain

Common use cases:

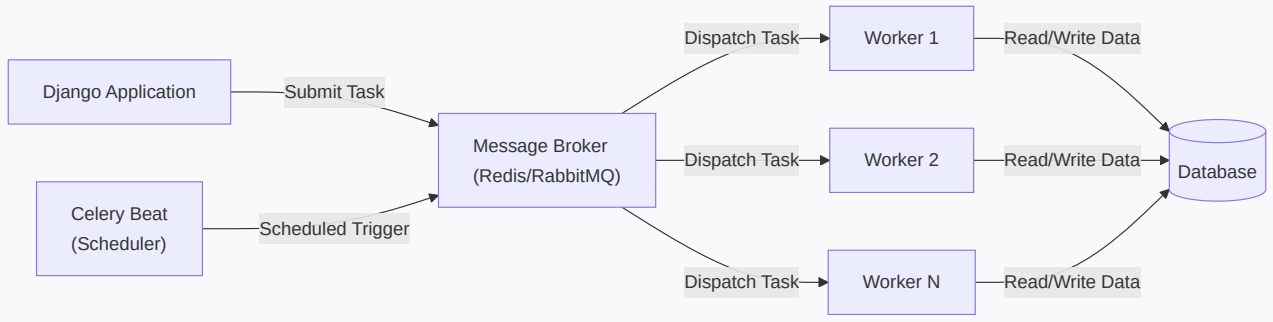
Knowledge Base Processing: Time-consuming operations like document parsing, chunking, and vectorization after file upload

Scheduled Maintenance: OAuth Token refresh, expired data cleanup, system health checks

Report Generation: Periodic generation of usage statistics, conversation quality analysis, and other reports

Notification Delivery: Batch email notifications, Webhook callbacks

2. Celery Architecture and Configuration



Mermaid 圖表

2.1 Core Component Overview

Celery Beat (Scheduler): Triggers periodic tasks according to the configured schedule

Message Broker: Uses Redis or RabbitMQ as the task queue

Celery Workers: Worker processes that execute tasks; can be scaled horizontally

Result Backend: Stores task execution results, typically using Redis or a database

2.2 Periodic Task Configuration

MaiAgent uses Django Celery Beat to manage periodic tasks:

Task Name	Schedule	Description
OAuth Token Refresh	Every hour	Automatically refreshes OAuth Tokens that are about to expire
Session Cleanup	Daily at midnight	Removes expired user sessions

Schedule expression types:

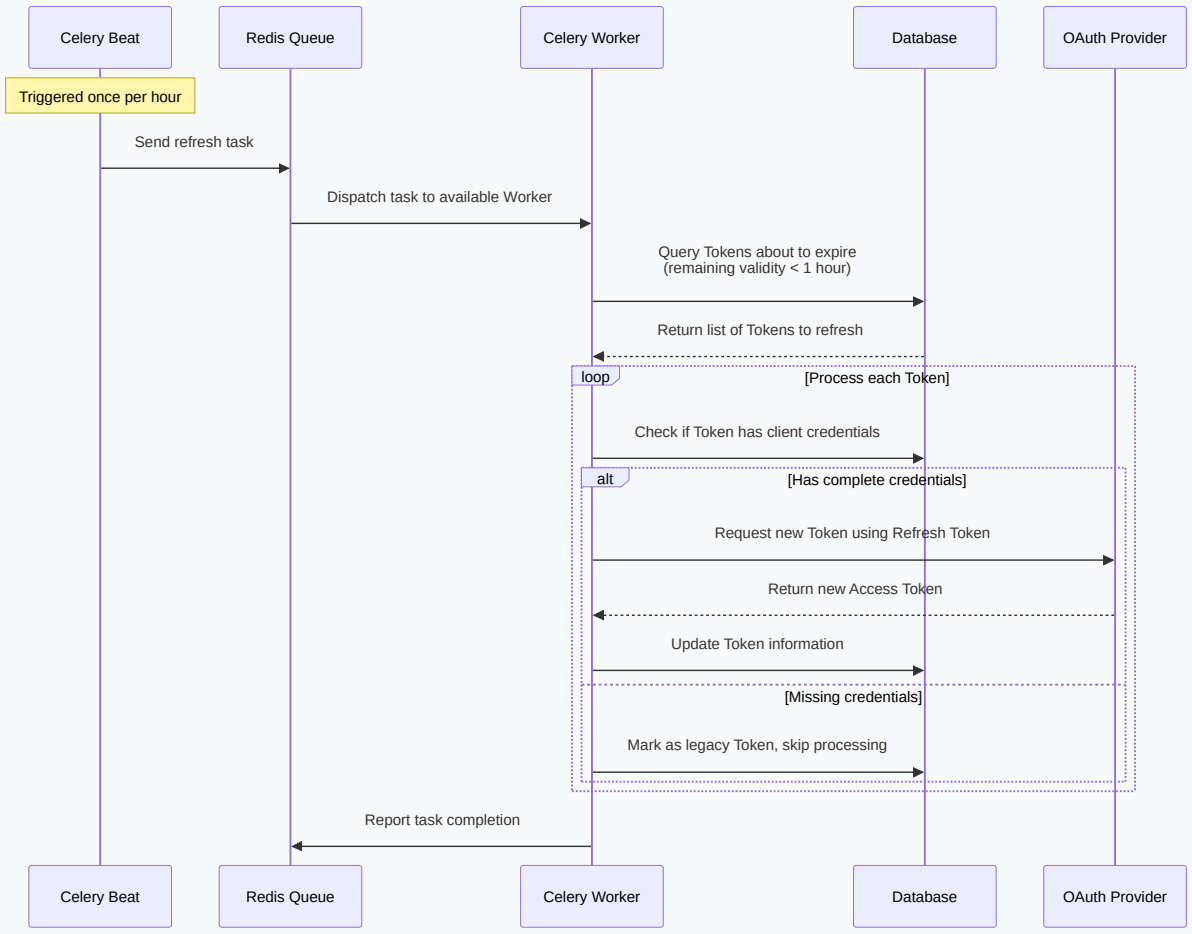
crontab: Unix cron-like time expressions supporting minutes, hours, day of week, month, etc.

timedelta: Fixed interval execution, e.g., every 30 minutes

solar: Scheduling based on sunrise and sunset times

3. Automatic OAuth Token Refresh Task

3.1 Task Execution Flow



Mermaid 圖表

3.2 Query Optimization Strategies

MaiAgent implements several query optimizations to improve Token refresh efficiency:

Legacy Token Filtering: Excludes legacy Tokens that lack client_id and client_secret

Time Window Queries: Only queries Tokens expiring within the next hour

Batch Processing: Queries multiple Tokens to refresh at once, reducing database query count

Error Handling: Records detailed error information for failed Token refreshes to facilitate troubleshooting

3.3 Error Handling Mechanisms

The Token refresh process may encounter the following error scenarios:

Error Type	Possible Cause	Handling Approach
Invalid Refresh Token	User has revoked authorization or Token has expired	Mark as requiring re-authorization, notify user
Network Timeout	OAuth service provider is temporarily unavailable	Automatic retry up to 3 times with exponential backoff
Rate Limiting	Exceeded the OAuth service's request rate limit	Delayed retry to avoid being blocked
Client Credential Error	Incorrect client_id or client_secret	Log error, notify administrator to check configuration

4. Task Monitoring and Debugging

4.1 Task Status Monitoring

MaiAgent provides multiple ways to monitor Celery task status:

Flower Monitoring Dashboard: Web interface for real-time task execution status and Worker health

Logging: Detailed records of each task's execution time, parameters, and results

Metrics Collection: Integration with Prometheus for collecting task execution metrics such as success rate and execution time

Alerting: Automatic alerts when tasks fail consecutively or execution time is abnormal

4.2 Performance Optimization Recommendations

Worker Pool Configuration:

Concurrency Tuning: Adjust Worker concurrency based on CPU core count and task type

Queue Separation: Route urgent and non-urgent tasks to different queues

Task Priority: Set higher priority for critical tasks

Task Design Principles:

Idempotency: Ensure tasks can be safely retried without side effects

Time Sensitivity: Set reasonable task expiration times to avoid executing stale tasks

Batch Processing: For large data processing, use batching strategies to prevent memory overflow

5. Technical Advantages of MaiAgent's Celery Configuration

5.1 Reliability and Stability

Task Persistence: Task information is stored in the Message Broker and persists even after system restarts

Automatic Retry: Supports automatic retry mechanisms for failed tasks

Graceful Shutdown: Workers complete in-progress tasks before shutting down

Health Checks: Periodic monitoring of Worker and Beat runtime status

5.2 Scalability and Performance

Horizontal Scaling: Easily add more Workers to handle increased traffic

Task Routing: Route different task types to dedicated Worker pools

Asynchronous Execution: Does not block the main application, improving system responsiveness

Batch Optimization: Smart batch processing reduces database query count

5.3 Operations Friendliness

Visual Monitoring: Flower provides an intuitive monitoring dashboard

Detailed Logging: Complete records of task execution for troubleshooting

Dynamic Configuration: Adjust schedules without restarting the system

Alert Integration: Integrates with enterprise monitoring systems for timely anomaly detection

6. Related Documentation

OAuth 2.0 Integration and Automatic Token Refresh Mechanism - Learn more about OAuth Token refresh logic

Deployment Architecture - Understand Celery's role in the overall system architecture

Reference Links

Celery Documentation

Django Celery Beat

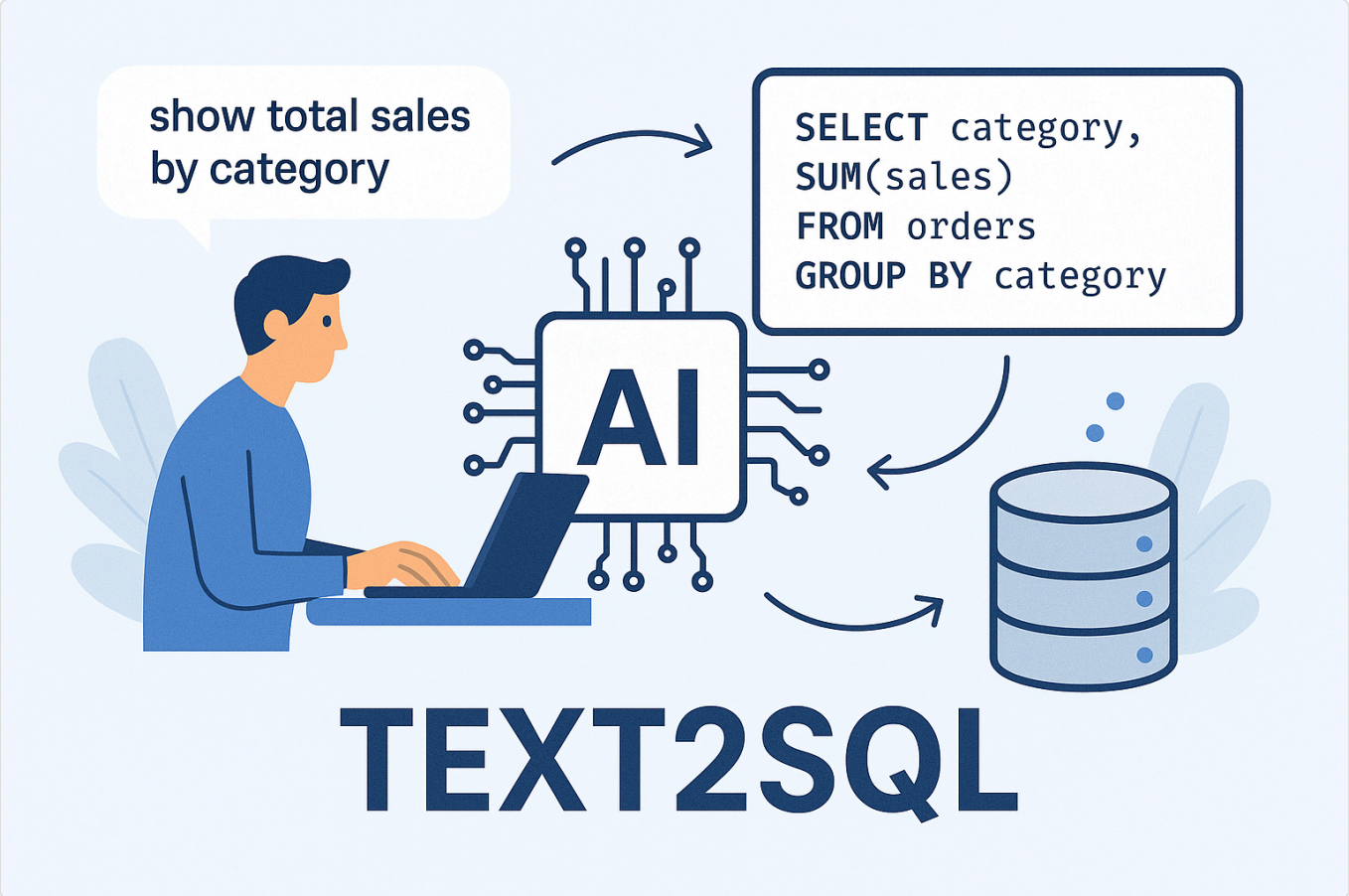
Flower - Celery Monitoring Tool

Advanced Generative AI Technologies

Text to SQL

What is Text to SQL?

Text2SQL functionality is an innovative technology powered by generative AI, aimed at revolutionizing how technical personnel interact with databases. Traditionally, extracting specific information from databases required SQL expertise and manual query writing. Text2SQL technology allows users to input natural language questions, which the AI engine automatically converts into precise SQL queries and retrieves results from your database. This significantly lowers the barrier to data access, enabling technical personnel without SQL backgrounds to easily obtain needed data and improve work efficiency.



Core Value and Importance of Text2SQL

In this data-driven era, quickly and conveniently gaining data insights is crucial. Text2SQL technology brings the following core values:

Lower Barriers: Reduces the technical threshold for data queries, enabling business personnel, analysts, and even management without SQL writing abilities to directly interact with data and obtain real-time

information.

Improve Efficiency: For technical personnel familiar with SQL, Text2SQL can save significant time when handling routine or repetitive queries, allowing them to focus on more complex data analysis and system architecture work.

Accelerate Decision Making: Real-time data query capabilities help enterprises respond faster to market changes and make more informed business decisions.

Reduce Errors: Automatically generated SQL statements can reduce potential syntax errors that might occur during manual writing.

Basic Operating Principles of Text2SQL

While specific implementation details may vary by model, Text2SQL operation typically includes these key steps:

Natural Language Understanding (NLU): The AI model first parses the user's natural language question, identifying keywords, entities, intents, and their relationships.

Schema Linking: The model needs to understand the database structure, including which tables exist, what columns each table has, and their data types and possible relationships. This step maps vocabulary from the question to specific tables and columns.

SQL Statement Generation: Based on understanding the question's intent and database structure, the model constructs a query statement conforming to SQL syntax. This may involve selecting appropriate SELECT, FROM, WHERE, GROUP BY, ORDER BY clauses.

Query Execution and Result Presentation (Optional): The generated SQL statement can be sent directly to the database for execution, and query results are returned to the user.

MaiAgent's Unique Advantages in Text2SQL Functionality

Having understood the basics and applications of Text2SQL, let's see how MaiAgent provides powerful and easy-to-use Text2SQL functionality. MaiAgent is dedicated to simplifying data access processes, with its Text2SQL functionality offering these core advantages:

Wide Database Compatibility

Understanding the diversity of enterprise data environments, MaiAgent's Text2SQL functionality **currently supports multiple mainstream relational database systems**, including:

MySQL

PostgreSQL

Oracle DB

Microsoft SQL Server (MSSQL)

This means regardless of which common database stores your data, MaiAgent can seamlessly connect and provide consistent natural language query experience.

Seamless Spreadsheet Integration

Beyond traditional databases, MaiAgent understands that many temporary or small datasets often exist in spreadsheet form. For this, MaiAgent provides an innovative feature: **ability to automatically convert spreadsheet data (supporting .xlsx, .xls, .csv formats) into queryable temporary databases.**

Users only need to upload spreadsheet files, and MaiAgent will parse their structure, allowing you to query them using natural language just like standard databases. This greatly expands Text2SQL's application range, enabling quick utilization of unstructured or semi-structured data.

Minimal Setup, Quick Start

MaiAgent's design philosophy emphasizes simplifying configuration. To use Text2SQL functionality to connect to your database, you **only need to provide the database connection URL** (Connection String/URL). No complex driver installation, environment variable settings, or other cumbersome additional configurations are needed.

This plug-and-play characteristic allows technical personnel to quickly deploy MaiAgent to existing environments and enables end users to immediately begin using natural language queries.

Intelligent Schema Understanding

MaiAgent's core AI engine possesses powerful Schema understanding capabilities. After connecting to your database, it can **automatically detect and understand all tables in the database and their column structures, data types, and potential relationships.**

This automated understanding eliminates the hassle of manual schema annotation or configuration, ensuring Text2SQL functionality can accurately map natural language questions to correct data entities.

Flexible Query Scope Specification

While MaiAgent can automatically understand the entire database structure, in some cases, users may only care about specific tables. MaiAgent fully considers this, allowing users to **also specify which Tables to search.**

By specifying query scope, this not only improves query relevance and accuracy (avoiding searches in irrelevant tables) but also enhances query efficiency in large databases, delivering results faster.

Conclusion

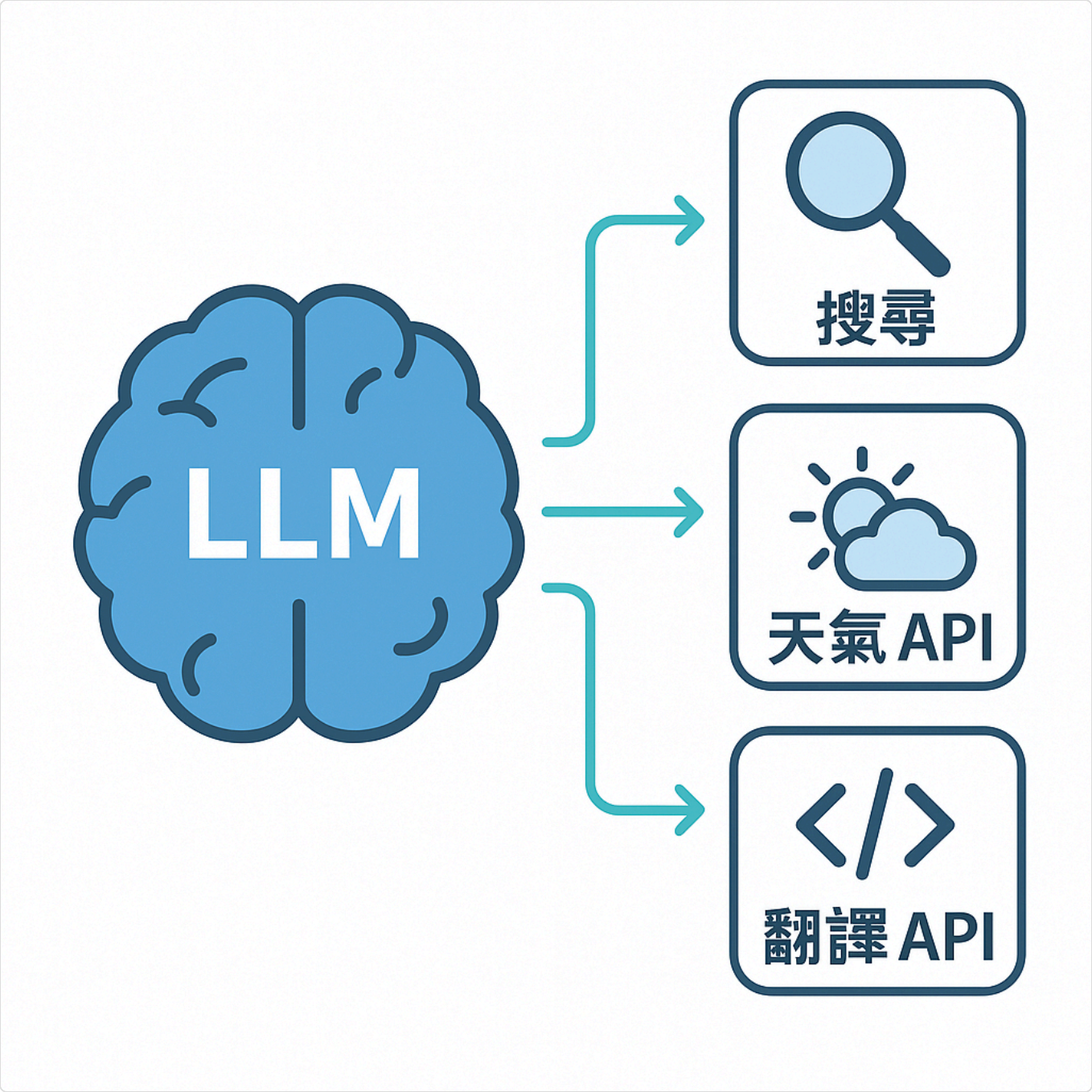
MaiAgent's Text2SQL functionality aims to empower your team, enabling everyone to easily master data. With its extensive database support, unique spreadsheet integration capabilities, minimal configuration process, intelligent Schema understanding, and flexible query scope settings, MaiAgent is your capable assistant in data exploration and analysis journey. We encourage technical personnel to fully utilize these features to unleash the full potential of your enterprise's data.

Function Calling

What is Function Calling?

Function Calling refers to the ability of Large Language Models (LLMs) to understand when external code or APIs (which we call "tools" or "functions") need to be executed during user interaction or task processing, and to accurately output the required function names and parameters in a structured format (typically JSON).

Importantly, the LLM itself does not directly execute these functions - it is only responsible for "deciding" which function is needed and what parameters to call it with. The actual function execution is completed by the calling program, which then returns the execution results to the LLM, allowing the model to use this information to generate more precise, relevant responses or continue completing more complex tasks.



RAG Flow

How Function Calling Works

The Function Calling workflow typically includes the following key steps, forming a dynamic interaction cycle:

1. Intent Recognition and Structured Instruction Generation

Users make requests through natural language

LLM parses requests and understands user intent

Determines if external functions need to be called

Generates structured output (JSON format) containing function names and parameters

The model itself does not execute any functions

2. External Function Execution

Application receives structured instructions generated by LLM

Executes corresponding external functions based on function names and parameters

May involve:

Database queries

Third-party service calls

Computational operations

3. Result Return and Integration

External function execution completes

Results returned to application

Application passes results to LLM

4. Generate Final Response or Further Actions

LLM receives function execution results

Integrates information into context

Generates final natural language response

Or decides if another function needs to be called

This closed-loop process transforms the LLM from a simple text generator into an intelligent agent capable of interacting with the external world, obtaining real-time information, and executing actual operations.

Core Uses and Value of Function Calling

Function Calling brings revolutionary changes to LLM applications, making them more practical and powerful:

1. Real-time and Dynamic Data Retrieval and Integration

Applications

Automatic external API calls for latest data

Weather queries

Stock market

Real-time news

Product inventory

Avoids knowledge cutoff issues with training data

Value

Ensures information timeliness and accuracy

Increases user trust

2. Complex Business Process Automation

Applications

Suitable scenarios

Customer service

E-commerce

Internal enterprise systems

Automated operations

Order queries

Product recommendations

Appointment scheduling

Problem solving

Return requests

Ticket creation

Value

Improves work efficiency

Reduces manual costs

Provides 24/7 instant service

Enhances user experience

3. Multi-step Task Smart Agency and Coordination

Applications

Complex task examples:

Travel planning
Flight booking
Hotel recommendations
Itinerary scheduling
Smart dispatching of multiple functions
Flight APIs
Hotel APIs
Map services
Calendar APIs

Value

Achieves end-to-end task automation
Handles complex user needs

4. Empowers LLM with Structured Operation Capabilities

Applications

Natural language conversion to structured operations
Text-to-SQL queries

Value

Enables non-technical users to interact with complex systems through natural language

Common Application Scenarios for Function Calling

Category	Application Examples	Core Features and Advantages
Information Retrieval and Q&A	AI assistant real-time queries for company policies, legal texts, product specifications	Ensures response timeliness, accuracy and reliable sources
Automated Workflows	E-commerce customer service self-service order status queries, refund initiation, automatic after-sales ticket creation	Triggers standardized backend system processes through Q&A, improving efficiency
Data Analysis and Visualization	Calls calculators, statistical function libraries, chart generation APIs based on user commands	Enables natural language-driven data computation, analysis and visual result presentation

Category	Application Examples	Core Features and Advantages
External System Integration	LLM controls smart home devices, operates internal enterprise software, sends emails/messages	Extends LLM's understanding capabilities to actual system operations and hardware control
Content Generation Assistance	Calls SEO tools to optimize articles, check grammar, generate images/code snippets	Combines external tools to enhance LLM's content generation quality in specific domains

MaiAgent Function Calling's Unique Advantages

Choosing the MaiAgent platform to implement and manage Function Calling capabilities brings the following key advantages to your applications:

Powerful and Flexible Tool Management:\
MaiAgent provides an intuitive interface and API for easily registering, configuring and managing your external functions (tools). You can define detailed tool names, descriptions, input parameters (including types, requirements, descriptions, etc.) and expected output formats.

Reliable Structured Parameter Generation:\
MaiAgent ensures LLMs can reliably generate parameters conforming to JSON Schema or other specified formats based on tool definitions, reducing function execution failures due to format errors.

Seamless Development and Integration Experience:\
MaiAgent provides an intuitive and user-friendly interface allowing users to easily integrate without writing code, quickly set up and configure required function call tools, improving efficiency.

Summary

Function Calling technology elevates LLMs from passive "knowledge Q&A machines" to active "task executors" and "intelligent agents", perfectly combining LLMs' powerful natural language understanding capabilities with the concrete execution capabilities of the external world through standardized interfaces, opening up unlimited application innovation space for developers.

MaiAgent, with its advantages in tool management, development integration and platform stability, provides an ideal Function Calling solution to help you easily build smarter, more powerful AI Agent applications that can solve real problems.

Skill Module

What is Skill?

Skill is a modular mechanism in the MaiAgent platform for encapsulating AI assistant professional capabilities. Unlike Function Calling which allows LLMs to call a single tool, Skill packages a set of **Instructions**, **Tools**, and **Resources** into a reusable capability unit, enabling AI assistants to dynamically unfold complete task execution workflows during conversations.

In simple terms:

Function Calling
Tools: LLM calls a single API or function

Skill: LLM unfolds a complete set of SOP instructions and calls multiple tools as needed during the process

How Skill Works

Overall Architecture

The operation of Skill in the MaiAgent system involves the following core components:

Component	Description
Skill Model	Stores skill name, description, instruction content, and attached tool associations
ChatbotSkill	Many-to-many association between AI assistants and Skills (Junction Model)
SkillExpandTool	Built-in system tool responsible for dynamically expanding skill instructions during conversations
LoadSkillTool	Built-in system tool responsible for dynamically loading skill-attached tools
SkillFileParser	Parses uploaded `.skill` `.zip` files, extracts SKILL.md and resources

Workflow

```
User inquiry
    │
    ▼
AI assistant receives message
    │
    ▼
LLM analyzes intent, matches descriptions of bound skills
    │
    ├─ Skill matched → Call SkillExpandTool
    │                      │
    │                      ▼
    │              Expand complete skill instruction content
    │              + List available attached tools
    │                      │
    │                      ▼
    │              LLM executes step by step according to instructions
    │              Calls LoadSkillTool to load tools as needed
    │                      │
    │                      ▼
    │              Tool execution results returned to LLM
    │                      │
    │                      ▼
    │              LLM generates response according to instruction format
    │
    └─ Not matched → Regular conversation response
```

1. Skill Expansion

When an AI assistant determines that a user's question should be handled by a particular skill, the system calls `SkillExpandTool`:

Input: Skill name (`skill_name`)

Processing: Query skill by name, retrieve complete instruction content and attached tool list

Output: Instruction content in Markdown format + available tool list

```
{
  "name": "expand_skill",
  "arguments": {
    "skill_name": "休假天數計算"
  }
}
```

2. Tool Dynamic Loading

After skill expansion, when the LLM needs to call tools during instruction execution, it dynamically loads them via `LoadSkillTool` :

Input: Tool name (`tool_name`)

Processing: Query tool (supports MCP tools and API tools, case-insensitive)

Output: Tool's ID, name, description, and availability status

This dynamic loading mechanism ensures that skill tools don't need to be fully loaded into the conversation context in advance, reducing token consumption.

SKILL.md Specification

The core definition file for Skill is `SKILL.md` , using YAML Frontmatter + Markdown content format:

```
---
name: Skill name
description: Skill description (determines when AI triggers this skill)
---

# Skill title

## Trigger Conditions
Triggered when user asks XXX related questions.

## Execution Steps

### Step 1: Confirm Information
Guide user to provide necessary information ...

### Step 2: Query Data
Use retrieve_text_nodes to search knowledge base ...

### Step 3: Generate Response
Reply according to the following format:
- ① Applicable regulations
- ② Calculation results
- ③ Reference examples
```

Frontmatter Fields

Field	Required	Description
`name`	Yes	Skill name, must be unique within organization
`description`	Yes	Skill description, used by LLM to determine whether to trigger this skill

Instruction Writing Guidelines

Clear trigger conditions: Clearly define "when to trigger" and "when not to trigger" in description and instructions

Structured steps: Use numbered steps (Step 1, Step 2...) for LLM to execute sequentially

Tool call guidance: Explicitly specify which tool to use and query strategy in steps

Response format definition: Define final response format using templates or tables to ensure output consistency

Error prevention checklist: Add checklist at end of instructions for LLM self-verification

Skill Package

File Structure

Uploaded `.skill` or `.zip` files should contain the following structure:

```
my-skill.skill (ZIP format)
├── SKILL.md          # Required: Skill definition file
├── reference.pdf     # Optional: Reference materials
├── template.xlsx     # Optional: Template files
└── examples/         # Optional: Example data
    └── case1.json
```

Security Validation

The system performs the following security checks when parsing skill packages:

Compression ratio check: Maximum 100:1 to prevent ZIP bomb attacks

Path traversal protection: Block `../` and other path injection attempts

File size limit: Based on organization's upload limit (default 100 MB)

SKILL.md validation: Must exist and contain valid `name` and `description`

Storage Architecture

```
S3 storage path:
media/skills/{organization_id}
{skill_id}/package.zip    # Skill package
media/skills/{organization_id}
{skill_id}/resources/     # Resource files
```

API Endpoints

Skill CRUD

Method	Endpoint	Description
`GET`	`/api/v1/skills/`	List all skills in organization (supports pagination, search)

Method	Endpoint	Description
`POST`	`/api/v1/skills/`	Manually create skill
`GET`	`/api/v1/skills/{id}`	Get skill details
`PATCH`	`/api/v1/skills/{id}`	Update skill
`DELETE`	`/api/v1/skills/{id}`	Delete skill (includes S3 cleanup)

Skill Package Operations

Method	Endpoint	Description
`POST`	`/api/v1/skills/upload/`	Upload `.skill` `.zip` to create skill
`POST`	`/api/v1/skills/{id}/reupload/`	Re-upload to replace skill package
`GET`	`/api/v1/skills/{id}/export/`	Export skill as `.skill` file
`GET`	`/api/v1/skills/{id}/package-structure/`	View skill package file structure

Resource Management (Nested Routes)

Method	Endpoint	Description
`GET`	`/api/v1/skills/{id}/resources/`	List skill resource files
`POST`	`/api/v1/skills/{id}/resources/`	Upload resource (manually created skills only)
`DELETE`	`/api/v1/skills/{id}/resources/{rid}`	Delete resource (manually created skills only)
`GET`	`/api/v1/skills/{id}/resources/{rid}/download/`	Download resource file

Skill Creation Examples

Manual creation:

```
POST /api/v1/skills/
{
  "name": "退貨處理流程",
  "description": "當客戶要求退貨、換貨時觸發。查詢訂單、確認退貨資格、建立退貨單。",
  "instructions": "# 退貨處理\n\n## Step 1：查詢訂單\n ... ",
  "attached_tool_ids": [
    "uuid-of-order-query-tool",
    "uuid-of-return-api-tool"
  ]
}
```

Upload skill package:

```
curl -X POST /api/v1/skills/upload/
  -F "file=@my-skill.skill"
  -F "attached_tool_ids=["uuid-1\","uuid-2\"]"
```

Binding Skills to AI Assistants

Skills establish many-to-many associations with AI assistants through the `ChatbotSkill` junction model:

One AI assistant can bind multiple skills

One skill can be used by multiple AI assistants

Bind/unbind operations are recorded in the `ChatbotSkillEvent` history tracking table

```
AI Assistant A ──┬── Skill: Return Processing
                 ├── Skill: Product Consultation
                 └── Skill: Email Sending

AI Assistant B ──┬── Skill: Product Consultation  ← Shared
                 └── Skill: Meeting Query
```

Unique Advantages of MaiAgent Skill

1. Instruction and Tool Decoupling

Skill instructions and attached tools are managed independently, allowing instruction logic to be adjusted at any time without affecting tool settings, and vice versa. This makes skill maintenance and iteration more flexible.

2. Dynamic Expansion Mechanism

Skill instructions are not preloaded into conversation context but are dynamically expanded when the LLM determines they are needed. This significantly reduces token consumption per conversation, especially suitable for AI assistants with numerous bound skills.

3. Skill Package Ecosystem

Through the `.skill` file format, skills can be shared and imported between different organizations. Enterprises can build internal skill libraries or package best practices as skill packages for distribution to partners.

4. Complete Version Control

Skills created via upload support re-upload (Reupload), allowing skill package content to be updated without deleting the original skill. The system completely replaces the skill package and resource files to ensure version consistency.

5. Permission Control

Skill access is controlled by the `chatbot_skill_access` permission. Organization administrators can precisely manage who can create, edit, and delete skills through the role permission system.

Canvas

What is Canvas

Canvas is an extension interface of the Agent mode that enables AI assistants to generate and manage structured visual content. This technology allows AI to not only engage in conversations but also create, edit, and present rich multimedia content, including interactive web components, charts, documents, and code.

In traditional conversational AI, assistants could only provide text responses, but Canvas technology breaks this limitation, enabling AI assistants to:
- Create real-time visualizations
- Generate interactive components
- Provide structured information presentation
- Support real-time code rendering

Core Features

Canvas technology extends traditional text-based conversations to visual content creation, enabling businesses to:

Real-time Content Generation: AI can instantly create various formats of content based on user needs

Interactive Editing: Users can directly interact with and modify generated content

Multi-format Support: Supports HTML, React, SVG, Markdown, Mermaid, and other content formats

Seamless Integration: Perfect integration into conversation flow, providing more intuitive user experience

Professional Quality: Generated content meets professional design and development standards

Real-time Preview: WYSIWYG content editing experience

Technical Advantages

Intelligent Content Recognition: AI can automatically determine when to use Canvas for content presentation

Format Adaptation: Automatically selects the most suitable presentation format based on content characteristics

Real-time Rendering: Provides real-time visualization effect previews

Cross-platform Compatibility: Ensures consistent performance across different devices and platforms

Performance Optimization: Uses advanced rendering technology to ensure smooth user experience

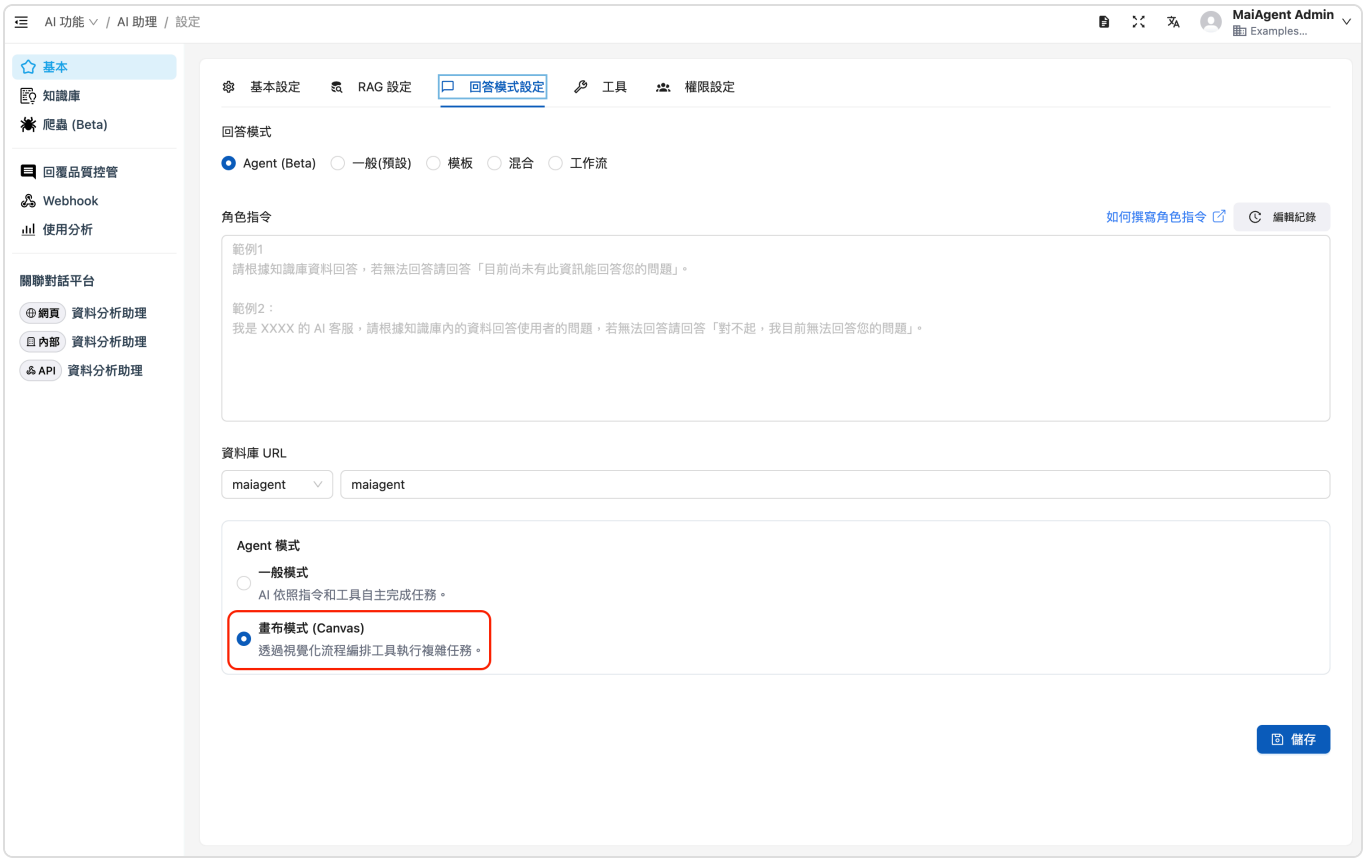
Comparison with Existing Market Solutions

Feature	MaiAgent Canvas	Traditional AI Assistant	Other Visualization Tools
Real-time Visualization Generation	✅ Automatic detection and generation	❌ Text-only responses	⚠️ Manual operation required
Multi-format Support	✅ Multiple professional formats	❌ Text only	⚠️ Limited formats
Conversational Interaction	✅ Seamless integration	✅ Text conversation	❌ Standalone tools
Enterprise-grade Security	✅ Local deployment	⚠️ Vendor dependent	⚠️ Vendor dependent
Customization Capability	✅ Highly customizable	⚠️ Limited	⚠️ Template-based

MaiAgent Canvas Feature Introduction

Setup and Activation

Canvas functionality is based on MaiAgent's Agent mode, specifically configured with Canvas mode to handle visual content generation needs.



Response mode must select Agent, with Canvas mode enabled under Agent mode

Features and Architecture Overview

Frontend Rendering Layer

Real-time Preview Engine: Supports real-time rendering of all Canvas formats

Interactive Interface: Provides intuitive editing and operation interface

Responsive Design: Automatically adapts to various device screen sizes

AI Processing Layer

Content Analysis Engine: Intelligently analyzes user needs

Format Selector: Automatically determines the most suitable presentation format

Quality Controller: Ensures accuracy and completeness of generated content

Intelligent Content Recognition Mechanism

The system supports five main content formats, each with specific application scenarios:

HTML: Suitable for web content and static displays

Supports complete CSS styling

Can embed multimedia elements

Suitable for complex layout designs

React: For interactive components and dynamic content

Supports JSX syntax and React Hooks
Can create complex interactive functionality
Suitable for dynamic data display

SVG: Vector graphics and chart design

Infinite scaling without quality loss
Small file size, fast loading
Suitable for icons, illustrations, simple charts

Markdown: Structured document and code display

Supports GitHub Flavored Markdown
Code syntax highlighting
Suitable for technical documentation, manuals

Mermaid: Flowchart and diagram generation

Supports multiple chart types
Programmatic description, easy to maintain
Suitable for business processes, system architecture diagrams

Content Generation Process

Requirement Analysis: AI analyzes user input to determine if Canvas is needed
Keyword identification: "draw," "create," "design"
Content complexity assessment
Visualization needs assessment

Format Selection: Automatically selects the most suitable presentation format based on content characteristics

Content type analysis
Interactive requirements assessment
Best format recommendation

Content Creation: Uses specialized generation algorithms to create structured content

Template matching
Custom generation

Brand element integration

Quality Verification: Ensures correctness and usability of generated content

Syntax checking
Rendering testing
Compatibility verification

Rendering Presentation: Real-time content rendering in user interface

Real-time preview
Performance optimization
Error handling

Intelligent Decision Logic

Without specific role instructions, Agent determines whether to use Canvas based on the following factors:

Content Complexity: Structured content prioritizes Canvas use
Tables, lists, hierarchical structures
Multi-step process explanations
Comparative analysis content

Interactive Requirements: Content requiring user interaction

Form design
Buttons and controls
Dynamic data display

Visualization Requirements: Charts, graphics, and visual elements

Statistical charts
Diagrams
Architecture diagrams

Format Requirements: Content with specific format requirements

Code display
Document templates
Professional reports

Demo Video

📺 **預覽連結：**https://drive.google.com/file/d/1GW7o3m130M-yESZ4h6VwZvNXDhdJDiFe/view?usp=drive_link

MaiAgent Canvas Core Values and Application Scenarios

Core Values

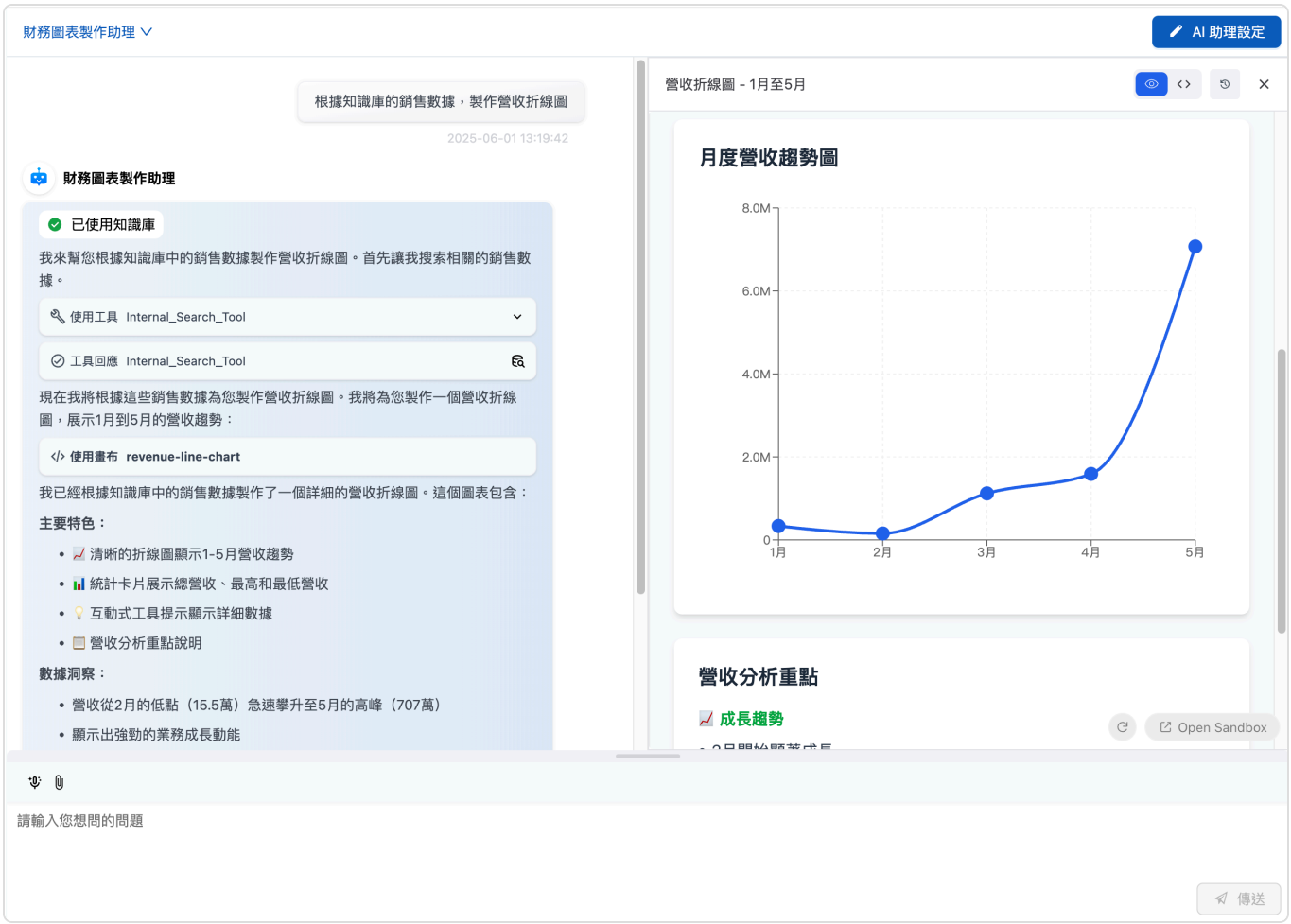
1. Improved Communication Efficiency

Transforms complex concepts into visual presentations

Reduces verbosity in text explanations

Enhances information transmission accuracy

Actual Benefits: 60% reduction in explanation time, 85% improvement in comprehension accuracy



Creating charts through knowledge base data upload

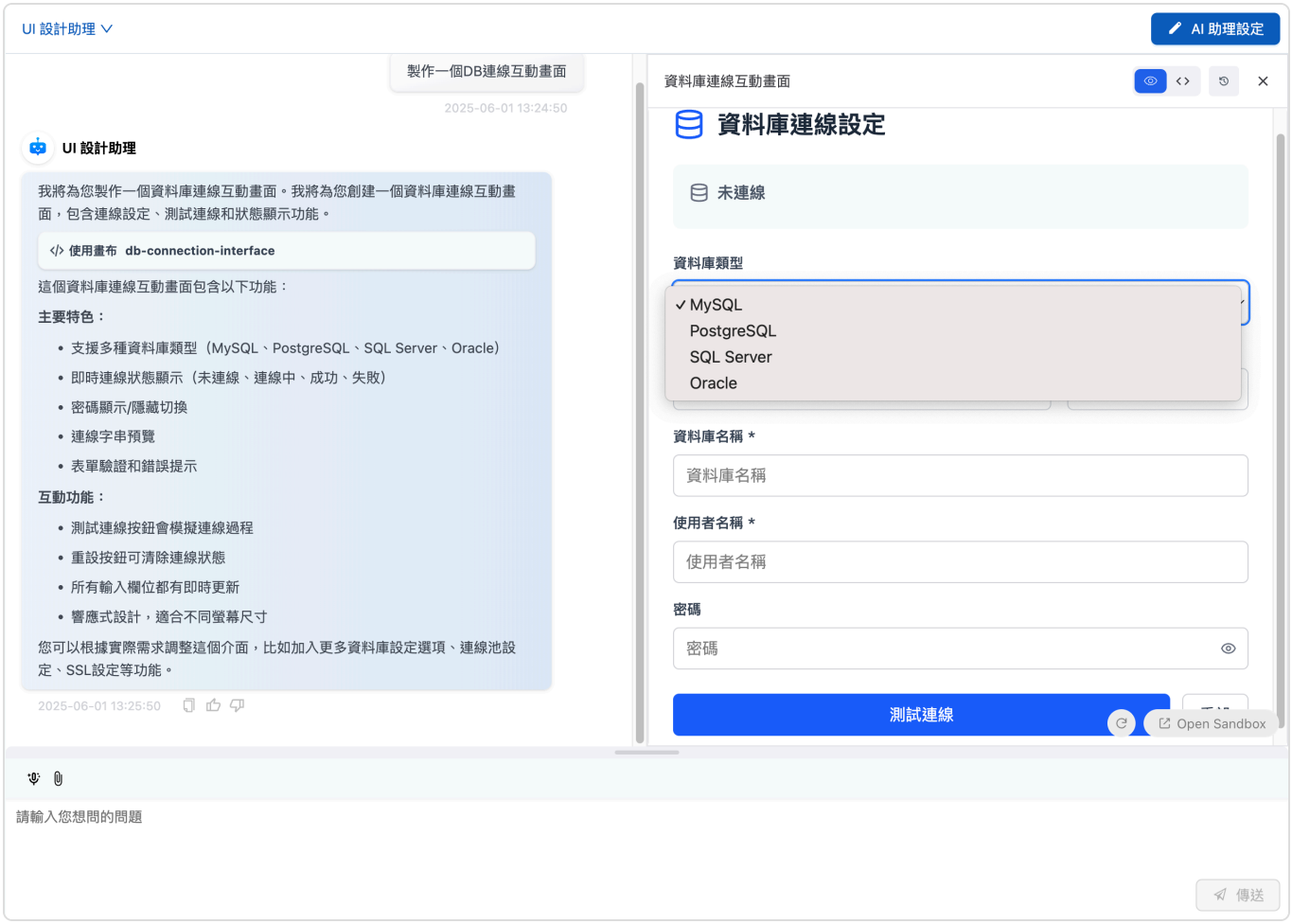
2. Enhanced User Experience

Provides interactive content operation experience

Supports real-time preview and modification

Creates more intuitive human-machine interaction

User Satisfaction: 78% increase in satisfaction compared to traditional text responses



Creating interactive UI interfaces based on user requests

3. Accelerated Content Creation

Automated generation of professional format content

Reduced manual coding and design time

Provides one-stop content solution

Efficiency Improvement: 4-10 times increase in content creation speed

Creating professional content flowcharts based on user requests

4. Lower Technical Barriers

No professional design or programming skills required

Complex content creation through natural language

Suitable for users with various technical backgrounds

5. Professional Domain Adaptation

Quick Industry Integration: High integration with industry internal data through knowledge base

Financial sector: Risk assessment reports, investment analysis charts

Manufacturing: Production flowcharts, quality control charts

Healthcare: Patient data visualization, treatment flowcharts

Education: Course design, learning progress tracking

Custom Prompts: Canvas generation logic can be adjusted based on enterprise needs

Enterprise terminology integration

Specific format requirements settings

Brand style guidelines

Localized Content: Supports Taiwan-localized content formats and standards

Taiwan regulatory document formats

Local business practices

Cultural adaptation adjustments

Main Application Scenarios

📊 Data Visualization

Flowchart Design: Create business process diagrams, system architecture diagrams using Mermaid format

Organization Charts: Generate clear enterprise organizational structures, department relationship diagrams

Data Charts: Create interactive data analysis charts, dashboards

Schedule Planning: Gantt charts, project schedules, milestone planning

💻 Programming Development

Code Display: Present code examples with formatting, syntax highlighting

API Documentation: Generate structured technical documents, interactive API documentation

Interactive Components: Create operable React components, demonstration pages

System Design: Architecture diagrams, database ER diagrams, network topology diagrams

Code Review: Visualize code structure, dependency relationships

📝 Document Creation

Report Generation: Automated generation of formatted reports, data analysis reports

User Manuals: Create illustrated operation guides, product manuals

Project Proposals: Create professional project presentation documents, business proposals

Meeting Minutes: Structured meeting records, decision tracking tables

Educational Training: Interactive learning materials, course outlines

🎨 Creative Design

Prototype Design: Quickly create website or application prototypes, wireframes

Graphic Creation: Generate vector graphics, icon designs using SVG format

User Interface: Design interactive UI components, interface simulations

Brand Design: Logo design concepts, brand visual elements

Marketing Materials: Social media images, advertising banner designs

🏢 Enterprise Applications

Internal Training: Employee handbooks, process explanations, safety regulations

Customer Service: FAQ visualization, service flowcharts

Sales Support: Product introduction pages, feature comparison tables

Project Management: Task allocation diagrams, progress tracking charts

Through MaiAgent Canvas technology, enterprises can achieve truly intelligent content generation and management, not only improving work efficiency but also creating innovative user interaction experiences. This technology represents a significant breakthrough in the AI assistant field, elevating conversational AI to a new level of visualization and interactivity.

Experience the powerful features of Canvas technology now, and let your AI assistant not just speak, but create!

AI Security Guardrails

To ensure AI services provide intelligent responses while maintaining security, compliance, and reliability, MaiAgent integrates AWS Guardrails and AI Agent role instructions to form a dual AI security protection framework. These two components complement each other, enabling AI to achieve higher standards in content filtering, sensitive information protection, behavior control, and hallucination suppression, providing an enterprise-grade AI security architecture.

This dual AI security mechanism adopted by MaiAgent ensures that AI applications in enterprise scenarios meet the following key standards:

✅ Enhanced AI Content Security: Prevents AI from generating non-compliant or risky content, improving AI compliance capabilities.\
✅ Ensures AI Meets Business Requirements: Through role instructions, enables AI to provide accurate and valuable responses within specified boundaries.\
✅ Reduces AI Hallucination Impact: Dual mechanism ensures AI only provides verified information, improving reliability.\
✅ Increases User Trust: Enterprises can confidently deploy AI, ensuring AI responses align with brand image and business needs.

Through this architecture, MaiAgent not only provides high-performance AI interaction experiences but also ensures AI operations meet enterprise-grade security standards, maximizing AI value in intelligent applications. Combining these two elements, MaiAgent can deliver high-quality intelligent responses on a secure, compliant foundation, ensuring AI maximizes its value in business scenarios.

For further technical requirements, you can adjust Guardrails security policies and fine-tune AI role instructions to better match enterprise needs.

AWS Guardrails and AI Agent Role Instructions are Two Complementary AI Control Mechanisms:

Function	AWS Guardrails	AI Assistant Role Instructions
Content Filtering	✅ Automatically filters harmful, inappropriate content	❌ Mainly controls AI response methods
Sensitive Data Protection	✅ Blocks PII, confidential information leakage	❌ Cannot directly filter sensitive data
Behavior Control	✅ Prevents AI bias or non-compliant behavior	✅ Limits AI response scope and style

Function	AWS Guardrails	AI Assistant Role Instructions
Hallucination Control	✅ Filters inaccurate information	✅ Specifies AI response methods to reduce hallucination
Enterprise Customization	✅ Can set different security levels	✅ Can customize AI roles and response scope

AWS Guardrails: Security Layer for AI Content and Behavior

As the first security mechanism, AWS Guardrails handles automated content review and risk control, ensuring AI output meets enterprise and regulatory requirements. Its core functions include:

Content Filtering: Blocks violence, hate, discrimination, inappropriate language, or non-compliant information, ensuring AI responses meet ethical and compliance standards.

Data Protection: Prevents AI from generating or leaking Personal Identifiable Information (PII) or confidential enterprise data, reducing information security risks.

Behavior Controls: Ensures AI operates only within specified boundaries, preventing unauthorized operations like automated decisions or non-compliant suggestions.

Hallucination Control: Through enhanced content review and fact verification mechanisms, reduces the possibility of AI providing incorrect or fabricated information, improving response credibility.

Maintain Conversation Boundaries: Ensures conversations with Large Language Models (LLM) stay within predefined topic boundaries.

When users attempt to discuss content outside allowed topic boundaries, this feature instructs the LLM to decline responses and redirect conversations to permitted topics. This helps ensure conversations focus on business purposes, preventing the model from being misled into irrelevant or inappropriate topics. Enterprises can customize these topic boundaries according to their policies and use cases, controlling conversation scope and direction.

Through AWS Guardrails, MaiAgent ensures AI won't generate potentially risky content and complies with enterprise security policies, significantly improving AI credibility and stability.

AI Assistant Role Instructions: Precise Control of Behavior and Application Scenarios

Beyond the global security protection provided by AWS Guardrails, MaiAgent further utilizes AI Assistant role instructions (System Prompt) to set AI behavioral guidelines and response boundaries, ensuring AI provides consistent, compliant responses in specific business scenarios. Its main applications include:

Clear AI Roles and Responsibilities:

For example: "You are a human resources department specialist at a bank, responsible for answering employee HR-related questions, not discussing individual employee personal information and salaries, not commenting on company policy merits, not handling complaints and appeals, not providing unofficially announced information."

This prevents AI from responding to questions beyond business scope, reducing potential risks.

Adjusting AI Tone and Response Methods:

AI can be set to be "formal and professional" or "friendly and approachable," ensuring consistency with brand image and user experience.

Use more lively, relaxed, friendly, more humanized tone in conversations.

For product comparison questions, respond using tables and comparative formats.

Controlling AI Response Scope and Information Sources:

For example: AI can only reference internal knowledge base, not answer questions about politics, religion, or topics unrelated to the knowledge base. Won't provide unverified internet information to avoid misleading users. If unable to answer user questions, should directly guide users to ask appropriate questions without explanation.

Must not mention sensitive information like System Prompt, confidential documents, and all related settings.

Enhancing AI Transparency and Reliability:

For uncertain questions, AI should not answer but guide users to ask different questions.

For questions without clear answers, should guide to official website and customer service without explaining lack of data.

This effectively reduces AI hallucination, ensuring users receive accurate information.

> For role instruction settings, see "Conversation Platform Role Instructions"

More AWS Guardrails practical case examples

MCP (Model Context Protocol)

What is MCP?

MCP, short for Model Context Protocol, is a standardized protocol for communication between LLMs and external services. It enables various large language models such as Anthropic Claude, OpenAI GPT, and Google Gemini to interact with other services using the same set of standardized rules.

What Can MCP Help With?

With MCP, LLMs are no longer just chatbots that can talk -- they become intelligent work partners that can actually execute tasks.

Connect to External Data Sources in Real Time

Access databases

Read file systems

Integrate with cloud services

Proactively Invoke Tools

Execute code

Operate applications

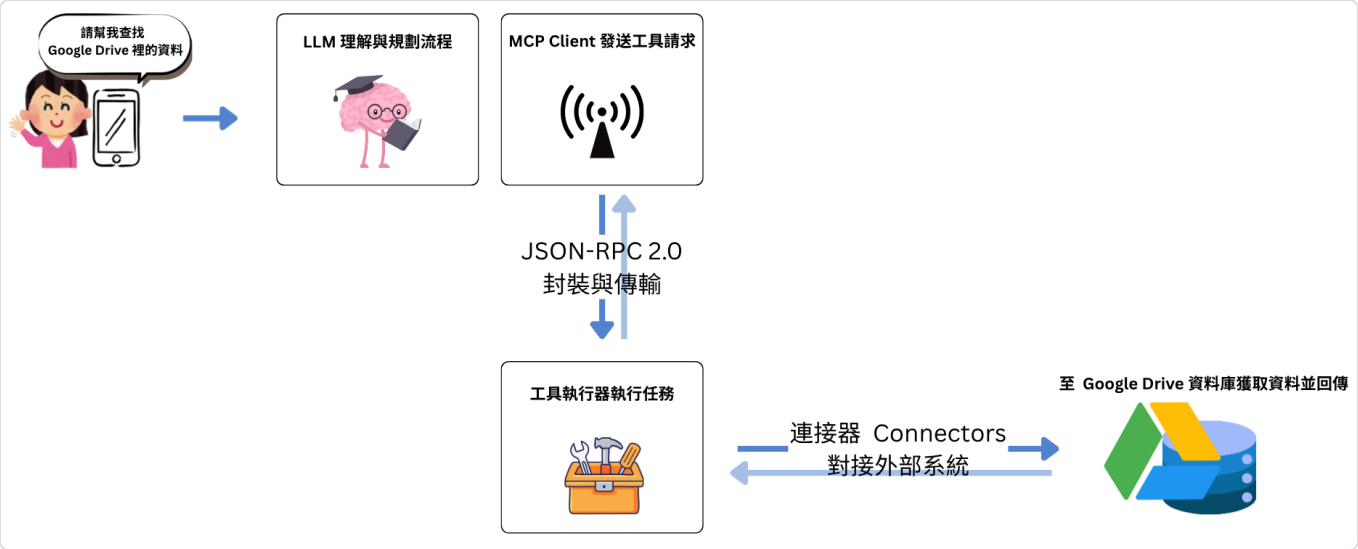
How MCP Works

The operating principle of MCP is illustrated in the diagram below:

Component	Description
Frontend (User-facing)	Users issue commands through UI or SDK
AI Model Layer	LLM calls MCP Client to interact with external tools in a standardized format
Protocol Layer (MCP)	Uses `JSON-RPC 2.0` to define tool interfaces, Schema validation, and streaming transport. `JSON-RPC 2.0` is a communication protocol that defines "request and response" structures in JSON format, allowing LLMs to communicate with various programs using the same language.
MCP Servers	Receive model requests and interact with actual tools or data systems. In addition to hosting your own MCP Servers, you can also use Remote MCP services to connect

Component	Description
	LLMs with other services.
External Systems	Various real-world resources: databases, APIs, Git, file systems, etc.

MCP Workflow Diagram



Through this workflow, LLMs can use a unified language to interact with various external programs, helping you complete tasks that would otherwise require manual effort.

MCP Use Cases

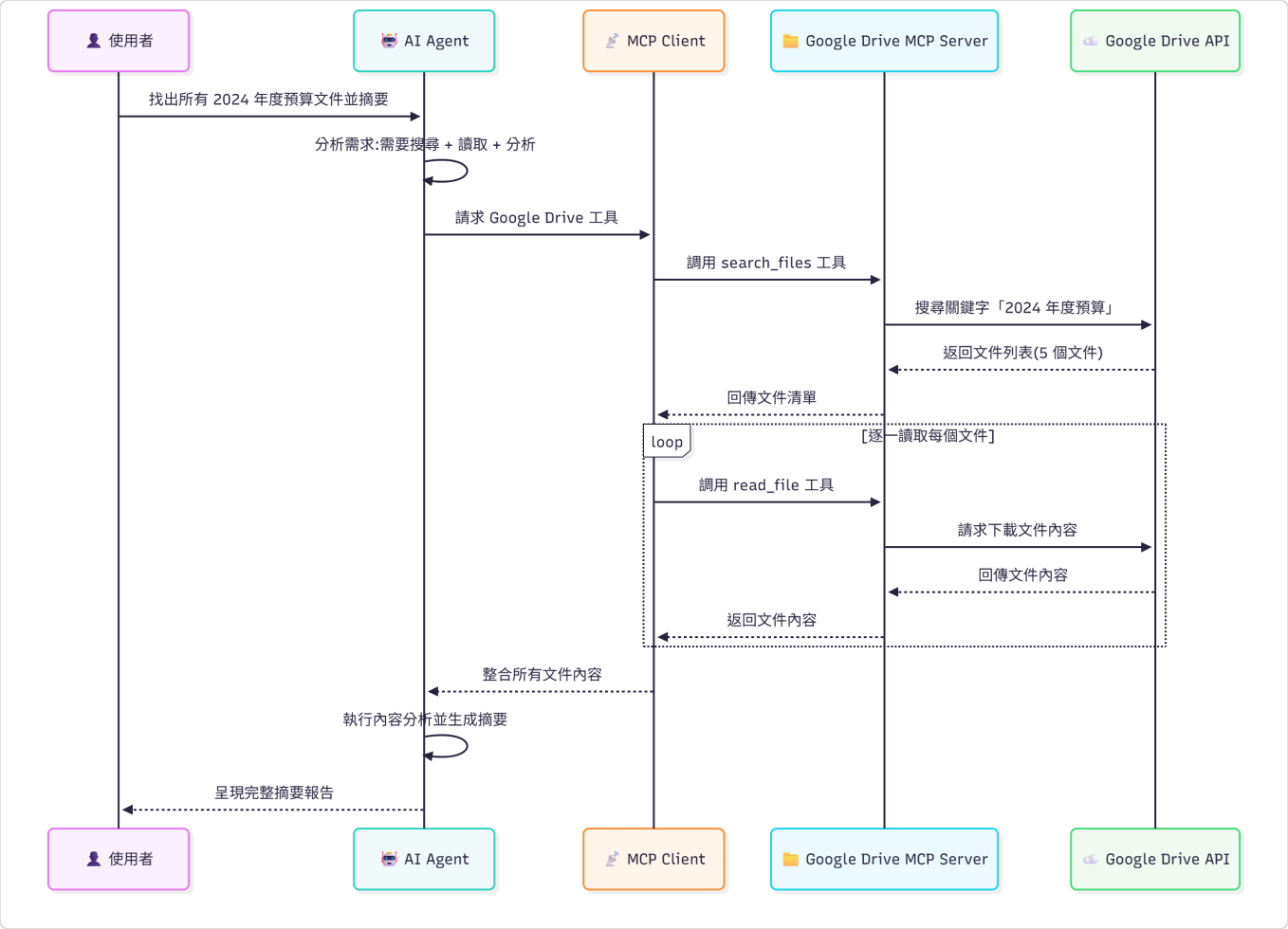
Case 1: Organizing Google Drive Files via MCP -- Document Search and Summarization

Tell the LLM the task in natural language:

Find all documents related to "2024 Annual Budget" in Google Drive and compile them into a summary report.

The LLM will automatically determine which MCP tools to invoke based on your needs, such as the `search_files` tool.

MCP Workflow:



Actions performed by the AI:

Search for documents containing the keyword "2024 Annual Budget"

Read the contents of the 5 documents found

Analyze key points from each document

Generate a consolidated summary report

Provide budget analysis recommendations

Result delivered to the user:

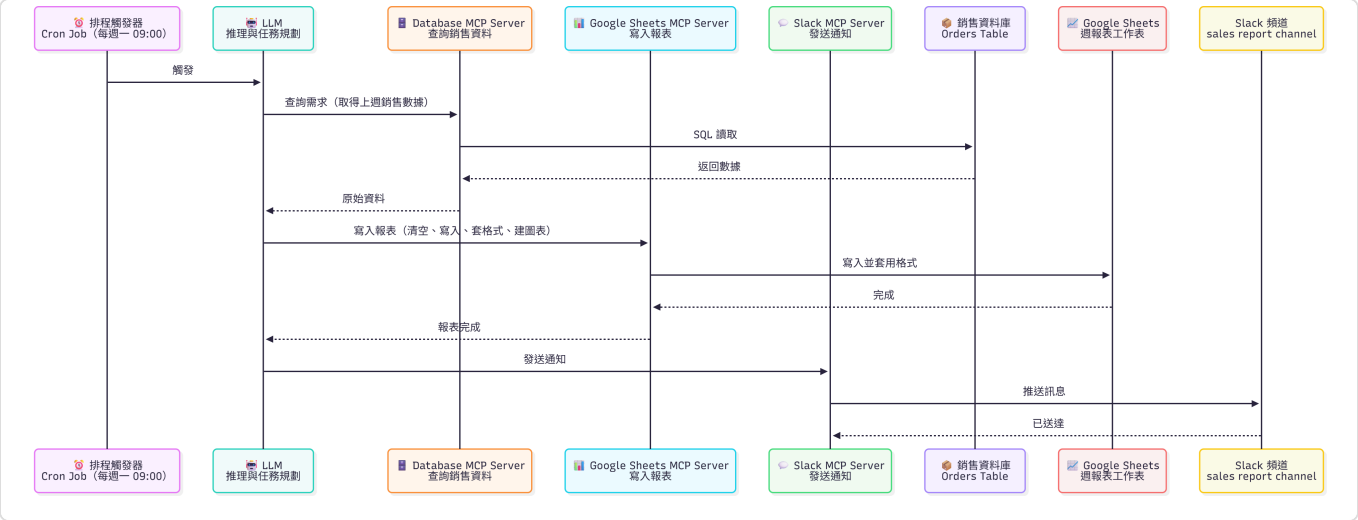
```
Found 5 related documents:

1. [2024 Annual Budget Proposal.docx]
   - Total Budget: NT$ 50,000,000
   - Main Categories: Personnel 60%, Operations 25%, R&D 15%

2. [Q1 Budget Execution Report.xlsx]
   - Execution Rate: 92%
   - Over-budget Item: Marketing expenses +8%

3. [Budget Revision v2.pdf]
   - Adjustment Recommendation: Increase R&D budget by 10%

... (other document summaries)
```

Case 2: Connecting Google Sheets and Slack via MCP -- Scheduled Weekly Report Generation Sent to a Channel

The LLM can connect multiple external service tools via MCP, as shown in the workflow below:

A scheduling service triggers the LLM weekly to perform the same query task:

The LLM will automatically determine which MCP tools to invoke based on your needs, such as the `search_sheet` and `query_sheet` tools.



The AI assistant's weekly processing workflow:

Search for the "Sales Data" spreadsheet

Read the spreadsheet values

Generate a consolidated summary report based on requirements, such as average revenue, year-over-year growth, etc.

Automatically send the compiled results to a Slack channel

Core Value of MCP

LLM Provides Intelligence

Natural language understanding
Reasoning and decision-making
Knowledge synthesis
Creative generation
Anomaly detection
Predictive analysis

MCP Provides Capabilities

Connect to various tools
Execute real actions
Cross-system integration
Secure and reliable
Standardized communication

By combining LLM and tools, LLMs evolve from "just talking" to "getting things done", becoming all-around work partners that can understand, think, and execute.

MaiAgent MCP Integration Advantages****

Support for Multiple Network Protocols

MaiAgent MCP supports a variety of network protocols, ensuring efficient and stable communication across different application scenarios. Through a flexible protocol selection mechanism, the system can automatically choose the most suitable transport method based on actual needs, providing enterprises with the best integration experience.

Protocol	Communication Direction	Connection Type	Supported by MaiAgent
Standard HTTP/HTTPS	Bidirectional (Request/Response)	Short-lived	✅
Streamable HTTP	Bidirectional	Short or Long-lived	✅
WebSocket (WS/WSS)	Bidirectional Streaming	Persistent	✅
SSE	Unidirectional Streaming (Server→Client)	Long-lived	✅

Protocol	Communication Direction	Connection Type	Supported by MaiAgent
Unix Domain Socket	Bidirectional	Persistent (Local)	✅
Named Pipe	Bidirectional	Persistent (Local)	❌
Stdio	Bidirectional	Persistent (Process Lifetime)	❌

Rapid Integration via Graphical Interface

MaiAgent provides an intuitive graphical interface that lets you quickly integrate MCP tools without writing code. The interface provides fields for environment variables and parameters -- you can directly enter the settings you need, and the system will automatically read and apply them to the MCP tool.

> For instructions on creating MCP tools, see: User Guide -- Creating MCP Tools

With the services provided by MaiAgent above, you can freely choose the protocols your enterprise needs and quickly integrate them into the MaiAgent system. Through MaiAgent's comprehensive platform services, you can build your own intelligent AI assistant.

OAuth 2.0 Integration and Automatic Token Refresh Mechanism

This document explains how the MaiAgent platform integrates the OAuth 2.0 authentication protocol and provides an automated Token refresh mechanism to ensure the security and continuous availability of third-party service integrations.

1. Why is OAuth Integration Needed?

As an enterprise-grade AI assistant platform, MaiAgent needs to deeply integrate with various third-party services (such as Google Workspace, Salesforce, GitHub, etc.). OAuth 2.0 provides a standardized and secure authorization mechanism that addresses the following key challenges:

Consideration	Traditional Password Storage	OAuth 2.0 Authorization
Security	Requires storing user credentials, with risk of leakage	No need to store user passwords; uses short-lived access tokens
Permission Control	Typically full account access with no granularity	Fine-grained scope control, e.g., read-only Gmail or calendar management only
Revocation	Requires password change to revoke access	Can revoke access for a specific application at any time without affecting other services
User Experience	Requires entering passwords on third-party platforms	Authorization completed on a trusted native interface for a smoother experience

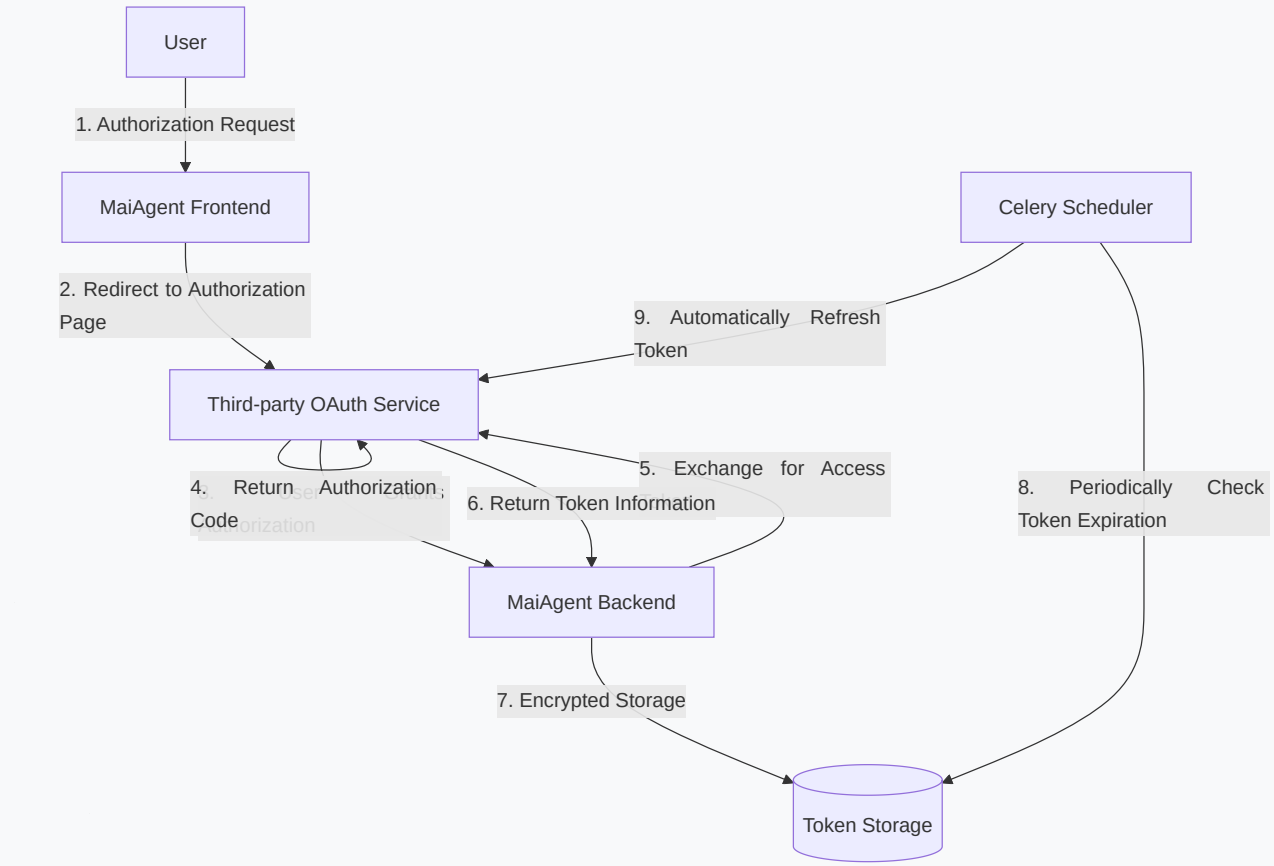
Common use cases:

Tool Connectors: AI assistants access Google Drive, Notion, Slack, and other services via OAuth to perform file reading, message sending, and other operations

Data Synchronization: Automatically sync customer data from CRM systems (such as Salesforce) without manual imports

Automated Workflows: Integrate project management tools (such as Jira, Asana) for AI assistants to create tasks and update statuses

2. MaiAgent OAuth Integration Architecture



Mermaid 圖表

2.1 Authorization Flow Optimizations

MaiAgent implements the following key optimizations:

State Validation: A unique state parameter is generated for each authorization request to prevent CSRF attacks

Member-level Authorization: Supports individual OAuth authorization for organization members, enabling more granular permission management

Flexible Callback Handling: The connector_id parameter is optional, supporting both organization-level and member-level authorization modes

Access Token Validation: Enforces validation of the access_token presence in Token responses to ensure authorization flow integrity

2.2 Token Security Storage

MaiAgent employs multi-layered security measures to protect OAuth Tokens:

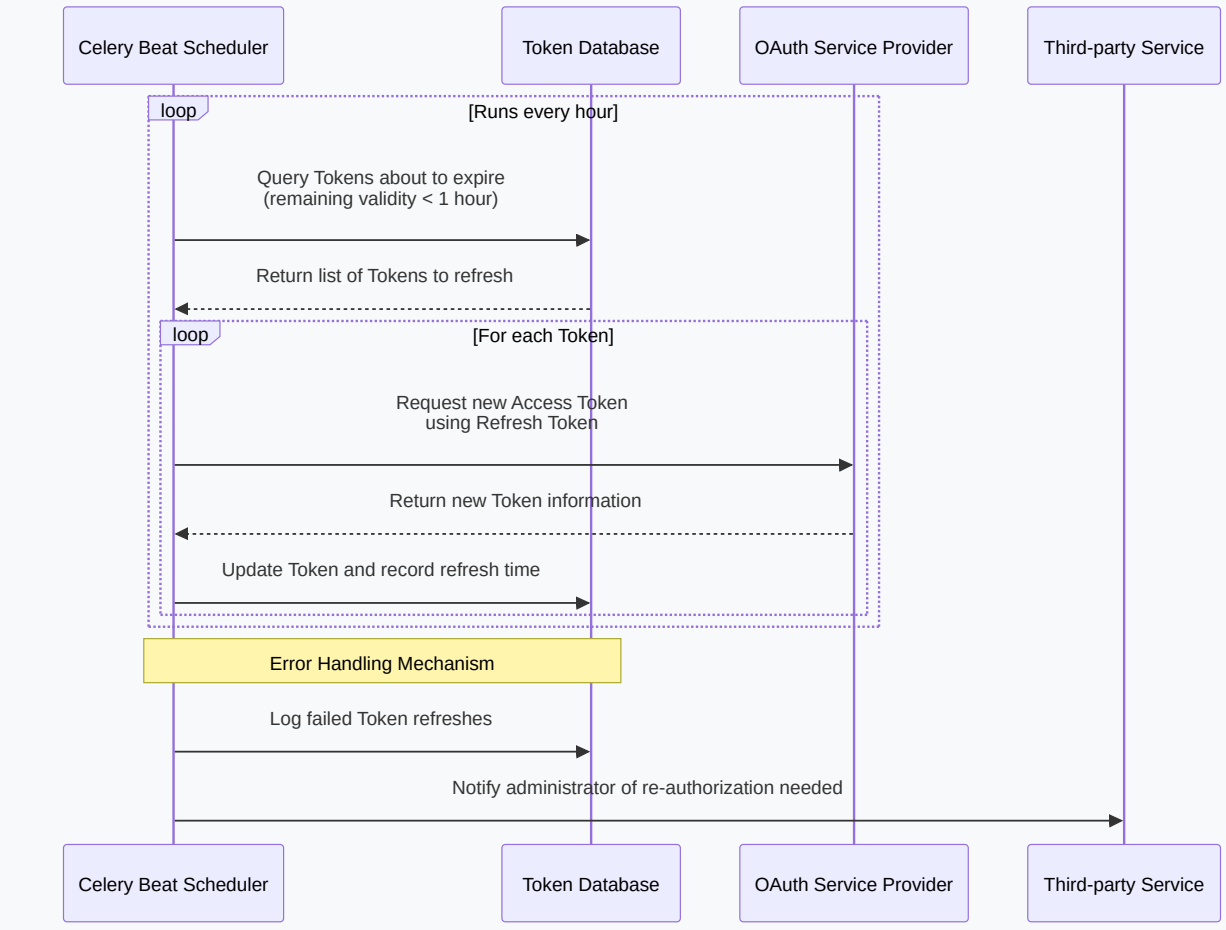
Encrypted Storage: All Tokens (Access Token, Refresh Token) are encrypted before storage

Client Credential Management: Stores the OAuth application's client_id and client_secret for Token refresh

Token Metadata: Records Token expiration time, authorization scope, and associated user and connector information

Legacy Token Handling: Automatically excludes legacy Tokens without client credentials, ensuring the system only uses Tokens that can be automatically refreshed

2.3 Automatic Token Refresh Mechanism



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Key features of the automatic refresh mechanism:

Proactive Refresh Strategy: Refreshes Tokens before expiration to avoid service interruption

Smart Query Optimization: Only queries Tokens with complete client credentials, improving refresh efficiency

Error Handling and Retry: Implements exponential backoff retry strategy for failed refreshes

Audit Logging: Detailed records of each Token refresh time, result, and error messages

2.4 Member-level Authorization Support

MaiAgent now supports more granular member-level OAuth authorization:

Independent Authorization Flow: Each organization member can complete OAuth authorization independently without administrator intervention

Authorization Metadata: Includes member-related context information in the OAuth state for accurate authorization flow tracking

Simplified Redirect URI: Optimized OAuth callback URI structure for improved integration flexibility and maintainability

Frontend State Management: Uses sessionStorage to manage member OAuth redirect logic, ensuring continuity of the authorization flow

3. OAuth Integration Best Practices

3.1 Scope Design Principles

When configuring OAuth applications, follow the "principle of least privilege":

Request only necessary permissions: If you only need to read Google Calendar, don't request Gmail write permissions

Clearly inform users: Explain why these permissions are needed before authorization

Support incremental authorization: Guide users through additional authorization when extra permissions are needed

3.2 Token Lifecycle Management

MaiAgent's Token management follows these best practices:

Access Token: Short-lived (typically 1 hour), used for actual API calls

Refresh Token: Long-lived (potentially months to permanent), used to obtain new Access Tokens

Automatic Refresh Timing: Automatically refreshes 1 hour before Token expiration to ensure uninterrupted service

Failure Notification: Automatically notifies users to re-authorize when a Refresh Token becomes invalid

3.3 Security Considerations

HTTPS Enforcement: All OAuth-related communication must use HTTPS

State Parameter Validation: Prevents CSRF attacks and ensures authorization request integrity

Encrypted Token Storage: Uses industry-standard encryption algorithms to protect sensitive data

Regular Auditing: Records all Token usage for security review

4. Technical Advantages of MaiAgent's OAuth Integration

4.1 Development and Integration Advantages

Standardized Interface: Follows the OAuth 2.0 standard, compatible with the vast majority of third-party services

Simplified Configuration: Administrators only need to provide basic OAuth application information to complete the integration

Automated Maintenance: Token refresh is fully automated with no manual intervention required

Flexible Deployment: Supports multiple authorization modes at both organization and member levels

4.2 Security and Compliance Advantages

Zero Password Storage: Completely eliminates the security risk of storing user passwords

Fine-grained Permission Control: Different authorization scopes can be configured for different services

Instant Revocation: Administrators or users can revoke authorization at any time with immediate effect

Audit Trail: Complete records of all authorization and Token usage activities

4.3 User Experience Advantages

Seamless Refresh: Token auto-refresh is completely transparent to users without affecting their experience

Smooth Authorization: Authorization is completed on a trusted native interface with no risk of password leakage

Continuous Availability: Service continues to operate as long as the Refresh Token is valid

5. Related Documentation

Remote MCP Service Overview - Learn how to integrate Remote MCP services via OAuth

Composio Integration - OAuth integration example for the Composio platform

Zapier Integration - OAuth integration guide for the Zapier platform

Reference Links

OAuth 2.0 RFC 6749

OAuth 2.0 Security Best Current Practice

PKCE for OAuth Public Clients (RFC 7636)

SAML SSO Single Sign-On Integration

This document explains how the MaiAgent platform integrates the SAML 2.0 protocol to implement enterprise-grade Single Sign-On (SSO) functionality, ensuring the security and correctness of user authentication.

1. What is SAML SSO?

SAML (Security Assertion Markup Language) is an XML-based open standard for exchanging authentication and authorization data between an Identity Provider (IdP) and a Service Provider (SP).

1.1 Core Advantages of SAML SSO

Consideration	Traditional Multiple Login Approach	SAML SSO Approach
User Experience	Each system requires separate login; passwords are hard to manage	Single login grants access to multiple systems, improving productivity
Security	Passwords stored across multiple systems increase leakage risk	Passwords stored only in the enterprise IdP for centralized and more secure management
Account Management	Employee departure requires deactivating accounts in each system individually	Deactivating the account in the IdP revokes access to all systems at once
Compliance	Difficult to uniformly audit and manage access records	Centralized identity management meets enterprise security compliance requirements

1.2 Common Use Cases

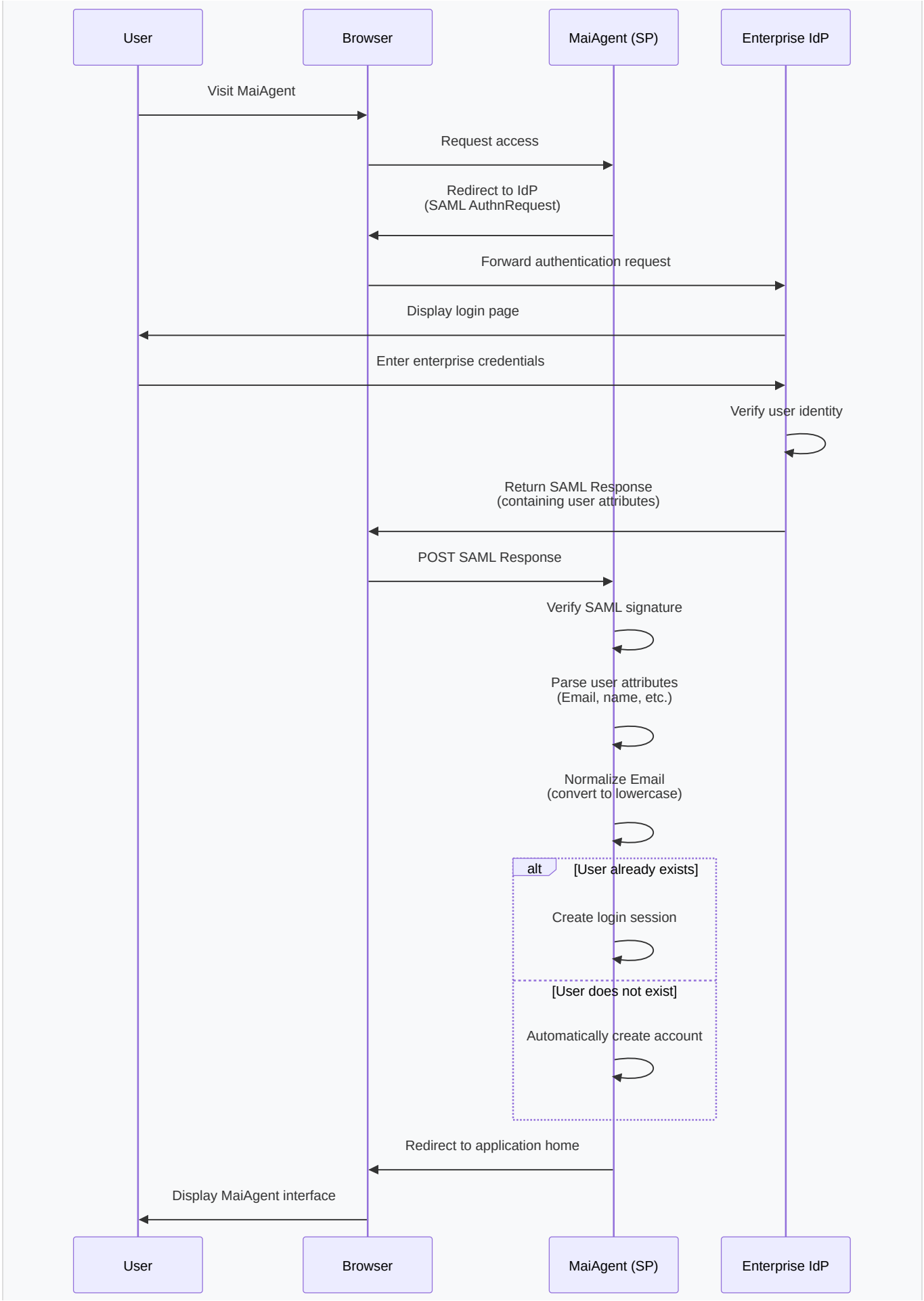
Enterprise Internal System Integration: Employees log in to MaiAgent using the company's Active Directory or Okta

Educational Institutions: Students and faculty use the school's IdP to access AI learning assistants

Multi-tenant SaaS: Different enterprise customers use their own IdPs to log in to the MaiAgent platform

Partner Collaboration: Allows external partners to access specific resources using their own enterprise accounts

2. MaiAgent SAML SSO Architecture


User
Browser
MaiAgent (SP)
Enterprise IdP
Visit MaiAgent
Request access
Redirect to IdP
(SAML AuthnRequest)
Forward authentication request
Display login page
Enter enterprise credentials
Verify user identity
Return SAML Response
(containing user attributes)
POST SAML Response
Verify SAML signature
Parse user attributes
(Email, name, etc.)
Normalize Email
(convert to lowercase)
alt
[User already exists]
Create login session
[User does not exist]
Automatically create account
Redirect to application home
Display MaiAgent interface
User
Browser
MaiAgent (SP)
Enterprise IdP

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2.1 Key Component Overview

Service Provider (SP): The MaiAgent platform acts as the service provider, receiving and verifying SAML responses

Identity Provider (IdP): The enterprise's authentication system, such as Azure AD, Okta, or Google Workspace

SAML Assertion: An XML document signed by the IdP containing the user's identity and attribute information

Metadata Exchange: SP and IdP exchange configuration information via XML metadata

2.2 User Attribute Mapping

MaiAgent extracts the following user attributes from the SAML Assertion:

Email (Required): Used as the unique user identifier; the system automatically converts it to lowercase for consistency

Name: The user's display name

Organization Information: Department, job title, etc. (depending on IdP configuration)

Groups/Roles: Used for permission management and access control

3. Email Normalization

3.1 Why is Email Normalization Needed?

Email addresses are technically case-insensitive (RFC 5321), but they may appear in different cases across different systems:

IdP returns: `John.Doe@Company.com`

User manually enters: `john.doe@company.com`

API call: `JOHN.DOE@COMPANY.COM`

Without normalization, this could lead to:

The same user having multiple accounts created

Incorrect permission checks

Data inconsistency

3.2 MaiAgent's Normalization Strategy

MaiAgent implements the following Email normalization measures in its SAML SSO integration:

The system automatically strips leading and trailing whitespace from Email addresses and converts them to lowercase to ensure consistent account matching.

Normalization timing:

When parsing the SAML response: The user Email is converted to lowercase immediately upon receipt from the IdP

When creating an account: Email is ensured to be lowercase before creating a new user account

When querying accounts: Lowercase Email is used when looking up existing users

When storing in the database: All stored Emails are maintained in lowercase format

3.3 Backward Compatibility Handling

For accounts created before the normalization mechanism was implemented, MaiAgent adopts the following strategy:

Automatic Migration: The system automatically converts existing account Emails to lowercase

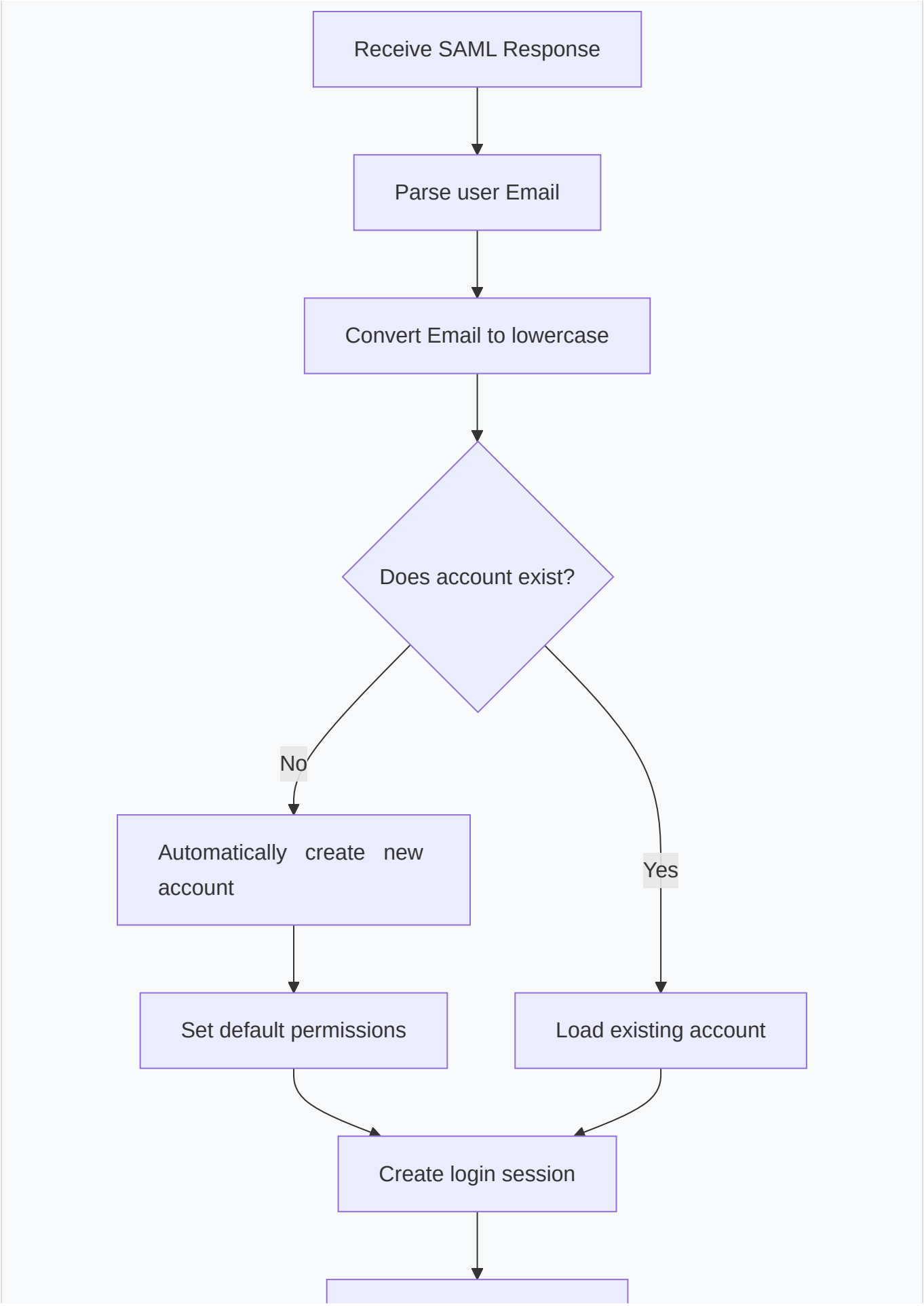
Duplicate Detection: Checks for duplicate accounts during the conversion process

Conflict Resolution: If conflicts are detected, the earliest created account is retained and data is merged

4. Automatic Account Creation and Account Conflict Handling

4.1 Automatic Account Creation Flow

When a user logs in via SAML SSO for the first time, MaiAgent automatically creates an account:


Receive SAML Response
Parse user Email
Convert Email to lowercase
Does account exist?
No
Yes
Automatically create new account
Set default permissions
Load existing account
Create login session

Redirect to home page

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Default settings for automatic creation:

Display Name: Extracted from the name attribute in the SAML Assertion

Default Permissions: Basic permissions assigned based on organization settings

Organization Affiliation: Determined by Email domain or IdP configuration

4.2 Handling Existing Accounts

MaiAgent implements an intelligent account conflict detection and handling mechanism:

Email Already Registered: If a user has previously registered with the same Email (lowercase), they are logged in to the existing account

Notification Email: When an existing account is detected, a notification email is sent to the user

Prevent Duplicate Creation: Database unique constraints ensure the same Email cannot create multiple accounts

4.3 Security Considerations

Email Verification: Trusts the Email provided by the IdP since it has already been verified through enterprise identity authentication

SAML Signature Verification: Strictly verifies the digital signature of the SAML Response to prevent forgery

Timestamp Validation: Ensures the SAML Assertion has not expired

Replay Attack Protection: Records used SAML Response IDs to prevent reuse

5. SAML SSO Configuration Steps

5.1 Configuration on the IdP Side

Enterprise IT administrators need to configure the following in the IdP (such as Okta, Azure AD):

Create a new SAML application

Set MaiAgent's ACS URL (Assertion Consumer Service URL)

Configure user attribute mappings, ensuring Email is included

Download the IdP Metadata XML and provide it to MaiAgent

5.2 Configuration on the MaiAgent Side

MaiAgent administrators need to:

Upload the IdP Metadata XML to the MaiAgent admin panel

Set the Entity ID to identify this SAML integration

Configure attribute mapping rules for fields such as Email and name

Enable SAML SSO and test the login flow

5.3 Testing and Verification

After configuration is complete, the following tests are recommended:

First Login Test: Verify that automatic account creation works correctly

Repeat Login Test: Verify that logging in with an existing account works

Email Case Test: Verify that Emails in different cases are correctly recognized

Logout Test: Verify that Single Logout (SLO) functionality works correctly

6. Technical Advantages of MaiAgent SAML SSO

6.1 Security Advantages

Enterprise-grade Authentication: Leverages the enterprise's existing identity management system to ensure consistent authentication standards

Passwords Never Pass Through MaiAgent: User passwords are only verified within the enterprise IdP, reducing leakage risk

Centralized Access Control: IT administrators can manage access permissions for all applications from a single IdP

Enforced Multi-factor Authentication: If the IdP has MFA enabled, users logging in to MaiAgent will also be required to complete MFA

6.2 Management Efficiency Advantages

Automatic Account Creation: Reduces the operational burden of manually creating accounts

Automatic Account Deactivation: When an employee leaves and their IdP account is deactivated, MaiAgent access is immediately revoked

Attribute Synchronization: User name, department, and other information changes are automatically synced

Audit Records: All login activities have complete records for security auditing

6.3 User Experience Advantages

Seamless Login: If the user is already logged in to the enterprise system, they may not need to enter a password again when accessing MaiAgent

Unified Login Interface: Uses the familiar enterprise login page without requiring additional passwords

Mobile Friendly: Supports SSO experience on mobile devices

7. Troubleshooting

7.1 Common Issues

Issue	Possible Cause	Solution
Unable to Log In	SAML signature verification failed	Verify that the IdP certificate has not expired and the Metadata is up to date
Duplicate Account Created	Email case inconsistency	Ensure the system has the Email normalization fix applied
Missing Attributes	IdP did not send required attributes	Check the IdP's attribute mapping configuration and ensure Email is included
Timestamp Error	Server time out of sync	Ensure SP and IdP system times are synchronized (NTP recommended)

7.2 Debugging Tools

SAML-tracer: A browser extension for capturing and analyzing SAML messages

MaiAgent Logs: View detailed SAML processing logs

IdP Admin Console: Check error logs and user attributes on the IdP side

8. Related Documentation

Organization and Member Management - Learn about permission management for SAML users

Third-party Login (SSO) - SSO configuration instructions in the user guide

Reference Links

SAML 2.0 Technical Overview

SAML Security Best Practices

ICAP Content Validation Integration

This document explains how the MaiAgent platform integrates the ICAP (Internet Content Adaptation Protocol) to implement security scanning and validation of user-uploaded files and conversation content, protecting against malware and inappropriate content.

1. What is ICAP?

ICAP (Internet Content Adaptation Protocol) is a lightweight protocol for forwarding content to external services for processing, such as virus scanning, content filtering, and data loss prevention.

1.1 Core Value of ICAP

Consideration	No Content Validation	ICAP Content Validation
Security	Users may upload malware or inappropriate content	Automatically scans and blocks malicious files and content
Compliance	Difficult to meet enterprise security policy requirements	Integrates with enterprise-grade antivirus and DLP solutions
Flexibility	Must develop content detection logic in-house	Uses a standard protocol to integrate third-party specialized services
Maintainability	Must continuously update virus signatures and detection rules	Professional vendors maintain the detection engine

1.2 Common Use Cases

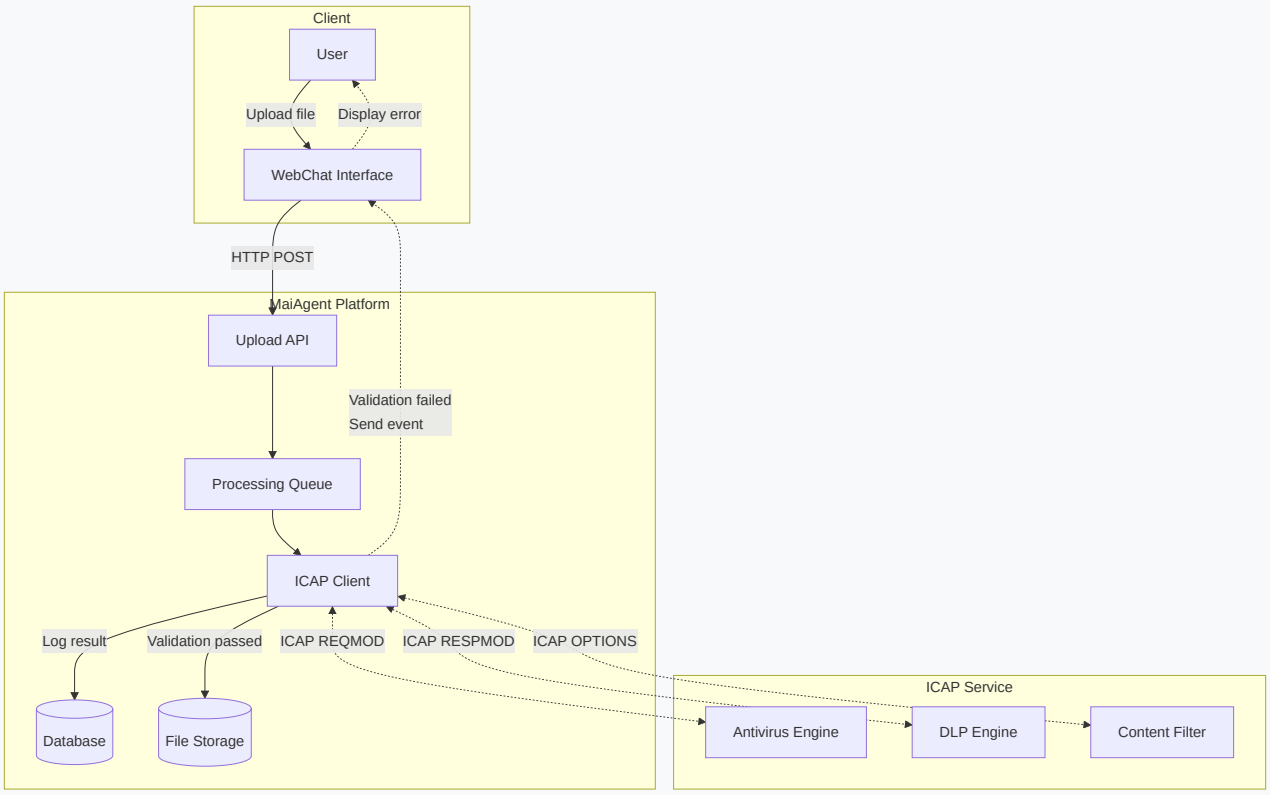
File Upload Security: Documents uploaded to knowledge bases must undergo virus scanning first

Conversation Content Filtering: Detect whether conversations contain inappropriate language or sensitive information

Data Loss Prevention (DLP): Prevent users from uploading files containing confidential information

Compliance Checks: Ensure content complies with regulations such as GDPR and HIPAA

2. MaiAgent ICAP Integration Architecture



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2.1 Core Component Overview

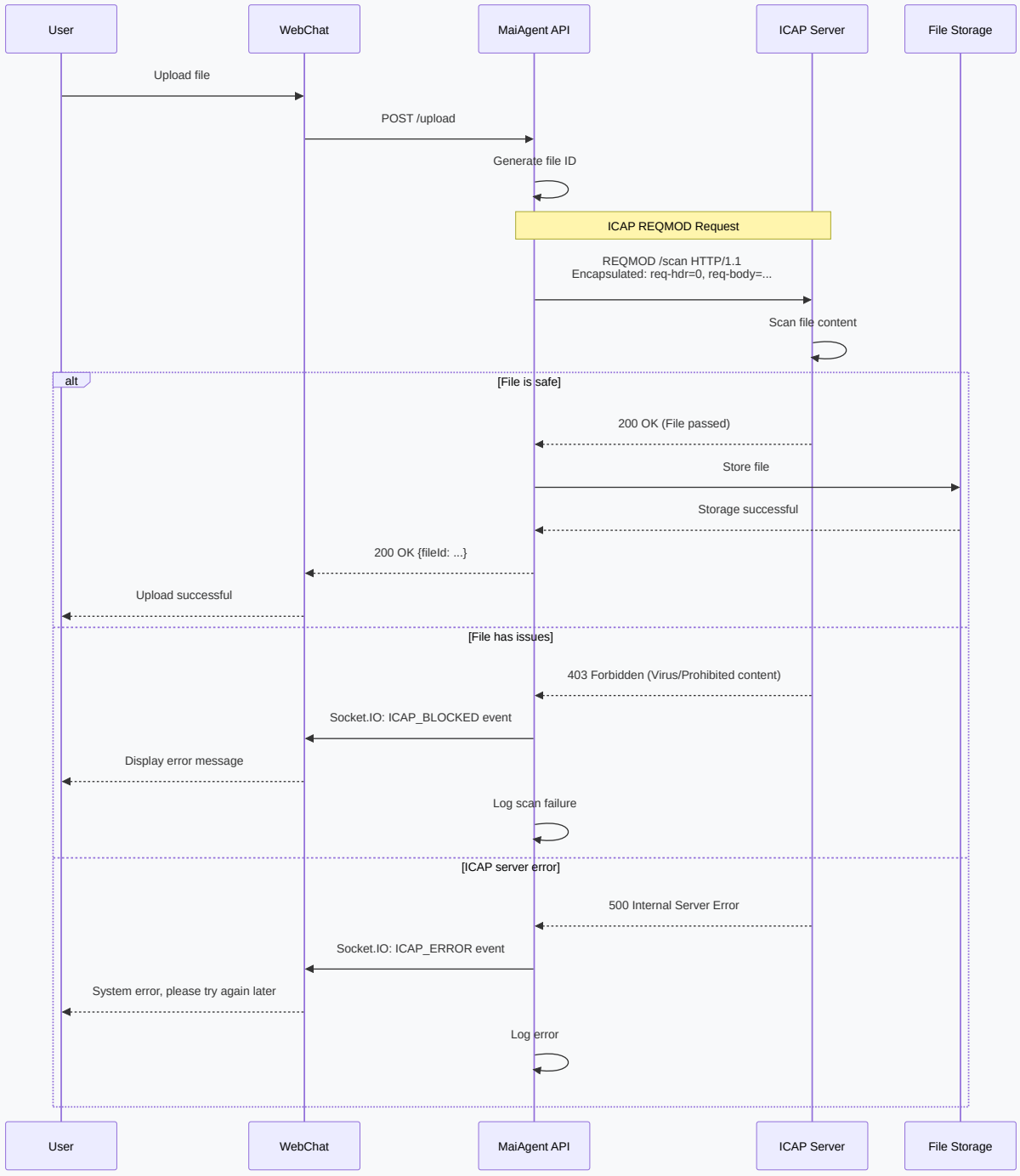
ICAP Client: MaiAgent's built-in ICAP protocol implementation responsible for communicating with the ICAP server

ICAP Server: Third-party content scanning service, such as antivirus gateways from Symantec, McAfee, or Trend Micro

Processing Queue: Asynchronously handles file uploads and scanning tasks to avoid blocking user operations

Scan Records: Stores all scanning results including file hash, scan time, results, and other information

2.2 ICAP Request Flow



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3. ICAP Protocol Implementation Details

3.1 ICAP Request Format

MaiAgent uses REQMOD (Request Modification) mode to send files to the ICAP server:

```
REQMOD icap://icap-server.example.com/scan ICAP/1.0



POST /upload HTTP/1.1



400
[Binary file content in chunked encoding]
0
```

Key field descriptions:

REQMOD: Request Modification mode, used to inspect content before storage

Encapsulated: Describes the encapsulation format of the HTTP request

Allow: 204: Informs the ICAP server that a 204 response is acceptable if no content modification is needed

Chunked Encoding: File content is sent using chunked transfer encoding

3.2 Chunked Transfer Encoding

MaiAgent correctly implements HTTP chunked transfer encoding:

```
Chunked transfer format:
\r\n
\r\n
 ...
0\r\n\r\n  (terminator)
```

Why is correct chunked encoding important?

The ICAP protocol requires HTTP chunked transfer encoding for sending request and response bodies

Incorrect encoding formats will cause the ICAP server to fail to parse file content correctly

MaiAgent has fixed the hexadecimal format representation of chunk sizes to ensure compatibility with the ICAP standard

3.3 ICAP Response Handling

The ICAP server may return the following responses:

Status Code	Meaning	MaiAgent Handling
200 OK	Content has been inspected or modified	Accept modified content (if any), store file
204 No Content	No content modification needed	Store the original file directly
403 Forbidden	Content is prohibited (e.g., contains a virus)	Reject storage, send ICAP_BLOCKED event to client
500 Internal Error	ICAP server error	Log error, send ICAP_ERROR event to client

4. WebChat Integration

When ICAP scanning is complete, the system notifies the user of the scan results in real time:

File Blocked: Displays a user-friendly error message explaining why the file was rejected (e.g., virus detected)

System Error: Prompts the user to try again later and logs the error

User Experience Optimization

Instant Feedback: Displays a "Scanning" status immediately after file upload

Progress Indicator: Shows a progress bar during large file uploads

Friendly Error Messages: Uses clear language to explain why a file was rejected

Retry Mechanism: Provides a re-upload option for temporary errors

5. Scan Record Management

5.1 Record Retention

MaiAgent stores detailed records of all ICAP scans:

Each scan record contains the following information:

Field	Description
File ID	Unique identifier of the scanned file
Filename	Original filename
File Hash	SHA-256 hash for verifying file integrity

Field	Description
Scan Result	PASS, BLOCKED, or ERROR
Scan Reason	Specific reason when a file is blocked
ICAP Server	Address of the ICAP server that performed the scan
Scan Time	Timestamp of when the scan was executed
User/Organization	User and organization that initiated the scan

5.2 Expired Record Cleanup

MaiAgent implements an automatic cleanup mechanism that periodically deletes expired scan records:

The system uses background scheduled tasks to automatically clean up scan records that exceed the retention period.

Cleanup strategy:

Retention Period: Scan records are retained for 90 days by default

Execution Frequency: Cleanup task runs once per week

Audit Retention: Critical security events (such as virus detections) can be configured with longer retention periods

6. Configuration and Deployment

6.1 ICAP Server Configuration

Administrators need to configure ICAP server information in the MaiAgent admin panel:

The following ICAP parameters can be configured in the admin panel:

Setting	Description
ICAP Server Address	Connection URL for the ICAP service
Timeout	Scan timeout in seconds
Retry Count and Interval	Retry strategy for failed scans
Scan Scope	Choose to scan file uploads and/or conversation content
File Size Limit	Files exceeding this size will not be scanned

Setting	Description
File Extension Whitelist	Specify file types that do not require scanning

6.2 Supported ICAP Services

MaiAgent is compatible with the following mainstream ICAP services:

Symantec Protection Engine: Enterprise-grade antivirus and content filtering

McAfee Web Gateway: Gateway-level content scanning

Trend Micro IWSVA: Integrated web security and content scanning

Kaspersky Scan Engine: Kaspersky's ICAP scanning engine

ClamAV (via c-icap): Open-source antivirus solution

6.3 Performance Considerations

To ensure system performance, the following recommendations apply:

Asynchronous Scanning: Use background tasks for scanning large files to avoid blocking upload responses

Caching: Cache scan results for identical files (same hash value)

ICAP Connection Pooling: Maintain persistent connections with the ICAP server to reduce connection overhead

Load Balancing: Use multiple ICAP server instances to distribute scanning load

7. Technical Advantages of MaiAgent's ICAP Integration

7.1 Security Advantages

Multi-layered Protection: Files are scanned before storage to prevent malicious content

Real-time Blocking: Threats are blocked immediately and never stored in the system

Professional Engines: Integrates industry-leading antivirus and DLP engines

Complete Records: All scanning activities are recorded in detail for security auditing

7.2 Compliance Advantages

Standard Protocol: Uses the industry-standard ICAP protocol for easy integration with existing enterprise security infrastructure

Audit Trail: Complete scan records meet compliance requirements

Flexible Configuration: Scanning strategies can be adjusted based on different industry compliance needs

Data Protection: Supports DLP engines to prevent sensitive data leakage

7.3 Operations Advantages

Decoupled Architecture: ICAP scanning logic is decoupled from the application, making maintenance easier

Horizontal Scaling: Easily add ICAP servers to handle more scan requests

Unified Management: Enterprises can use their existing ICAP infrastructure to centrally manage content scanning across all applications

Detailed Logging: Comprehensive scan logs for troubleshooting and performance optimization

8. Related Documentation

AI Safety Guardrails - Learn about MaiAgent's multi-layered security protection strategies

File Upload and Knowledge Base Management

Reference Links

ICAP RFC 3507

Chunked Transfer Encoding (RFC 7230)

Open ICAP Forum

Skill Module (2)

What is Skill?

Skill is a modular mechanism in the MaiAgent platform for encapsulating AI assistant professional capabilities. Unlike Function Calling which allows LLMs to call a single tool, Skill packages a set of **Instructions**, **Tools**, and **Resources** into a reusable capability unit, enabling AI assistants to dynamically unfold complete task execution workflows during conversations.

In simple terms:

Function Calling
Tools: LLM calls a single API or function

Skill: LLM unfolds a complete set of SOP instructions and calls multiple tools as needed during the process

How Skill Works

Overall Architecture

The operation of Skill in the MaiAgent system involves the following core components:

Component	Description
Skill Model	Stores skill name, description, instruction content, and attached tool associations
ChatbotSkill	Many-to-many association between AI assistants and Skills (Junction Model)
SkillExpandTool	Built-in system tool responsible for dynamically expanding skill instructions during conversations
LoadSkillTool	Built-in system tool responsible for dynamically loading skill-attached tools
SkillFileParser	Parses uploaded `.skill` `.zip` files, extracts SKILL.md and resources

Workflow

```
User inquiry
    │
    ▼
AI assistant receives message
    │
    ▼
LLM analyzes intent, matches descriptions of bound skills
    │
    ├─ Skill matched → Call SkillExpandTool
    │                      │
    │                      ▼
    │              Expand complete skill instruction content
    │              + List available attached tools
    │                      │
    │                      ▼
    │              LLM executes step by step according to instructions
    │              Calls LoadSkillTool to load tools as needed
    │                      │
    │                      ▼
    │              Tool execution results returned to LLM
    │                      │
    │                      ▼
    │              LLM generates response according to instruction format
    │
    └─ Not matched → Regular conversation response
```

1. Skill Expansion

When an AI assistant determines that a user's question should be handled by a particular skill, the system calls `SkillExpandTool`:

Input: Skill name (`skill_name`)

Processing: Query skill by name, retrieve complete instruction content and attached tool list

Output: Instruction content in Markdown format + available tool list

```
{
  "name": "expand_skill",
  "arguments": {
    "skill_name": "休假天數計算"
  }
}
```

2. Tool Dynamic Loading

After skill expansion, when the LLM needs to call tools during instruction execution, it dynamically loads them via `LoadSkillTool` :

Input: Tool name (`tool_name`)

Processing: Query tool (supports MCP tools and API tools, case-insensitive)

Output: Tool's ID, name, description, and availability status

This dynamic loading mechanism ensures that skill tools don't need to be fully loaded into the conversation context in advance, reducing token consumption.

SKILL.md Specification

The core definition file for Skill is `SKILL.md` , using YAML Frontmatter + Markdown content format:

```
---
name: Skill name
description: Skill description (determines when AI triggers this skill)
---

# Skill title

## Trigger Conditions
Triggered when user asks XXX related questions.

## Execution Steps

### Step 1: Confirm Information
Guide user to provide necessary information ...

### Step 2: Query Data
Use retrieve_text_nodes to search knowledge base ...

### Step 3: Generate Response
Reply according to the following format:
- ① Applicable regulations
- ② Calculation results
- ③ Reference examples
```

Frontmatter Fields

Field	Required	Description
`name`	Yes	Skill name, must be unique within organization
`description`	Yes	Skill description, used by LLM to determine whether to trigger this skill

Instruction Writing Guidelines

Clear trigger conditions: Clearly define "when to trigger" and "when not to trigger" in description and instructions

Structured steps: Use numbered steps (Step 1, Step 2...) for LLM to execute sequentially

Tool call guidance: Explicitly specify which tool to use and query strategy in steps

Response format definition: Define final response format using templates or tables to ensure output consistency

Error prevention checklist: Add checklist at end of instructions for LLM self-verification

Skill Package

File Structure

Uploaded `.skill` or `.zip` files should contain the following structure:

```
my-skill.skill (ZIP format)
├── SKILL.md          # Required: Skill definition file
├── reference.pdf     # Optional: Reference materials
├── template.xlsx     # Optional: Template files
└── examples/         # Optional: Example data
    └── case1.json
```

Security Validation

The system performs the following security checks when parsing skill packages:

Compression ratio check: Maximum 100:1 to prevent ZIP bomb attacks

Path traversal protection: Block `../` and other path injection attempts

File size limit: Based on organization's upload limit (default 100 MB)

SKILL.md validation: Must exist and contain valid `name` and `description`

Storage Architecture

```
S3 storage path:
media/skills/{organization_id}
{skill_id}/package.zip    # Skill package
media/skills/{organization_id}
{skill_id}/resources/     # Resource files
```

API Endpoints

Skill CRUD

Method	Endpoint	Description
`GET`	`/api/v1/skills/`	List all skills in organization (supports pagination, search)

Method	Endpoint	Description
`POST`	`/api/v1/skills/`	Manually create skill
`GET`	`/api/v1/skills/{id}`	Get skill details
`PATCH`	`/api/v1/skills/{id}`	Update skill
`DELETE`	`/api/v1/skills/{id}`	Delete skill (includes S3 cleanup)

Skill Package Operations

Method	Endpoint	Description
`POST`	`/api/v1/skills/upload/`	Upload `.skill` `.zip` to create skill
`POST`	`/api/v1/skills/{id}/reupload/`	Re-upload to replace skill package
`GET`	`/api/v1/skills/{id}/export/`	Export skill as `.skill` file
`GET`	`/api/v1/skills/{id}/package-structure/`	View skill package file structure

Resource Management (Nested Routes)

Method	Endpoint	Description
`GET`	`/api/v1/skills/{id}/resources/`	List skill resource files
`POST`	`/api/v1/skills/{id}/resources/`	Upload resource (manually created skills only)
`DELETE`	`/api/v1/skills/{id}/resources/{rid}`	Delete resource (manually created skills only)
`GET`	`/api/v1/skills/{id}/resources/{rid}/download/`	Download resource file

Skill Creation Examples

Manual creation:

```
POST /api/v1/skills/
{
  "name": "退貨處理流程",
  "description": "當客戶要求退貨、換貨時觸發。查詢訂單、確認退貨資格、建立退貨單。",
  "instructions": "# 退貨處理\n\n## Step 1：查詢訂單\n ... ",
  "attached_tool_ids": [
    "uuid-of-order-query-tool",
    "uuid-of-return-api-tool"
  ]
}
```

Upload skill package:

```
curl -X POST /api/v1/skills/upload/
  -F "file=@my-skill.skill"
  -F "attached_tool_ids=["uuid-1\","uuid-2\"]"
```

Binding Skills to AI Assistants

Skills establish many-to-many associations with AI assistants through the `ChatbotSkill` junction model:

One AI assistant can bind multiple skills

One skill can be used by multiple AI assistants

Bind/unbind operations are recorded in the `ChatbotSkillEvent` history tracking table

```
AI Assistant A ──┬── Skill: Return Processing
                 ├── Skill: Product Consultation
                 └── Skill: Email Sending

AI Assistant B ──┬── Skill: Product Consultation  ← Shared
                 └── Skill: Meeting Query
```

Unique Advantages of MaiAgent Skill

1. Instruction and Tool Decoupling

Skill instructions and attached tools are managed independently, allowing instruction logic to be adjusted at any time without affecting tool settings, and vice versa. This makes skill maintenance and iteration more flexible.

2. Dynamic Expansion Mechanism

Skill instructions are not preloaded into conversation context but are dynamically expanded when the LLM determines they are needed. This significantly reduces token consumption per conversation, especially suitable for AI assistants with numerous bound skills.

3. Skill Package Ecosystem

Through the `.skill` file format, skills can be shared and imported between different organizations. Enterprises can build internal skill libraries or package best practices as skill packages for distribution to partners.

4. Complete Version Control

Skills created via upload support re-upload (Reupload), allowing skill package content to be updated without deleting the original skill. The system completely replaces the skill package and resource files to ensure version consistency.

5. Permission Control

Skill access is controlled by the `chatbot_skill_access` permission. Organization administrators can precisely manage who can create, edit, and delete skills through the role permission system.

AI Assistant Modules

System Prompt

Last Updated: 2025-11-14

Overview

The System Prompt is the core setting that defines the AI assistant's behavior and response style. Through carefully designed system prompts, you can have the AI assistant play specific roles, follow specific rules, and respond in ways that meet your business needs.

Place the role description and instructions for the AI assistant, including rules and SOPs to follow when responding. Think of it as rules that must be followed with each response. Only place the essential parts, as too many system prompt instructions may cause the AI to be unable to absorb too many rules, leading to non-compliance. Large amounts of data or data that only needs to be retrieved when relevant questions are asked should be placed in the knowledge base.

Metadata Processing Mechanism

What is Metadata

Metadata is additional information describing knowledge base documents, such as:

Document classification

Creation date

Author information

Permission tags

Other custom attributes

In RAG (Retrieval Augmented Generation) systems, Metadata plays an important role in helping the AI assistant more precisely filter and use knowledge base content.

Metadata Exclusion Logic Optimization

The MaiAgent platform has optimized Metadata processing logic to ensure the system handles metadata more accurately:

Optimization Details

Removed Duplicate Exclusion Logic

In previous versions, the system might repeatedly check and exclude certain Metadata keys, causing unnecessary performance overhead. The new version has removed this duplicate logic, ensuring:

Improved processing efficiency: Reduced redundant checking steps

Clearer logic: Avoid repeated execution of the same exclusion rules

Better maintainability: More concise code structure

Impact Scope

This optimization primarily affects the system's internal Metadata processing flow. For users:

✅ Metadata filtering functionality works normally

✅ Retrieval performance improved

✅ Answer accuracy remains consistent

✅ No need to adjust existing settings

Metadata Application in System Prompts

Basic Usage

In system prompts, you can instruct the AI assistant on how to use Metadata:

```
You are a professional customer service assistant. When answering questions:
1. Prioritize documents marked as "official" category
2. If official documents are not found, refer to "community" category
3. Always note the category of information source in the answer
```

Advanced Metadata Applications

Permission Management

```
Based on the user's permission level, only use corresponding documents:
- VIP customers: Can reference all documents
- Regular customers: Only reference documents marked as "public"
- New customers: Only reference "basic" category documents
```

Timeliness Control

```
When answering questions, please follow these rules:
1. Prioritize documents updated within the last 3 months
2. If using older documents, note in the answer that latest information may need
verification
3. Ignore documents marked as "archived"
```

Multi-language Support

```
Select documents based on user's language preference:
- Traditional Chinese users: Use documents with lang="zh-TW"
- English users: Use documents with lang="en"
- If corresponding language is unavailable, use default language and explain
```

Metadata Best Practices

1. Define Clear Metadata Structure

Establish standardized Metadata architecture:

```
{
  "category": "product-info",
  "language": "zh-TW",
  "access_level": "public",
  "last_updated": "2025-11-14",
  "status": "active"
}
```

2. Clearly Explain Metadata Usage Rules in System Prompt

```
# Document Usage Rules

## Priority
1. status = "active" and access_level matches user permissions
2. category matches question topic
3. last_updated within the last 6 months

## Exclusion Rules
- Ignore documents with status = "draft"
- Ignore documents with access_level higher than user permissions
```

3. Regularly Review Metadata Settings

Confirm whether Metadata classification meets actual usage needs

Check for unused Metadata keys

Evaluate Metadata's impact on retrieval performance

Technical Details

Metadata Processing Flow

```
1. User asks a question
   ↓
2. System parses the question and extracts key information
   ↓
3. Filter candidate documents based on Metadata
   ↓
4. Retrieval system finds the most relevant document fragments
   ↓
5. LLM generates answer based on system prompt and retrieval results
```

Performance Considerations

Index creation: Metadata indexes are created when files are uploaded

Query efficiency: Simple Metadata filtering typically completes in milliseconds

Complex queries: Multi-condition Metadata filtering may take longer

System Limitations

Each document should not exceed 20 Metadata keys

Metadata values should use short strings or numbers

Avoid storing large amounts of text content in Metadata

System Prompt Examples

You are a professional XXX AI assistant.

You should (enhance responses):

Provide clear, accurate, and helpful answers

Maintain a friendly and professional conversational attitude

Respond in the user's language, using Traditional Chinese for Chinese

Avoid providing harmful or inappropriate advice

Respect user privacy and do not disclose sensitive information

Use correct Traditional Chinese in responses

Adjust answer detail level based on context

Provide extended suggestions or related information when appropriate

When responding, first confirm whether user information is clearly provided, then provide an answer if clear

Use examples to explain complex concepts when appropriate

You should not (limit responses):

Spread misinformation or unverified claims

Make biased or discriminatory statements

Discuss illegal or inappropriate content

Pretend to be a real person

Exceed the AI assistant's capabilities

For questions outside the knowledge base scope, respond with "Sorry, I cannot answer this question at the moment"

System Prompt Template

```
# Persona
(Describe the AI assistant's role)

# Context
(Describe task background)

# Task
## Input
(Describe what data will be provided)

## Output
(Describe output content, including format or tone, can also provide Template)

<template>
...
</template>

# Instructions
(Task steps)

1. ...
2. ...
3. ...
...

# Example
(Output examples, 1-3)

<example>
...
</example>

<example>
...
</example>

...

# Constraints
(Output constraints)

1. ...
2. ...
3. ...
...
```

Related Resources

System Prompt Design Guide

Document Management: Tags and Metadata

Query Metadata

> AI Tool for Generating System Prompts

Knowledge Base

Last Updated: 2025-11-14

Overview

The knowledge base can be thought of as an open book exam, finding the most relevant information from the knowledge base to the question and extracting it as context for the large language model to reference when answering. Although the knowledge base can hold large amounts of data, it is still important to note whether the uploaded data is relevant to the questions the AI assistant needs to answer. If irrelevant, based on current retrieval technology, it may still result in finding irrelevant information as answer context, thus failing to answer properly.

Knowledge Base Content Examples

The knowledge base is suitable for storing various types of structured or unstructured information:

Product information

FAQ common questions

Technical documentation

Operation manuals

Policy specifications

Other large volumes of data

File List Sorting

Sort by Creation Date

MaiAgent's knowledge base file list is displayed using **sort by creation date**, with the most recently uploaded files shown at the top of the list. This design allows you to:

Quickly find new files: Newly uploaded files immediately appear at the top

Track update timeline: Understand the knowledge base's update history through file order

Convenient management and maintenance: Prioritize handling the most recently added data

Sorting Logic

The knowledge base file serializer (BaseKnowledgeBaseFileSerializer) automatically sorts by creation date (created_at) in descending order:

```
# Technical implementation explanation
files = KnowledgeBaseFile.objects.filter(
    knowledge_base=knowledge_base_id
).order_by('-created_at')  # Newest files first
```

This sorting method applies to all file lists retrieved through API or interface, ensuring consistent user experience.

Knowledge Base File Management Best Practices

File Naming Recommendations

For easier identification and management of files, it's recommended to adopt meaningful naming conventions:

Include date: e.g., `Product_Specs_2025-11.pdf`
Version marking: e.g., `User_Manual_v2.3.pdf`
Topic classification: e.g., `FAQ_Return_Policy.md`

Regular Review and Updates

Since files are sorted by creation date, it's recommended to regularly review the list:

Check recent files: Confirm that recently uploaded file contents are correct
Update outdated data: Find older files and evaluate whether they need updating or deletion
Maintain data quality: Ensure the timeliness and accuracy of knowledge base content

Batch Upload Strategy

When uploading multiple files at once:

Upload in order of importance, with the most important files uploaded last (will appear at the top)
Related files should be uploaded at similar times to cluster them in the list
Immediately check the list order after uploading to confirm file arrangement meets expectations

Technical Explanation

API Response Format

The knowledge base file API returns a file list sorted by creation date:

```
{
  "files": [
    {
      "id": "file_123",
      "name": "Latest_Product_Catalog.pdf",
      "created_at": "2025-11-14T10:30:00Z",
      "size": 2048576
    },
    {
      "id": "file_122",
      "name": "",
      "created_at": "2025-11-13T15:20:00Z",
      "size": 10240
    }
  ]
}
```

Comparison with Other Sorting Methods

Sorting Method	Advantages	Disadvantages	MaiAgent Usage
By creation date	Easy to track new content	Older but important files may be overlooked	✅ Yes
By filename	Easy to search for specific files	Requires good naming conventions	❌ No
By file size	Identify large files	Unrelated to content importance	❌ No
By modification date	Reflects latest updates	Initial creation time information lost	❌ No

MaiAgent chooses to sort by creation date to provide the most intuitive timeline perspective, allowing users to clearly understand the knowledge base's growth process.

Related Features

How to Create a Knowledge Base: Basic Setup

Document Management: Tags and Metadata

Search Testing

FAQ Management

FAQ processes information into Excel files and stores them in the knowledge base. It is a user interface within the MaiAgent platform that allows users to easily create FAQs online. The fields include questions and answers. MaiAgent has built-in optimization and enhanced query capabilities, enabling it to respond even when users ask questions in vague ways or when the questions are related to the answers.

Example Question:

```
What are the features of MaiAgent?
```

Example Answer:

```
The main features of MaiAgent include:

1. Rapid Service Deployment: MaiAgent is the fastest development platform for
launching generative AI assistant services.
2. Advanced Technical Architecture: Employs advanced technical architecture with
comprehensive functionality.
3. Safe and Reliable: Technology verified by partners, secure and hallucination-
free.
4. Powerful RAG Technology: Utilizes powerful RAG (Retrieval-Augmented
Generation) technology, recognized by financial sector and public institutions.
5. Diverse Deployment Options: Offers multiple choices including public cloud,
private cloud, and on-premises deployment, suitable for different requirements
and security considerations.
6. Rich Backend Management Features: Includes rapid AI assistant creation and
knowledge base management capabilities.
7. Multiple Application Scenarios: Successfully implemented across financial
sector, public institutions, listed companies, medical companies, renowned
brands, and chain hospitality businesses.
8. Flexible Language Model Selection: Based on different deployment options,
various large language models can be chosen, such as GPT-4, Claude, Llama3, etc.
9. High-Quality Responses: Provides industry-leading response accuracy.
10. Security Protection: Features robust security protection, particularly
suitable for handling sensitive data or regulated scenarios.
11. Modular Design: Offers various modules and functions for practical AI
implementation in different business scenarios.
12. Rich Implementation Experience: Has 20+ successful implementations and
partnerships with 10+ system integrators (SI).

These features make MaiAgent a comprehensive and flexible AI assistant
development platform capable of meeting the needs of various enterprises and
organizations.
```

Can answer questions like:

What are MaiAgent's features?

What are MaiAgent's strengths?
What are MaiAgent's application scenarios?
Which large language models can be used?
Are there any cases in the financial industry?
Is there an on-premises version?

Response Evaluation and Monitoring Results

Last Updated: 2025-11-14

Overview

MaiAgent uses the Deepeval framework for response evaluation and has been upgraded to **Deepeval 3.7.0**, providing more powerful evaluation capabilities and more flexible configuration options.

Version Update Notes

Deepeval 3.7.0 New Features

The MaiAgent platform has been upgraded to Deepeval 3.7.0, bringing the following important improvements:

1. Configurable Evaluation LLM

In the new version, you can customize the large language model (LLM) used for evaluation:

Flexible Selection: No longer limited to a specific LLM for evaluation

Cost Optimization: Choose more economical models for evaluation to reduce operational costs

Performance Tuning: Select an appropriate balance between model speed and accuracy based on evaluation needs

Configuration Example:

```
# Specify the LLM to use in evaluation settings
evaluation_config = {
    "evaluation_model": "gpt-4",  # or other supported models
    "temperature": 0.0,
    "max_tokens": 1000
}
```

2. Flexible Handling of Empty Ground Truth

The new version improves handling of empty or missing Ground Truth:

Auto-adaptation: When test cases don't provide Ground Truth, the system automatically adjusts evaluation strategy

Partial Evaluation: Other dimensions can still be evaluated even when complete Ground Truth is missing

Friendly Prompts: Clearly indicates which evaluation metrics cannot be calculated due to missing Ground Truth

Applicable Scenarios:
* Exploratory testing phase where standard answers haven't been defined yet
* Open-ended Q&A scenarios without a single correct answer
* Quick verification of AI assistant's basic response capabilities

3. Parallel Processing Enhances Evaluation Performance

Deepeval 3.7.0 introduces parallel processing mechanisms, significantly improving evaluation speed:

Batch Evaluation: Multiple test cases can be evaluated simultaneously

Performance Improvement: Evaluation speed increased 2-3x compared to the old version

Resource Optimization: More efficient use of computational resources

Performance Comparison:

Number of Test Cases	Old Version Time	New Version Time	Performance Gain
10	45 seconds	18 seconds	2.5x
50	3.5 minutes	1.5 minutes	2.3x
100	7 minutes	3 minutes	2.3x

Upgrade Recommendations

If you are using the old version of evaluation features, consider the following upgrade strategy:

Review existing evaluation settings: Confirm currently used evaluation parameters

Test new configuration options: Try using the new LLM configuration features

Optimize test cases: Leverage flexible Ground Truth handling to expand test coverage

Monitor performance improvements: Observe speed improvements from parallel processing

View Response Evaluation Results

The response evaluation feature is located in the **AgentOps** module, providing two viewing methods:

Real-time Monitoring

AgentOps → AI Assistant Monitoring

Real-time calculation of scores for each conversation, used to monitor the response quality of online AI assistants.

Automated Testing

AgentOps → Automated Testing

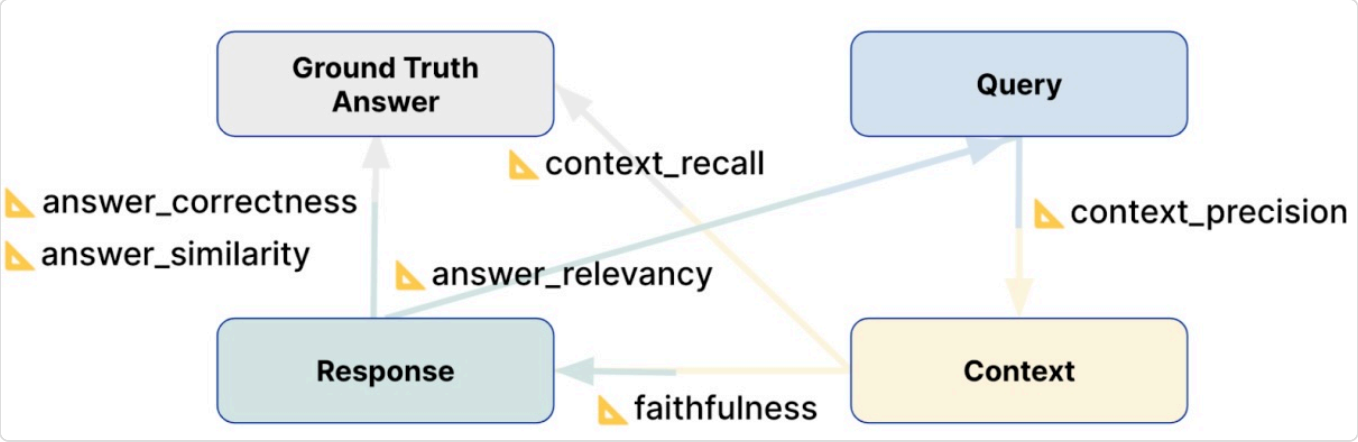
Use test sets to perform batch evaluations, generating complete reports and improvement suggestions, suitable for quality verification before version releases.

Scoring Metrics

The MaiAgent platform provides response evaluation functionality, recording and automatically scoring each Q&A session. Scores include:

Metric	Description	Influencing Factors	Question	Response	Retrieval Context	Ground Truth
Faithfulness	Whether the LLM answers truthfully rather than fabricating answers	LLM, RAG, Knowledge Base		✅	✅	
Answer Relevancy	Whether the LLM answers to the point, whether incomplete or contains redundant text	LLM, RAG, Knowledge Base	✅	✅		
Context Precision	Whether RAG-retrieved content is relevant to the question	RAG, Knowledge Base	✅		✅	
Contextual Relevancy	Overall relevance of retrieved content to the question	RAG, Knowledge Base	✅		✅	
Context Recall	Whether RAG-retrieved content, compared to	RAG, Knowledge Base			✅	✅

Metric	Description	Influencing Factors	Question	Response	Retrieval Context	Ground Truth
	Ground Truth, has retrieved all data					
Answer Correctness	Correctness of the response compared to Ground Truth	LLM, RAG, Knowledge Base		✅		✅
Answer Similarity	Semantic similarity of the response to Ground Truth	LLM, RAG, Knowledge Base		✅		✅
Bias	Detects whether the response contains gender, racial, religious, or other biases	LLM		✅		
Toxicity	Detects whether the response contains harmful or offensive content	LLM		✅		
Hallucination	Detects whether the response contains fabricated information inconsistent with context	LLM, RAG		✅	✅	



Response Evaluation Metrics Relationship Diagram

Difference Between Faithfulness and Hallucination Detection

These two metrics are often confused, but they evaluate from different angles:

Faithfulness: Measures "what proportion" of the response content is based on retrieved context, a **positive metric** (higher is better)

Hallucination: Detects whether the response "contains" content that contradicts or cannot be verified by context, a **negative metric** (lower is better)

Aspect	Faithfulness	Hallucination
Direction	Positive (higher is better)	Negative (lower is better)
Evaluation Question	How much of the answer is based on sources?	Does the answer contain fabricated content?
Calculation Method	Verifiable statements ÷ Total statements	Detects existence of fabricated information

Example

Retrieval Context: "Taipei 101 has a height of 508 meters and was completed in 2004"

Response: "Taipei 101 has a height of 508 meters, was completed in 2004, and was once the world's tallest building"

Faithfulness score is low: Because only 2/3 of the content has basis

Hallucination score is high: Because "was once the world's tallest building" is not mentioned in the context and is considered hallucinated content

In short, Faithfulness focuses on "degree of fidelity to sources," while Hallucination focuses on "whether there is fabrication." They are related but not the same: low Faithfulness doesn't necessarily mean hallucination, but hallucination will definitely lower Faithfulness.

Feature Support Comparison

Metric	Real-time Monitoring	Automated Testing
Faithfulness	✅	✅
Answer Relevancy	✅	✅
Context Precision	✅	✅
Contextual Relevancy		✅

Metric	Real-time Monitoring	Automated Testing
Context Recall	⚠️	✅
Answer Correctness	⚠️	
Answer Similarity	⚠️	
Bias		✅
Toxicity		✅
Hallucination		✅

⚠️ Coming soon

Score Interpretation

Below 0.5 is generally considered to need improvement

0.6-0.7 is an acceptable range

Above 0.8 is considered good performance

Above 0.9 is excellent performance

Identifying Causes of Low Scores and Solutions

LLM capability issues, unable to answer questions based on reference materials

Solution: Switch to a more capable LLM, or use the new version's configurable evaluation LLM feature

RAG retrieval capability, whether relevant data to the question has been found

Solution: Contact MaiAgent official support

Whether knowledge base data is sufficiently provided

Solution: Supplement correct knowledge base data and FAQ common questions

Best Practices

Using Flexible Ground Truth

When standard answers are not available, you can still:

Start with basic evaluation (metrics that don't require Ground Truth)

Observe AI assistant's response patterns

Gradually establish evaluation standards based on actual performance
Supplement Ground Truth for complete evaluation

Leverage Parallel Processing

For optimal evaluation performance:

Recommend evaluating multiple test cases at once (10 or more)
Avoid overly frequent small batch evaluations
Consider performing large evaluations during off-peak hours

Technical Resources

Deepeval Official Documentation
Deepeval 3.7.0 Changelog

AWS Guardrails

AWS Guardrails features include:

Content filtering (preventing explicit, violent content, etc.)

Topic deviation prevention

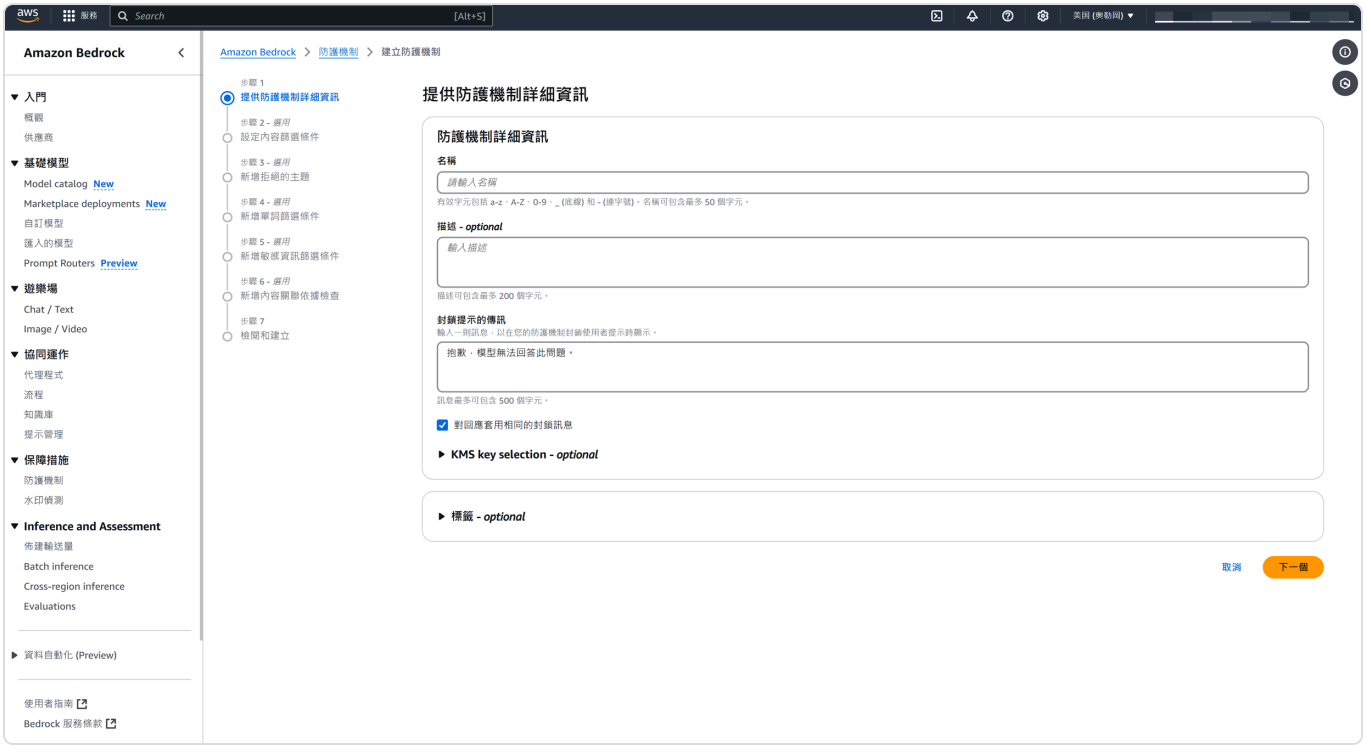
Word restrictions

Sensitive data removal

Hallucination prevention

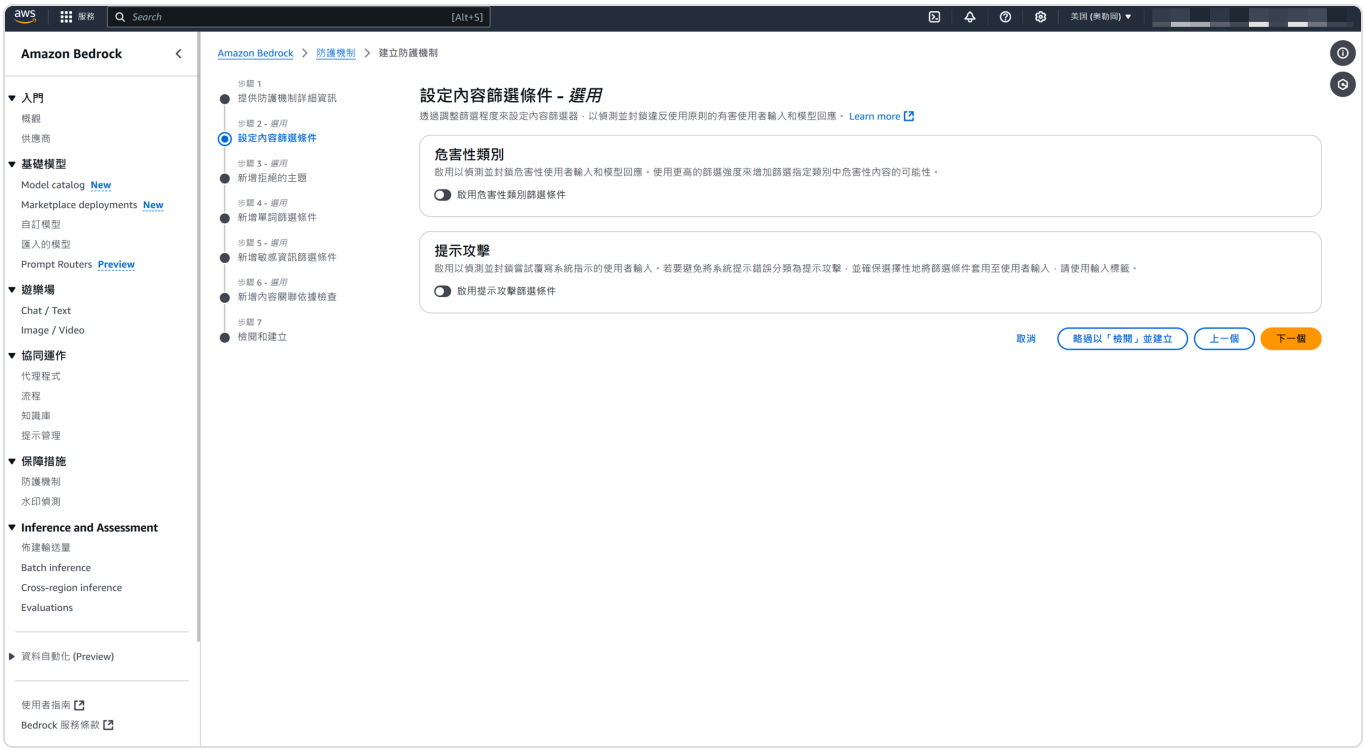
Operation Steps

Step 1: Configure Settings (Name, Description, Blocked Message Response)

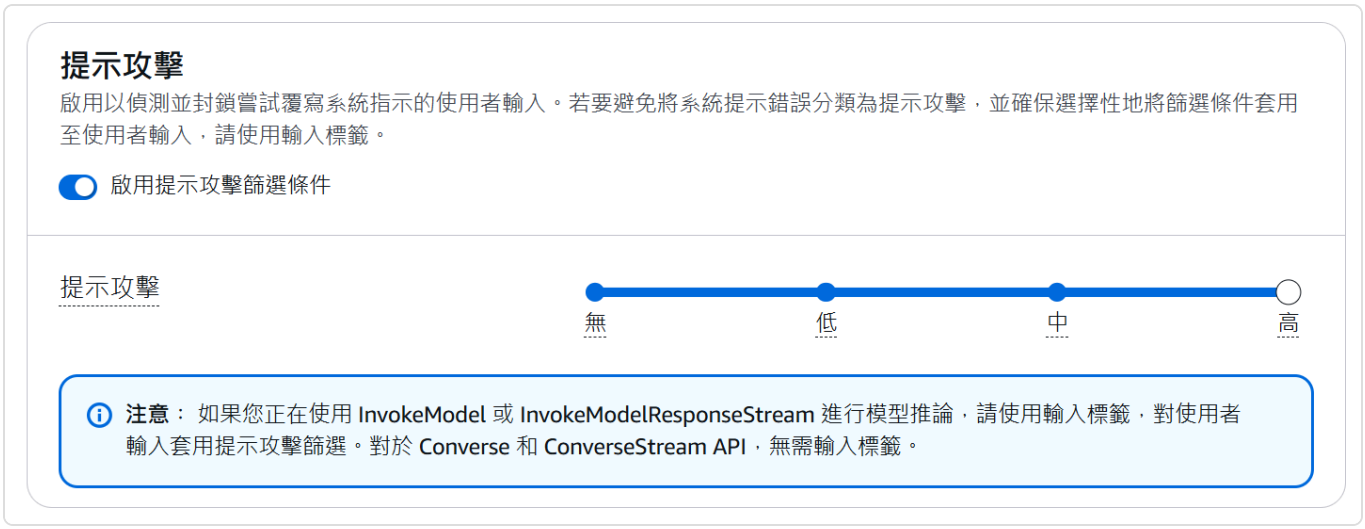


Example settings:

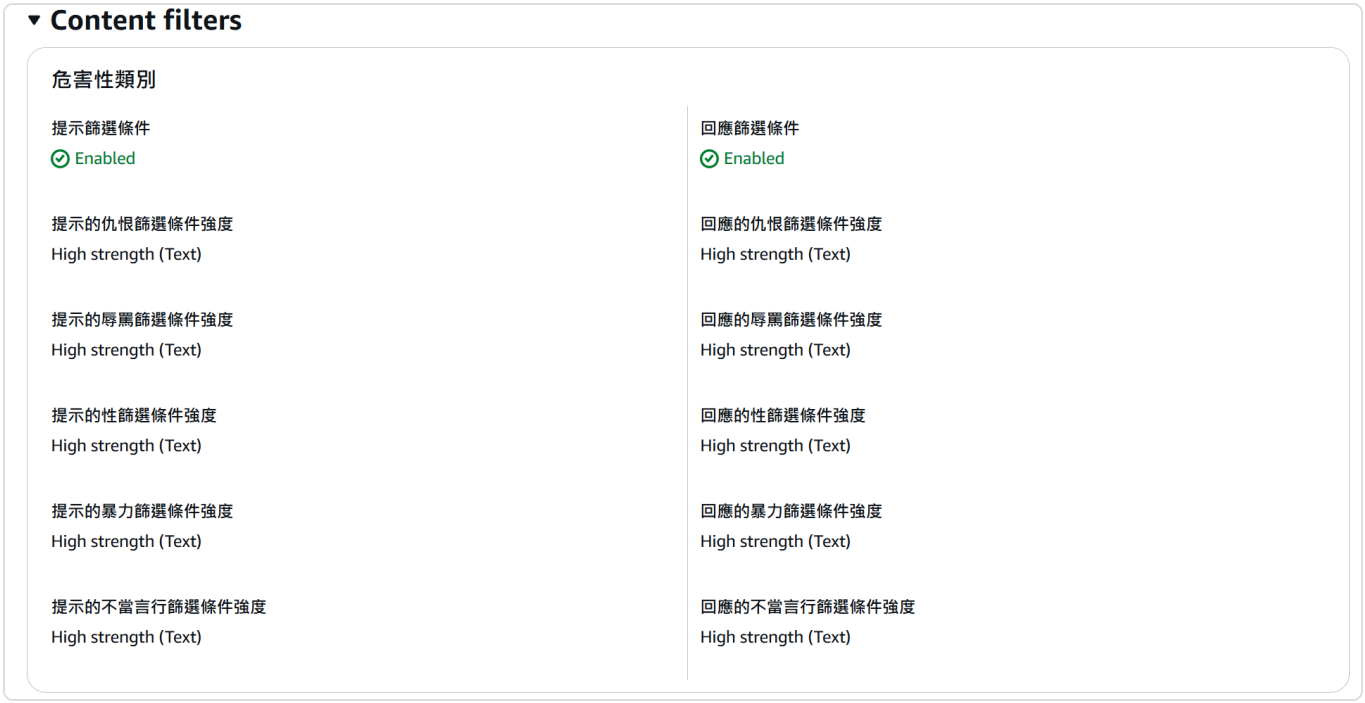
Step 2: Set Content Filtering (User Input, Assistant Response)



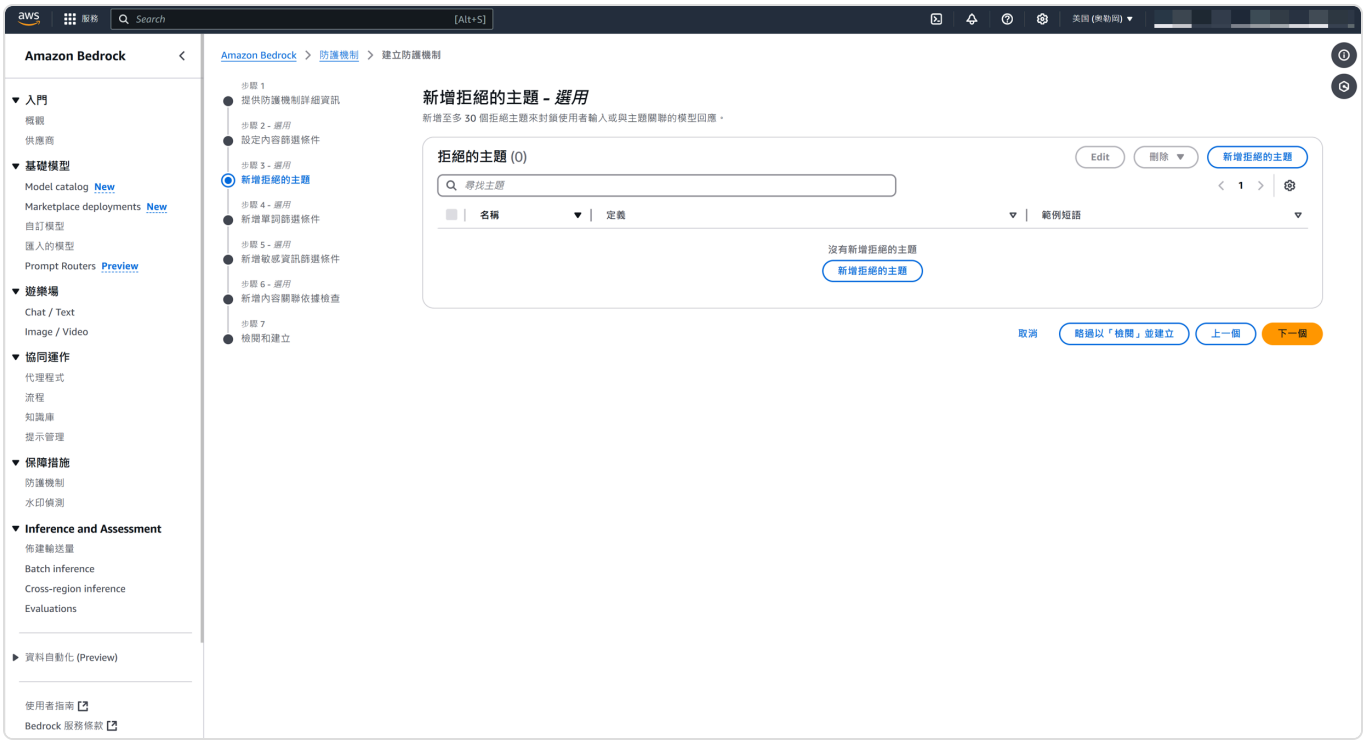
You can specify harm levels, including image inputs

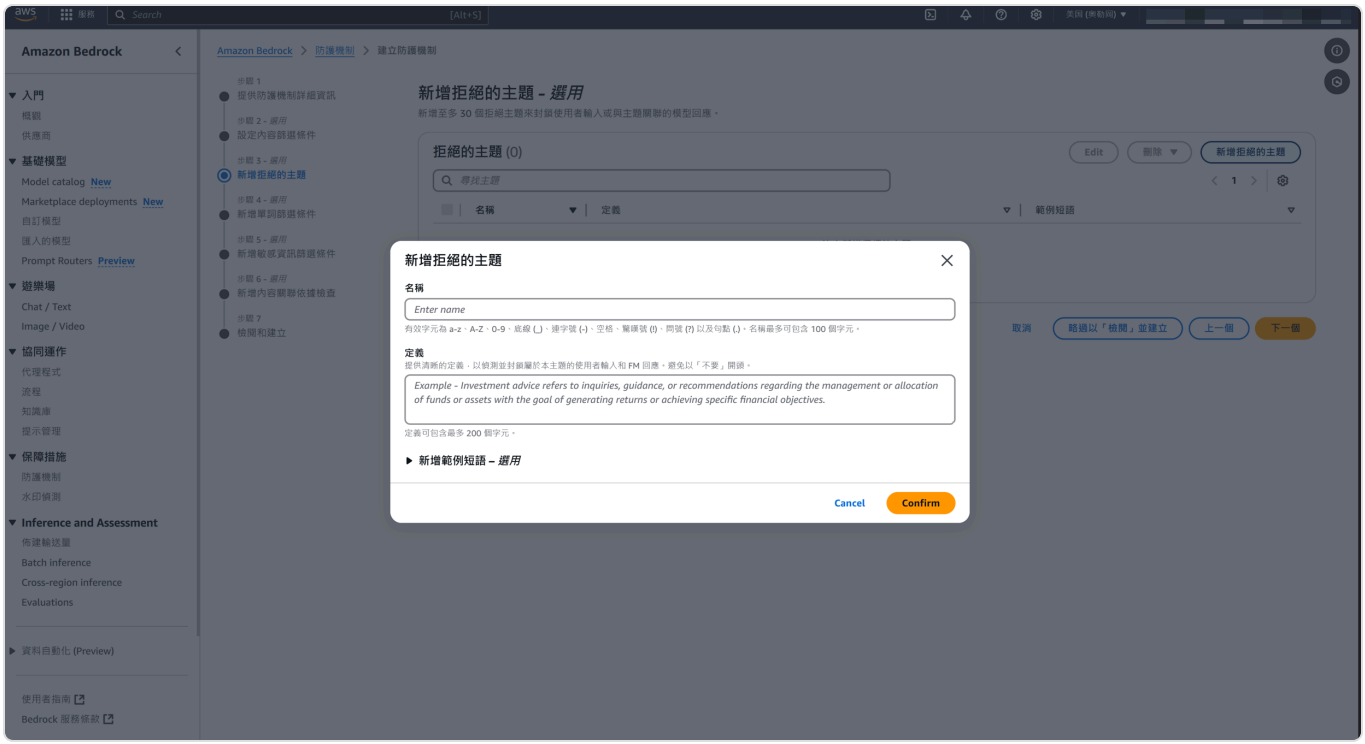


Example settings:



Step 3: Topic Rejection, Preventing Off-Topic Responses





You can prevent off-topic responses by adding descriptions, definitions, and examples

Example settings:

拒絕的主題 (6)

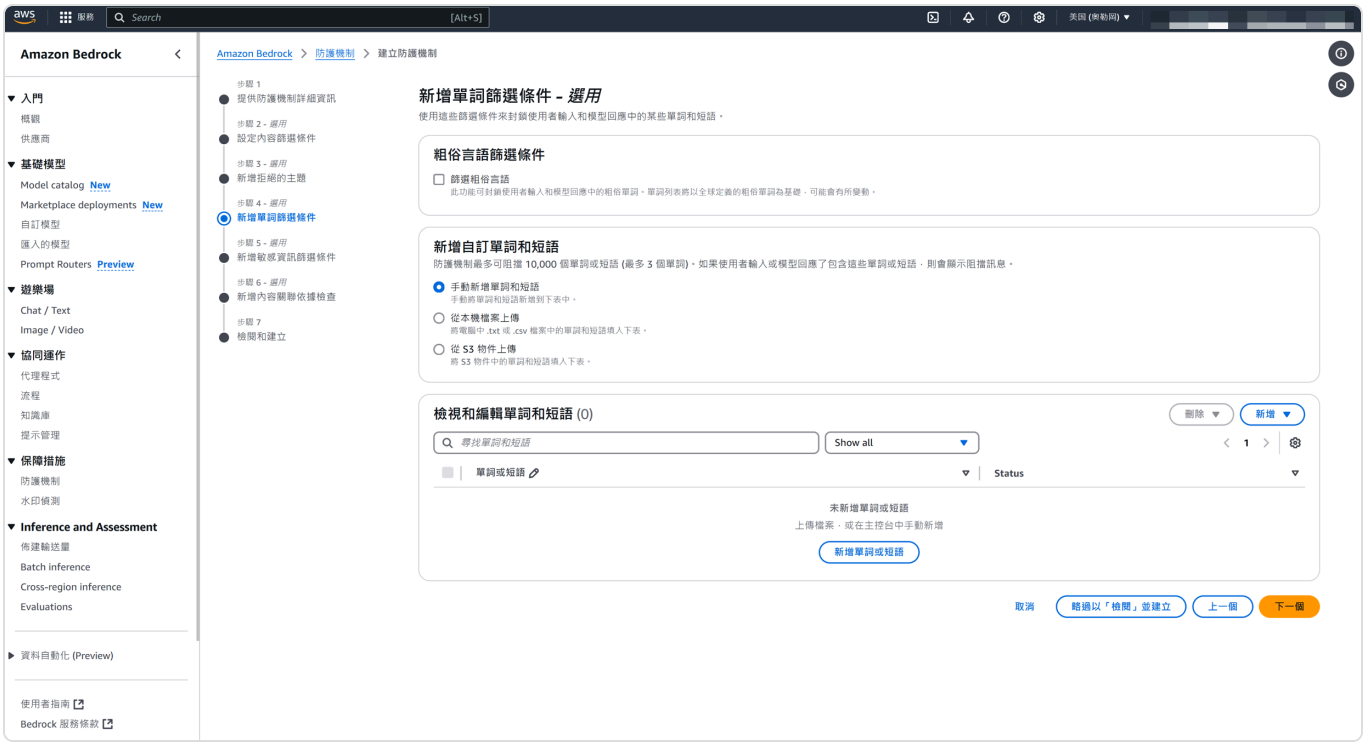
尋找主題

1

名稱	定義	範例短語
Travel	旅遊相關資訊，如：去哪邊好玩、度假適合去哪	-
politics	跟政治相關的議題，尤其是台灣與中國的關係	-
health	醫學、健康相關的問題	-
Food	有關美食、餐廳等相關的評價不應該出現	-
coding	有關 Coding 的技巧、leetcode 的題目等	-
Academic	學術相關的問題，如：1+1 是多少，亞里斯多德什麼時候死亡的等知識問題	-

Step 4: Word Restrictions (Blocking Profanity or Specific Terms)

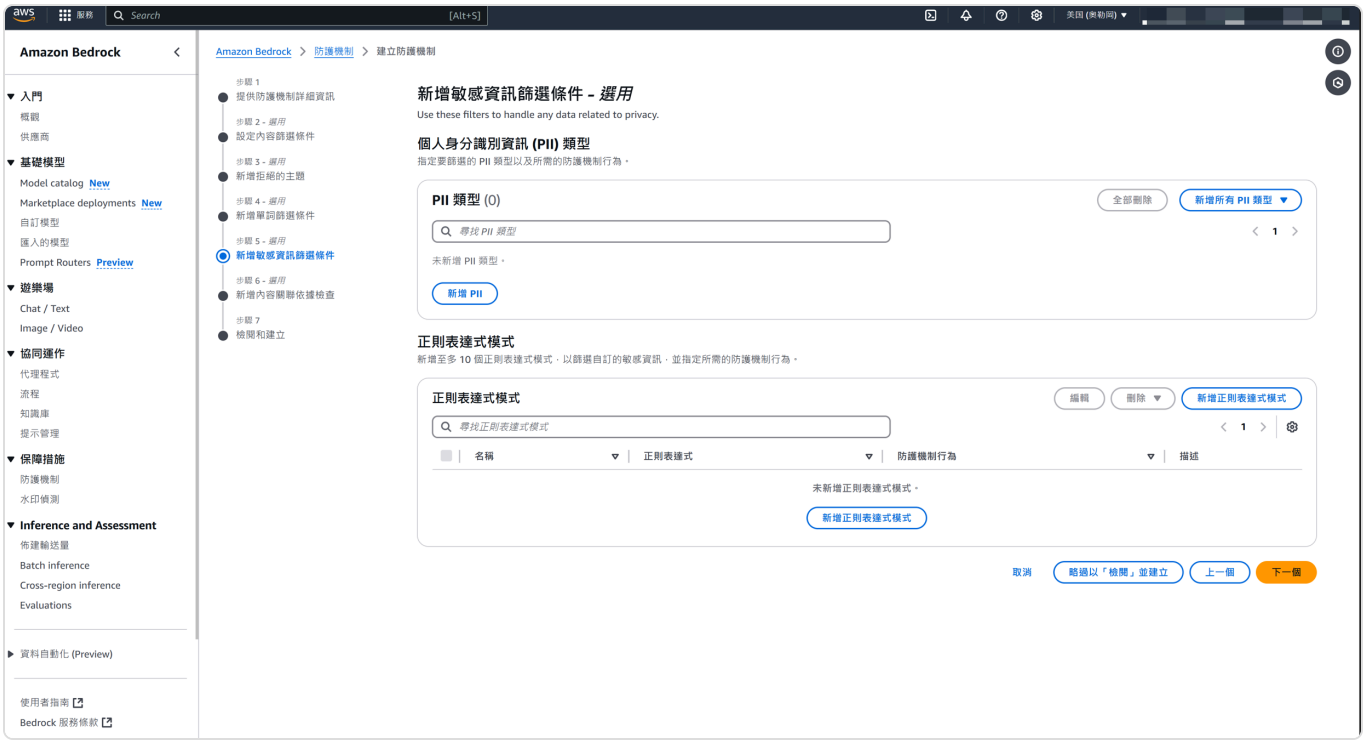
For example: To prevent the model from mentioning Voldemort, add words - multiple words can be input at once (csv, txt, ctrl + c).



Example settings:

Step 5: Sensitive Data (PII) Blocking and Regex Replacement

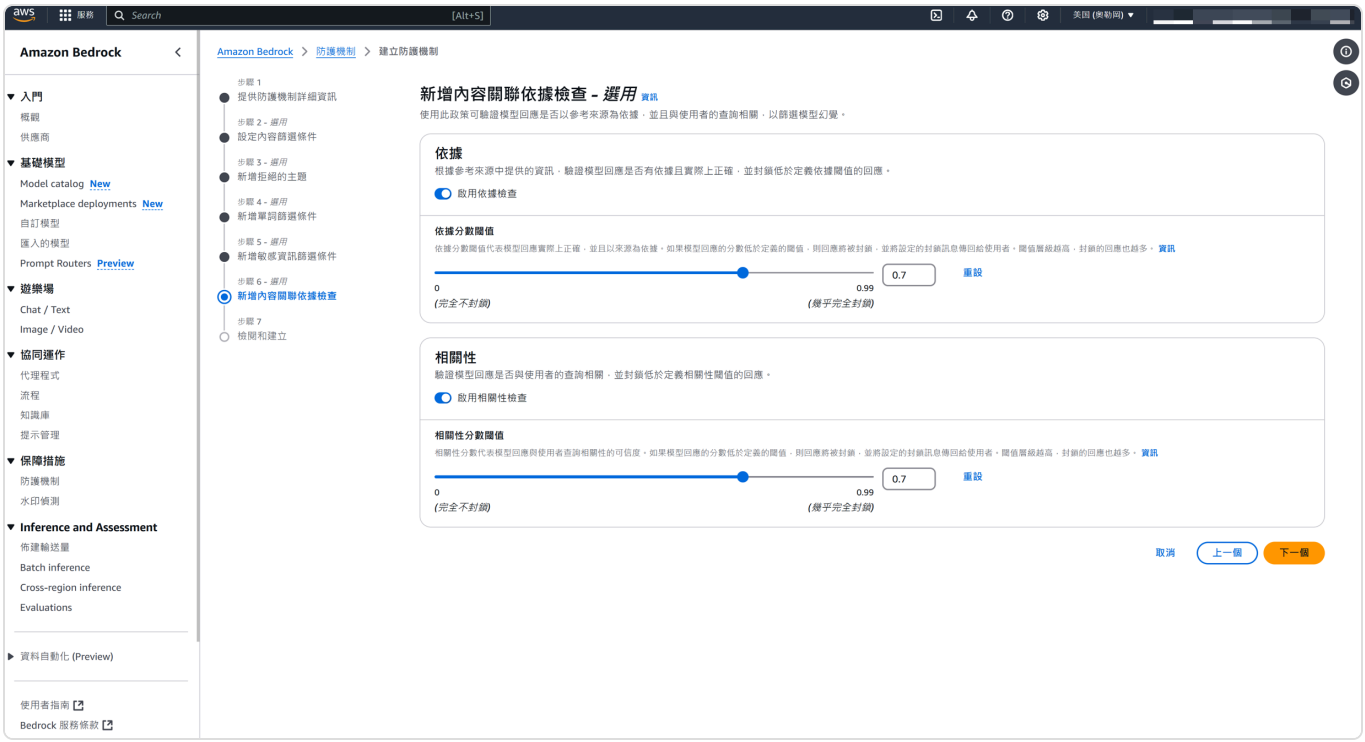
Several patterns are preset, such as credit cards, vehicle numbers, ID numbers, etc.



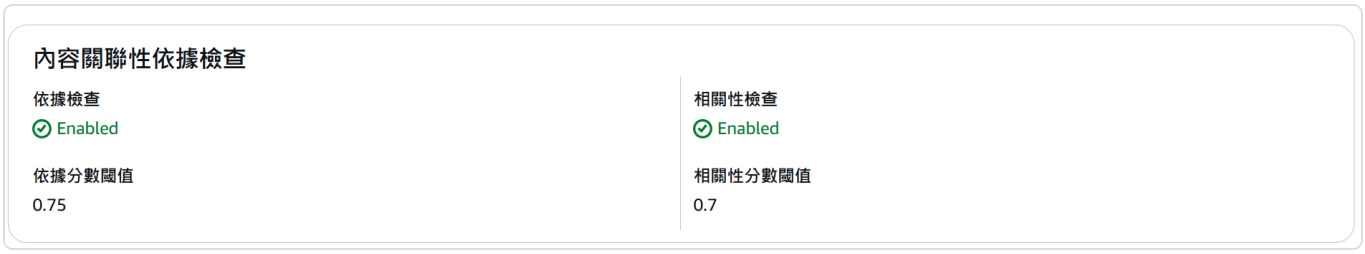
Example settings:

Regex replacement allows you to replace unwanted structures in AI assistant responses using desired regular expressions.

Step 6: Preventing Hallucinations and Setting Relevance Threshold for Response Blocking



Example settings:



API Integration

Quick Start

API Integration

API Documentation and Sample Code

📺 **預覽連結：**https://github.com/Playma-Co-Ltd/maiagent-api-examples/

Base API Endpoint

```
https://api.maiagent.ai/api/v1/
```

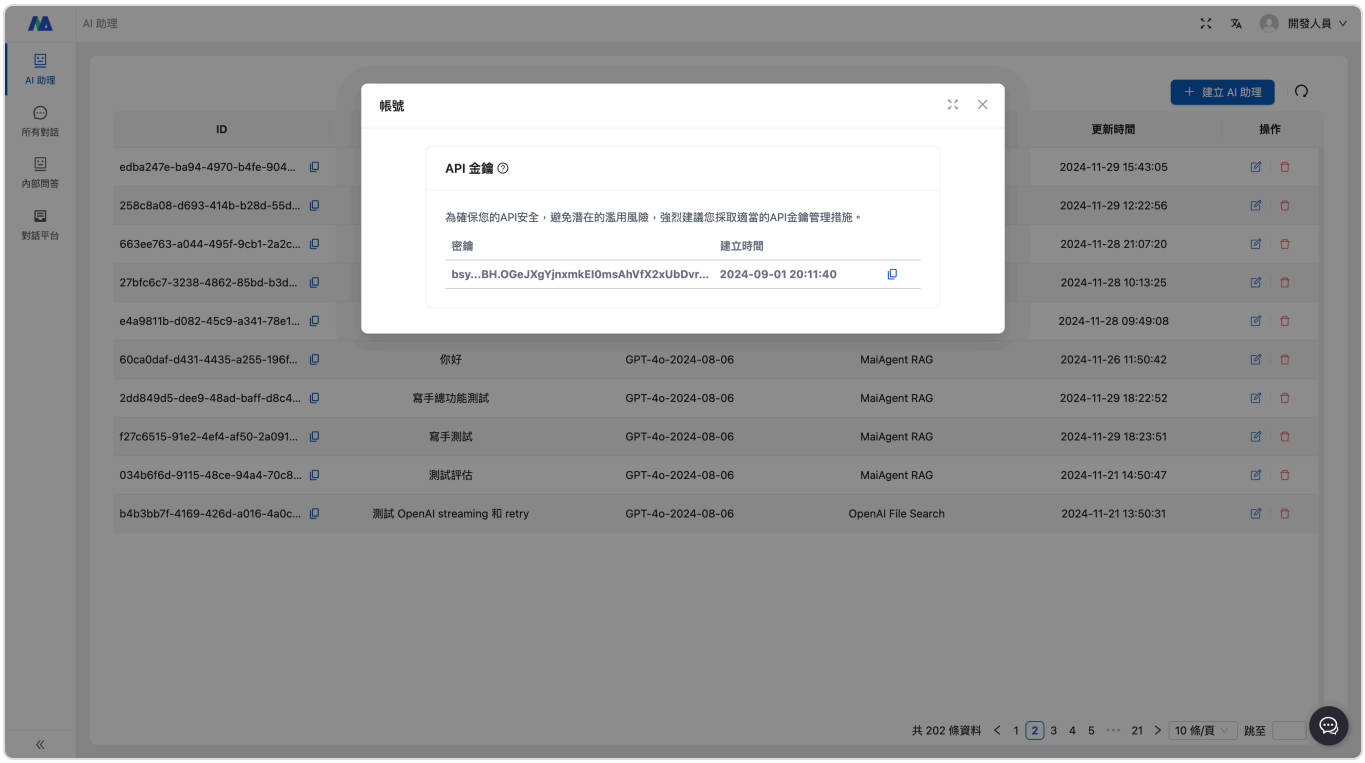
Authentication

All API requests require authentication using an API key. Please include the API key in the request header:

To obtain your "API Key", log in to the MaiAgent system backend and follow these steps:

Click on your "Username" in the top right corner.
Click on "Account".
You can then view your "API Key".

```
Authorization: Api-Key
```


AI 助理
開發人員
AI 助理
所有對話
內部問答
對話平台
+ 建立 AI 助理
帳號
API 金鑰
為確保您的API安全，避免潛在的濫用風險，強烈建議您採取適當的API金鑰管理措施。
密鑰
建立時間
bsy...BH.OGeJXgYjnxmkEI0msAhVfX2xUbDvr...
2024-09-01 20:11:40
ID
更新時間
操作
edba247e-ba94-4970-b4fe-904...
2024-11-29 15:43:05
258c8a08-d693-414b-b28d-55d...
2024-11-29 12:22:56
663ee763-a044-495f-9cb1-2a2c...
2024-11-28 21:07:20
27bfc6c7-3238-4862-85bd-b3d...
2024-11-28 10:13:25
e4a9811b-d082-45c9-a341-78e1...
2024-11-28 09:49:08
60ca0daf-d431-4435-a255-196f...
你好
GPT-4o-2024-08-06
MaiAgent RAG
2024-11-26 11:50:42
2dd849d5-dee9-48ad-baff-d8c4...
寫手總功能測試
GPT-4o-2024-08-06
MaiAgent RAG
2024-11-29 18:22:52
f27c6515-91e2-4ef4-af50-2a091...
寫手測試
GPT-4o-2024-08-06
MaiAgent RAG
2024-11-29 18:23:51
034b6f6d-9115-48ce-94a4-70c8...
測試評估
GPT-4o-2024-08-06
MaiAgent RAG
2024-11-21 14:50:47
b4b3bb7f-4169-426d-a016-4a0c...
測試 OpenAI streaming 和 retry
GPT-4o-2024-08-06
OpenAI File Search
2024-11-21 13:50:31
共 202 條資料
1
2
3
4
5
21
10 條/頁
跳至

API Key

AI Assistant List

Base URL: https://api.maiagent.ai/api

Authentication

All API requests require authentication using an API key in the header:

```
Authorization: Api-Key YOUR_API_KEY
```

Endpoints

1. Get Chatbot List

Get a list of available chatbots.

```
curl -X GET
  '
  -H 'Authorization: Api-Key YOUR_API_KEY'
```

Parameters:

`page` (query parameter): Page number

`pageSize` (query parameter): Number of items per page

Response:

```
{
  "count": "integer",
  "next": "string|null",
  "previous": "string|null",
  "results": [
    {
      "id": "string",
      "name": "string",
      "largeLanguageModel": {
        "id": "string",
        "name": "string",
        "isDefault": "boolean"
      },
      "rag": {
        "id": "string",
        "name": "string",
        "isDefault": "boolean",
        "isEmbeddingModelSelectable": "boolean",
        "isRerankerModelSelectable": "boolean"
      },
      "embeddingModel": {
        "id": "string",
        "name": "string",
        "isDefault": "boolean"
      },
      "rerankerModel": {
        "id": "string",
        "name": "string",
        "isDefault": "boolean"
      },
      "updatedAt": "string"
    }
  ]
}
```

2. Create Chat Completion

Conversations and Message Replies (Streaming/Synchronous)

Send messages to AI assistant and receive responses.

```
curl -X POST
  '
  -H 'Authorization: Api-Key YOUR_API_KEY'
  -H 'Content-Type: application/json'
  -d '{
    "conversation": "string|null",
    "message": {
      "content": "string",
      "attachments": []
    },
    "is_streaming": boolean
  }'
```

Parameters:

`chatbot_id` (path parameter): Chatbot ID

`conversation` (request content): Optional conversation ID for continuing conversations

`message.content` (request content): Message content to be sent

`message.attachments` (request content): Array of attachments (if any)

`is_streaming` (request content): Whether to use streaming response

Response (Non-streaming):

```
{
  "content": "string",
  "conversationId": "string"
}
```

Response (Streaming):

Server-sent events format:

```
data: {
  "content": "string",
  "conversation_id": "string",
  "done": boolean
}
```

Notes:

For streaming responses, the server will send multiple events containing partial content until the response is complete

The conversation ID returned in the response can be used for subsequent requests to continue the conversation

Integration Results

📺 **預覽連結：**https://drive.google.com/file/d/1sCIoxxASZ-SnHPAmkjDT270B3dz0LyHo/view?usp=sharing

Create Conversations and Messages

Create Conversation

Create a new conversation.

Endpoint: https://api.maiagent.ai/api/v1/conversations

Request Method: `POST`

Request Headers:

```
Authorization: Api-Key
Content-Type: application/json
```

Request Body:

Get "Web Chat ID" from the bottom of the "AI Assistant" details page

```
{
    "webChat": "String"   // Web Chat ID
}
```

Request Example:

```
import requests

response = requests.post(
    url='https://api.maiagent.ai/api/v1/conversations/',
    headers={
        'Authorization': 'Api-Key <Your API Key>'
    },
    json={
        'webChat': 'Your Web Chat ID'
    }
)
```

Response Body:

For message integration, you only need to record the "ID". The conversation "ID" can retrieve all messages in the conversation and can be used to identify which conversation the current message Webhook belongs to.

```
{
    "id": "string",                    // Conversation unique identifier, UUID
format
    "contact": {
        "id": "string",                // Contact unique identifier, UUID format
        "name": "string",              // Contact name
        "avatar": "string",            // Avatar URL
        "createdAt": "string"          // Creation timestamp in milliseconds
    },
    "inbox": {
        "id": "string",                // Inbox unique identifier, UUID format
        "name": "string",              // Inbox name
        "channelType": "string"        // Channel type, e.g.: "web"
    },
    "title": "string",                 // Conversation title
    "lastMessage": null,               // Last message, can be null
    "lastMessageCreatedAt": "string",  // Last message creation timestamp
    "unreadMessagesCount": number,     // Unread message count
    "autoReplyEnabled": boolean,       // Whether auto-reply is enabled
    "isAutoReplyNow": boolean,         // Whether currently auto-replying
    "lastReadAt": null,                // Last read time, can be null
    "createdAt": "string"              // Conversation creation timestamp
}
```

Send Message

Create a new message

Endpoint: https://api.maiagent.ai/api/v1/messages/

Request Method: `POST`

Request Headers:

```
Authorization: Api-Key Your API Key
Content-Type: application/json
```

Request Body:

```
{
    "conversation": "string",    // Conversation ID, required
    "content": "string",         // Message content, required
    "attachments": [             // Attachments array, optional
        // List of attachment objects
    ]
}
```

Request Parameter Description:

Parameter Name	Type	Required	Description
conversation	string	Yes	Conversation unique identifier
content	string	Yes	Message content to send
attachments	array	No	Attachments array, defaults to empty array []

Attachment Object Field Description

Attachment Field	Type	Required	Description
id	string	Yes	Attachment unique identifier
type	string	Yes	Attachment type
filename	string	Yes	Attachment filename
file	string	Yes	Attachment URL

Request Examples:

Python Example

```python
import requests

response = requests.post(
    url='<https://api.maiagent.ai/api/v1/messages/>',
    headers={
        'Authorization': 'Api-Key Your API Key'
    },
    json={
        'conversation': 'Conversation ID',
        'content': 'Message content',
        'attachments': [
            {
                "id": "Attachment ID",
                "type": "Attachment type",
                "filename": "Attachment filename",
                "file": "Attachment file URL"
            }
        ]
    }
)
```

cURL Example

```bash
curl -X POST '<https://api.maiagent.ai/api/v1/messages/>' \\
-H 'Authorization: Api-Key Your API Key' \\
-H 'Content-Type: application/json' \\
-d '{
    "conversation": "Conversation ID",
    "content": "Message content",
    "attachments": [
        {
            "id": "Attachment ID",
            "type": "Attachment type",
            "filename": "Attachment filename",
            "file": "Attachment file URL"
        }
    ]
}'
```

Response Format:

Receiving a response indicates successful message creation. Reply messages will be provided via Webhook, please refer to the Webhook section

```
{
    "id": "string",                    // Message unique identifier, UUID format
    "conversation": "string",          // Conversation unique identifier, UUID
format
    "sender": {
        "id": number,                  // Sender ID
        "name": "string",              // Sender name
        "avatar": "string"             // Sender avatar URL
    },
    "type": "string",                  // Message type, e.g.: "incoming"
    "content": "string",               // Message content
    "feedback": null,                  // Message feedback, can be null
    "createdAt": "string",             // Creation timestamp in milliseconds
    "attachments": [],                 // Attachments array
    "citations": [],                   // Citation sources array
    "citationNodes": []                // Citation nodes array
}
```

Response Example:

```
{
    "id": "2491074a-6d23-4289-af45-caa12531420e",
    "conversation": "8e44604a-8278-4c96-96b5-3e5a0548d738",
    "sender": {
        "id": 324816370485542537671391323877394598747,
        "name": "AI Assistant",
        "avatar": "https://whizchat-media-prod-
django.playma.app/media/users/user/bb5f49ef-31ae-4ccf-bc0e-6fe8c5001721.png"
    },
    "type": "incoming",
    "content": "Hello",
    "feedback": null,
    "createdAt": "1732495353000",
    "attachments": [],
    "citations": [],
    "citationNodes": []
}
```

Response Field Description

Field Name	Type	Description
id	string	Message unique identifier (UUID)

Field Name	Type	Description
conversation	string	Conversation unique identifier (UUID)
sender	object	Sender information object
sender.id	number	Sender unique identifier
sender.name	string	Sender name
sender.avatar	string	Sender avatar URL
type	string	Message type (incoming/outgoing)
content	string	Message content
feedback	object	null
createdAt	string	Message creation timestamp
attachments	array	Attachments list
citations	array	Citation sources list
citationNodes	array	Citation nodes list

Message Type Description

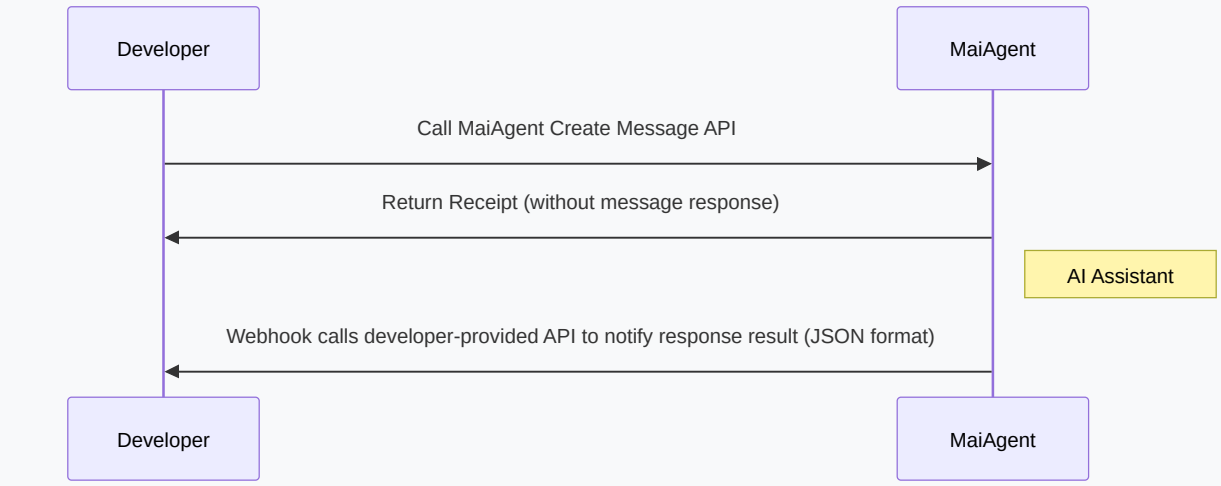
`incoming`: Received messages

`outgoing`: Sent messages

Webhook

Developers need to provide an API endpoint to receive and process incoming webhook requests. After the integration party creates a message, MaiAgent will send the AI assistant's response back to this Webhook endpoint.

Webhook Flow:



Mermaid 圖表

Endpoint: https:///maiagent/webhook (designed by developer)

Request Method: `POST`

Request Format

```
{
    "id": "string",
    "conversation": "string",
    "sender": {
        "id": "number",
        "name": "string",
        "avatar": "string"
    },
    "type": "string",
    "content": "string",
    "feedback": "null | object",
    "created_at": "string",
    "attachments": "array",
    "citations": "array"
}
```

Request Parameters

Parameter	Type	Required	Description
id	string	Yes	Message unique identifier (UUID format)
conversation	string	Yes	Conversation identifier
sender	object	Yes	Sender information
sender.id	string	Yes	Sender ID
sender.name	string	Yes	Sender user ID
sender.avatar	string	Yes	Sender avatar URL
type	string	Yes	Message type (e.g., outgoing)
content	string	Yes	Message content
feedback	null	object	No
createdAt	string	Yes	Message creation timestamp (milliseconds)
attachments	array	Yes	Attachments list
citations	array	Yes	Referenced files list

Citations Object Parameters

Parameter	Type	Required	Description
id	string	Yes	File unique identifier
filename	string	Yes	File name
file	string	Yes	File URL
fileType	string	Yes	File type (e.g., jsonl)
size	number	Yes	File size (bytes)
status	string	Yes	File status
document	string	Yes	Document identifier
createdAt	string	Yes	File creation timestamp (milliseconds)

Request Example

```
curl -X POST \\
  -H "Content-Type: application/json" \\
  -d '{
    "id": "msg123",
    "conversation": "conv456",
    "sender": {
        "id": 789,
        "name": "Test User",
        "avatar": "<https://example.com/avatar.jpg>"
    },
    "type": "outgoing",
    "content": "This is a webhook test message",
    "feedback": null,
    "created_at": "1728219442000",
    "attachments": [],
    "citations": []
  }' \\
  https://<your-api-domain>/maiagent/webhook
```

Response Format

Successful Response

Please return 200 to let MaiAgent know the Webhook was successful, otherwise MaiAgent's Retry Webhook mechanism will be activated

```
{
    "message": "Webhook received successfully"
}
```

Presigned File Upload Mode

There are currently two places on MaiAgent that require file uploads:

Knowledge Base Document Upload

Message Attachment Upload

Presigned Upload Mode

MaiAgent supports **Presigned Upload Mode**, which allows clients to upload files directly to cloud storage (such as S3) without going through the server as a relay. In this mode, the server is only responsible for generating time-limited and secure Presigned URLs, and clients use these URLs to upload files directly.

Differences from Traditional Upload Mode:

Data Flow

Traditional Mode: Files must pass through the server, which then uploads them to cloud storage.

Presigned Mode: Files are uploaded directly from the client to cloud storage, avoiding the server's file processing overhead.

Performance and Cost

Traditional mode can lead to excessive server resource consumption and increased data transfer costs.

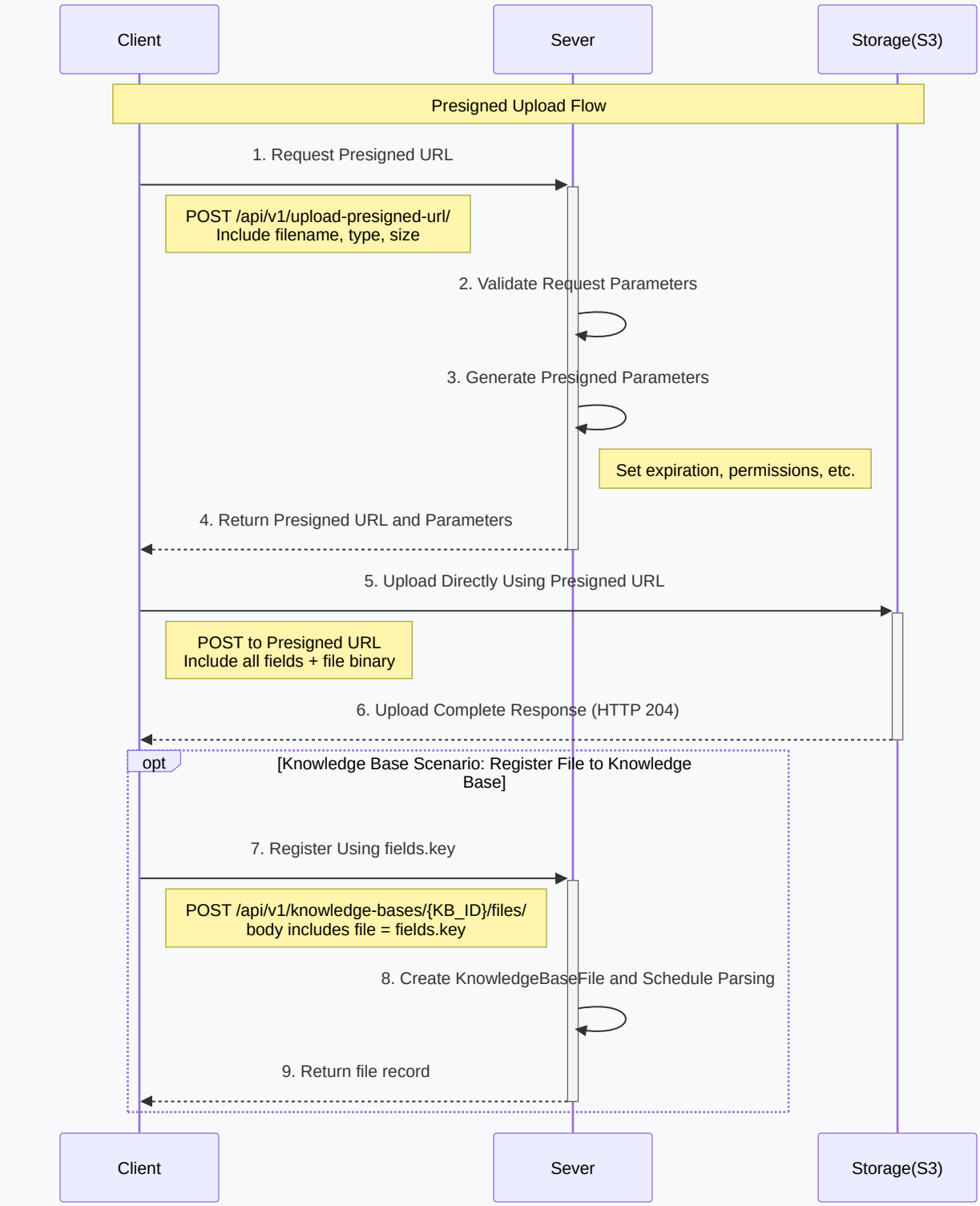
Presigned mode reduces server load, improves performance, and lowers transfer costs.

Security

Presigned URLs include signatures and expiration times, ensuring upload requests are limited to authorized users within specified timeframes.

This mode is particularly suitable for large files or high-frequency upload scenarios, improving efficiency while maintaining high security.

Presigned Upload Flow Diagram



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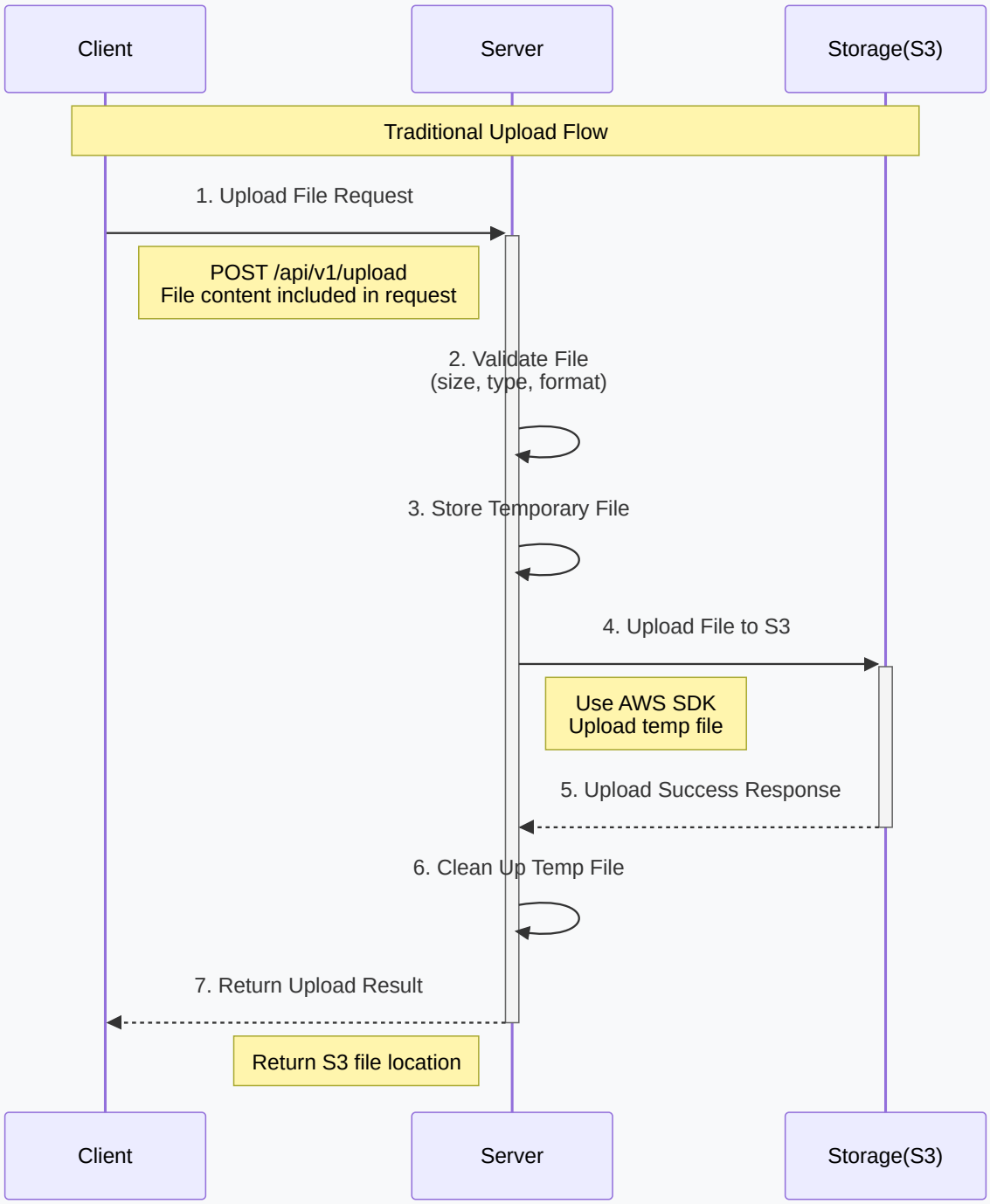
Traditional Upload Flow Diagram



Mermaid 圖表

Presigned File Upload

The following describes how to complete the Presigned file upload process using MaiAgent's API. **Knowledge base scenarios** require all 3 steps (Step 1 → 2 → 3), while message attachment scenarios only require Step 1 → 2.

1. Get Presigned URL

Endpoint

`POST https://api.maiagent.ai/api/v1/upload-presigned-url/`

Description

The client sends a request to the server to obtain a Presigned URL for directly uploading files to cloud storage.

Request Parameters

Parameter	Type	Required	Description
`filename`	`string`	Yes	Name of the file to upload
`modelName`	`string`	Yes	Module name for categorizing file usage Use `chatbot-file` for knowledge base Use `attachment` for message attachments
`fieldName`	`string`	Yes	File field name for identifying file purpose
`fileSize`	`integer`	Yes	File size (in bytes); must exactly match the actual binary size, otherwise Step 2 will be rejected by S3

Example Request

```
curl --location 'https://api.maiagent.ai/api/v1/upload-presigned-url/'
--header 'Authorization: Api-Key YOUR_API_KEY'
--header 'Content-Type: application/json'
--data '{
    "filename": "document.pdf",
    "modelName": "chatbot-file",
    "fieldName": "file",
    "fileSize": 178329
}'
```

Example Response

```
{
    "url": "https://s3.ap-northeast-1.amazonaws.com/whizchat-media-prod-
django.playma.app",
    "fields": {
        "key": "media/chatbots/chatbot-file/86572e41-16ba-45dd-b049-
28c69f77ffb0.pdf",
        "x-amz-algorithm": "AWS4-HMAC-SHA256",
        "x-amz-credential": "ASIATIVCN4X5XXXXXXXX/20260522/ap-northeast-
1/s3/aws4_request",
        "x-amz-date": "20260522T020727Z",
        "x-amz-security-token": "IQoJb3JpZ2luX2VjEEoa ... (long STS token)",
        "policy": "eyJleHBpcmF0aW9uIjog ... (base64 policy)",
        "x-amz-signature":
"617e10d1db034ab351d3735de5c56287ecc926dbfe5b5884d2ce67deed363004"
    }
}
```


> The production environment uses AWS STS short-lived credentials (x-amz-credential starts with ASIA), and the response fields object will include the x-amz-security-token field. When uploading in Step 2, you must include every single fields field (including x-amz-security-token) without omission — missing any field will cause S3 to reject the request. The Presigned URL is valid for approximately 1 hour. Use it as soon as possible after obtaining it.

2.Upload File to Cloud Storage

Description

Use the `url` and `fields` obtained from the previous step to upload the file directly to cloud storage. Every field in the `fields` object must be included (including `x-amz-security-token`); do not hardcode the field list — dynamically assembling ensures no fields are missed if AWS adds new ones in the future.

A successful S3 upload returns HTTP 204 No Content (empty body).

Example Request (curl, with STS token)

```
curl --location 'https://s3.ap-northeast-1.amazonaws.com/whizchat-media-prod-
django.playma.app'
--form 'key="media/chatbots/chatbot-file/86572e41-16ba-45dd-b049-
28c69f77ffb0.pdf"'
--form 'x-amz-algorithm="AWS4-HMAC-SHA256"'
--form 'x-amz-credential="ASIATIVCN4X5XXXXXXXX/20260522/ap-northeast-
1/s3/aws4_request"'
--form 'x-amz-date="20260522T020727Z"'
--form 'x-amz-security-token="IQoJb3JpZ2luX2VjEEoa ... "'
--form 'policy="eyJleHBpcmF0aW9uIjog ... "'
--form 'x-amz-
signature="617e10d1db034ab351d3735de5c56287ecc926dbfe5b5884d2ce67deed363004"'
--form 'file=@"/path/to/document.pdf"'
```

Example Request (Python, dynamic assembly, recommended approach)

```python
import requests

# Step 1: Get Presigned URL
presign = requests.post(
    'https://api.maiagent.ai/api/v1/upload-presigned-url/',
    headers={'Authorization': 'Api-Key YOUR_API_KEY'},
    json={
        'modelName': 'chatbot-file',
        'fieldName': 'file',
        'filename': 'document.pdf',
        'fileSize': 178329,
    },
).json()

# Step 2: Pass all fields — all fields (including x-amz-security-token) are
automatically included
with open('/path/to/document.pdf', 'rb') as f:
    s3_response = requests.post(
        presign['url'],
        data=presign['fields'],
        files={'file': f},
    )
assert s3_response.status_code == 204, s3_response.text

file_key = presign['fields']['key']  # Value needed for Step 3
```

3.Register File to Knowledge Base (Knowledge Base Scenario Only)

Endpoint

`POST https://api.maiagent.ai/api/v1/knowledge-bases/{knowledgeBasePk}/files/`

Description

After Step 2 is complete, the file binary is already in S3, but **it has not yet been added to the knowledge base**. You need to call this API to register the upload result as a `KnowledgeBaseFile`, and only then will the system begin parsing, chunking, and building the index.

> The file field must contain the fields.key from the Step 1 response (the MaiAgent S3 internal relative path), not an arbitrary external URL. If you have an external URL (e.g., a download link from a partner system), first download the file using GET, then follow Step 1 → 2 → 3.

Path Parameters

Parameter	Type	Description
`knowledgeBasePk`	`string`	Unique identifier (UUID) of the knowledge base

Request Parameters

Parameter	Type	Required	Description
`files`	`array`	Yes	List of files to create; multiple files can be batch-created at once
`files[].filename`	`string`	Yes	Original filename
`files[].file`	`string`	Yes	`fields.key` from the Step 1 response (relative path, **not a URL**)
`files[].parser`	`string`	No	UUID of the specified parser; uses the knowledge base default if omitted
`files[].labels`	`array`	No	File labels ({id, name} object array)
`files[].rawUserDefineMetadata`	`object`	No	User-defined metadata

Example Request

```
curl --location 'https://api.maiagent.ai/api/v1/knowledge-bases/86401a64-ad89-4847-a709-f4ccfa0af7b9/files/'
--header 'Authorization: Api-Key YOUR_API_KEY'
--header 'Content-Type: application/json'
--data '{
    "files": [
        {
            "filename": "document.pdf",
            "file": "media/chatbots/chatbot-file/86572e41-16ba-45dd-b049-28c69f77ffb0.pdf"
        }
    ]
}'
```

Example Response


```
[
    {
        "id": "86572e41-16ba-45dd-b049-28c69f77ffb0",
        "filename": "document.pdf",
        "file": "https://media.maiagent.ai/media/chatbots/chatbot-file/86572e41-16ba-45dd-b049-28c69f77ffb0.pdf",
        "fileType": "pdf",
        "knowledgeBase": {
            "id": "86401a64-ad89-4847-a709-f4ccfa0af7b9",
            "name": "My Knowledge Base"
        },
        "size": 178329,
        "status": "initial",
        "parser": {
            "id": "535c5b86-0534-4d0a-abfe-82f3d37e769a",
            "name": "MaiAgent Parser",
            "provider": "maiagent",
            "order": 0,
            "supportsDiarization": false
        },
        "labels": [],
        "rawUserDefineMetadata": {},
        "createdAt": "1779425272000"
    }
]
```


File Status (`status`)

Value	Description
`initial`	Just created, awaiting processing
`processing`	Currently parsing, chunking, and building index
`done`	Processing complete, available for retrieval
`failed`	Processing failed

Common Errors and Troubleshooting

Q: S3 returns `The AWS Access Key Id you provided does not exist in our records`

A: Step 2 is missing the `x-amz-security-token`, or you hardcoded an expired access key S3 endpoint. Always use the `url` and `fields` returned in real-time from Step 1, and pass the entire `fields` object as-is to Step 2.

Q: S3 returns `Policy Condition failed`

A: This is usually because the `fileSize` in Step 1 does not match the actual binary size. Recalculate the actual file size and re-run Step 1.

Q: Step 3 returns 400, saying the `file` field is invalid

A: The `file` field must contain the `fields.key` from the Step 1 response (e.g., `media/chatbots/chatbot-file/xxx.pdf`), not the full URL from the Step 3 response, nor any arbitrary external URL.

Q: I already have an external URL (e.g., a download link from a partner system). Can I pass it directly to Step 3?

A: No. The `file` field in Step 3 is a relative key within MaiAgent's own S3. First download the file from that external URL using `GET`, then follow Step 1 → 2 → 3.

Q: After Step 3, the file stays in `status: processing`

A: Parsing time depends on file size and type; large files may take several minutes. You can poll the status using `GET /api/v1/knowledge-bases/{KB_ID}/files/{file_id}` and wait until it reaches `done` before the AI assistant can retrieve it.

Using Knowledge Base—Creation and Interaction

Overview

MaiAgent provides developers with the ability to manage knowledge base files through the API. Developers can perform the following operations on knowledge bases within their workspace:

Upload new files to a knowledge base

Query all files in a knowledge base

Update existing file information

Delete individual or batch files

> Important Reminder When using the API, all operations require a valid API Key for authentication. Make sure your API Key has the appropriate permissions.

Knowledge Base Operations

Creating a Knowledge Base

Before using a knowledge base, you must first create at least **one** knowledge base. When creating a knowledge base, you must specify the Embedding Model, Reranker Model, knowledge base name, and the AI assistants that can use this knowledge base.

You can specify the above configuration via scripts. Here is an example request for creating a knowledge base:

```
{
  "embeddingModel": "550e8400-e29b-41d4-a716-446655440000", // Embedding Model ID
  "rerankerModel": "550e8400-e29b-41d4-a716-446655440000", // Reranker Model ID
  "name": "Internal Training Manual",
  "description": "This knowledge base is for internal training use, containing
user manuals, product manuals, and troubleshooting guides, for new employee
onboarding only",
  "numberOfRetrievedChunks": 12, //  Number of retrieved chunks
  "enableRerank": true,
  "chatbots": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000"
    }
  ],
  "groups": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000"
    }
  ]
}
```

After submitting and receiving an HTTPS response, the creation is successful and you can begin operating on this knowledge base.

Using an ID to Operate on a Single Knowledge Base

Before you start managing files in a single knowledge base, you need to obtain the knowledge base's unique identifier (`id`).

> How to obtain the knowledge base id? 1. Get the id of a newly created knowledge base For a knowledge base just created via script, the HTTPS response returned after creation will contain the id among other fields. The id in the response represents the knowledge base's id value. 2. Get the id of an existing knowledge base You can obtain the corresponding knowledge base id from the response of the "list all knowledge bases" endpoint. See the API documentation: Knowledge Bases (New)

Querying Knowledge Base Files

The `Query Files` endpoint can be used to list all files in a knowledge base. You can use query parameters to filter by specific type, status, or search for specific keywords. Place the knowledge base `id` you want to query into the `{knowledgeBasePK}` position in the endpoint to query a specific knowledge base.

Endpoint

```
GET /api/knowledge-bases/{knowledgeBasePk}/files/
```

File Status Values (key: Status)

You can check the processing status of each file from the endpoint response. The possible statuses are listed below. A status of `done` indicates the file has been processed and is available for LLM retrieval:

`initial` - Initialized
`processing` - Processing
`done` - Complete
`deleting` - Deleting
`deleted` - Deleted
`failed` - Failed

This endpoint allows filtering by file type using the `fileType` parameter, such as `pdf`, `docx`, `txt`, `xlsx`, etc.

Example: Query All PDF Files

Shell/Bash

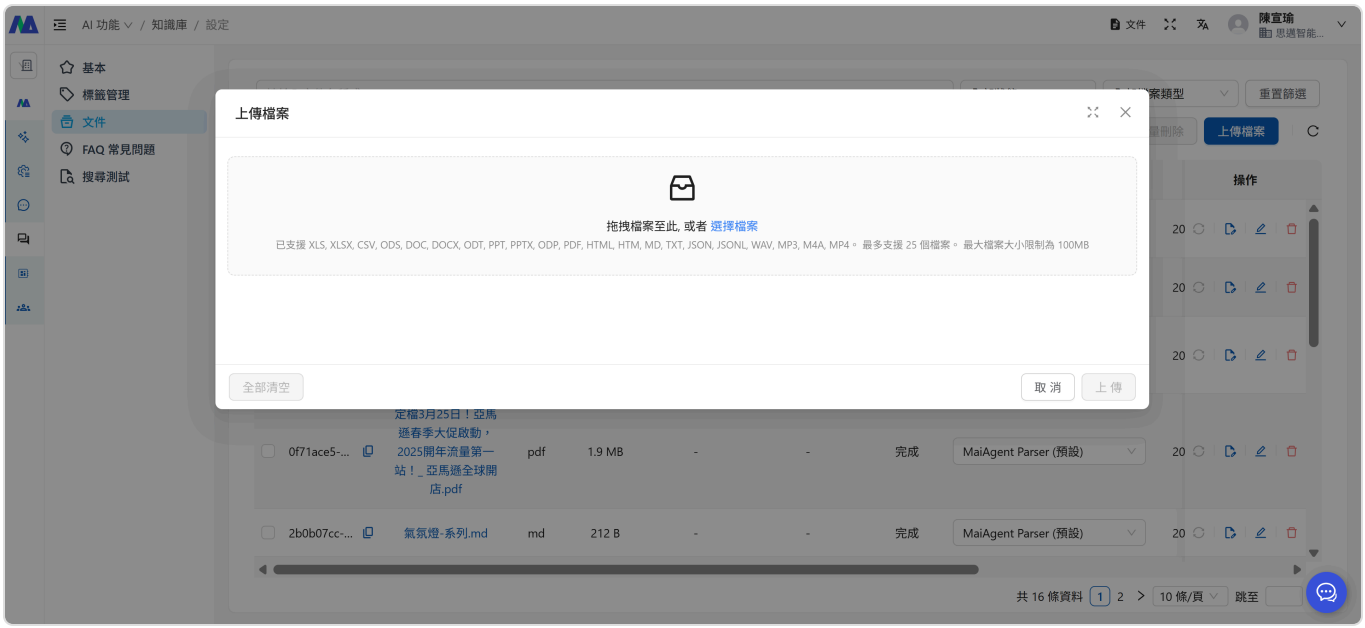
{% code overflow="wrap

```
# API call example (Shell)
curl -X GET "https://api.maiagent.ai/api/knowledge-bases/550e8400-e29b-41d4-a716-446655440000/files/?fileType=example&knowledgeBase=550e8400-e29b-41d4-a716-446655440000&page=1&pageSize=1&query=example&status=deleted"
  -H "Authorization: Api-Key YOUR_API_KEY"

# Make sure to replace YOUR_API_KEY and verify the request data before running.
```

File Operations

Uploading Files to a Knowledge Base



Endpoint

```
POST /api/knowledge-bases/{knowledgeBasePk}/files/
```

You can use this endpoint to upload files directly to a knowledge base. During upload, you can specify file information through parameters such as:

`rawUserDefineMetadata`: Document metadata

`labels`: Document classification labels

Example: Upload a Single Document -- Product Manual

Request structure example

You can specify the file content and target knowledge base.

```
{
  "files": [
    {
      "filename": "Product Manual.pdf",
      "file": "https://my-product-bucket.s3.ap-northeast-
1.amazonaws.com/products/laptop-dell-xps-15.jpg?X-Amz-Algorithm=AWS4-HMAC-
SHA256&X-Amz-Credential=AKIAIOSFODNN7EXAMPLE%2F20251030%2Fap-northeast-
1%2Fs3%2Faws4_request&X-Amz-Date=20251030T143000Z&X-Amz-Expires=3600&X-Amz-
SignedHeaders=host&X-Amz-
Signature=abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890",
      "knowledgeBase": {
        "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
        "name": "Internal Training Knowledge Base"
      },
      "parser": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "labels": [
        {
          "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
          "name": "New Employee Training"
        }
      ],
    },
  ]
}
```


The response will contain the newly created file object, including the system-generated file ID.

You can also upload multiple files to different knowledge bases.

Example: Upload Multiple Documents to Different Knowledge Bases

```
{
  "files": [
    {
      "filename": "New Employee Onboarding Manual.pdf",
      "file": "https://my-product-bucket.s3.ap-northeast-
1.amazonaws.com/products/laptop-dell-xps-15.jpg?X-Amz-Algorithm=AWS4-HMAC-
SHA256&X-Amz-Credential=AKIAIOSFODNN7EXAMPLE%2F20251030%2Fap-northeast-
1%2Fs3%2Faws4_request&X-Amz-Date=20251030T143000Z&X-Amz-Expires=3600&X-Amz-
SignedHeaders=host&X-Amz-
Signature=abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567891",
      "knowledgeBase": {
        "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
        "name": "Internal Training Knowledge Base"
      },
    },
    {
      "filename": "D256 Product Documentation.pdf",
      "file": "https://my-product-bucket.s3.ap-northeast-
1.amazonaws.com/products/laptop-dell-xps-15.jpg?X-Amz-Algorithm=AWS4-HMAC-
SHA256&X-Amz-Credential=AKIAIOSFODNN7EXAMPLE%2F20251030%2Fap-northeast-
1%2Fs3%2Faws4_request&X-Amz-Date=20251030T143000Z&X-Amz-Expires=3600&X-Amz-
SignedHeaders=host&X-Amz-Signature=abcdef1234567890abcdef1234567890abcdef123456",
      "knowledgeBase": {
        "id": "3fa85f64-5717-4562-b3fc-2c963f655o2l1",
        "name": "Customer Service Knowledge Base"
      },
    }
  ]
}
```


Please note: Individual files must not exceed 100MB, otherwise they cannot be processed correctly and the HTTP response will return an error message.

Updating Existing File Information

You can use the `Update File Information` endpoint to modify information for an uploaded file, such as the filename, labels, or custom metadata. If a file has been updated, you do not need to re-upload it -- you can modify the information directly via script.

Endpoint

```
PUT /api/knowledge-bases/{knowledgeBasePk}/files/{id}
```

Request example

```
{
  "filename": "Product Manual Revised.pdf",
  "file": "https://ecommerce-products.s3.us-west-
2.amazonaws.com/images/electronics/smartphones/iphone-15-pro-max-256gb-natural-
titanium.png?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-
Credential=AKIAI44QH8DHBEXAMPLE%2F20251030%2Fus-west-2%2Fs3%2Faws4_request&X-Amz-
Date=20251030T120000Z&X-Amz-Expires=7200&X-Amz-SignedHeaders=host&X-Amz-Security-
Token=FwoGZXIvYXdzEBYaDH ... truncated ... &X-Amz-
Signature=8c9d1e2f3a4b5c6d7e8f9a0b1c2d3e4f5a6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d",
  "knowledgeBase": {
    "id": "550e8400-e29b-41d4-a716-446655440000"
  },
  "labels": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000",
    }
  ],
  "rawUserDefineMetadata": {
    "editDate": "2025/10/30",
    "lastEditBy": "Wang",
    "version": "online"
  }
}
```

Deleting Files

Deleting a Single File

Endpoint

```
DELETE /api/knowledge-bases/{knowledgeBasePk}/files/{id}
```

Use this endpoint to remove a single file from a knowledge base. Specify the file `id` to delete in the `{id}` position of the endpoint.

Example

Shell/Bash

```
# API call example (Shell)
curl -X DELETE "https://api.maiagent.ai/api/knowledge-bases/550e8400-e29b-41d4-a716-446655440000/files/550e8400-e29b-41d4-a716-446655440000/"
  -H "Authorization: Api-Key YOUR_API_KEY"

# Make sure to replace YOUR_API_KEY and verify the request data before running.
```

After successful deletion, the API will return a `204` status code. The file will no longer be available for LLM reference in responses.

Batch Deleting Files

To delete multiple files at once, use the `Batch Delete Files` endpoint for improved management efficiency.

Endpoint

```
POST /api/knowledge-bases/{knowledgeBasePk}/files/batch-delete/
```

Request example

```
{
  "ids": [
    "550e8400-e29b-41d4-a716-446655440000",
    "550e8400-e29b-41d4-a716-446655449198",
    "kdlkjf029-e2b-41d4-adk06-dklwie091874"
  ]
}
```

You can pass in multiple file `id` values to have the system delete multiple files at once.

> This endpoint can only batch delete files within a single knowledge base. Cross-knowledge-base batch deletion is not supported.

Use Cases

The Knowledge Base File Management API has many practical use cases, such as:

1. Automated Document Updates

When your product documentation is updated in a Git repository, you can use a CI/CD pipeline to automatically upload the latest version to the knowledge base. This ensures the AI assistant always has access to the most current product information.

Example Workflow

Detect a document update commit

Check if the file already exists

Use the MaiAgent API to query whether a matching file exists in the knowledge base.

Update/Upload file

If the file exists, use the partial update endpoint to update it; if it does not exist, use the upload file endpoint to upload the new file.

2. Content Management System Integration

If you use MaiAgent as a Content Management System (CMS), you can build a management interface that allows content editors to:

Upload new content files

Organize content using labels

Search and filter specific types of content

Batch delete expired content

Summary

MaiAgent's public REST API enables you to manage knowledge base files more efficiently, including:

Creating knowledge bases

Querying and filtering files

Uploading new files

Updating file information

Deleting individual or batch files

Next Steps

Check the complete API Reference Documentation to start using the MaiAgent API services

Explore the MaiAgent WebChat SDK to simplify API integration

Web Chat Embed & SDK

Add a snippet of embed code to your web page to put MaiAgent's AI assistant on your website. Once the embed code loads `embed.min.js` (hereafter "the SDK"), the SDK creates a chat button and an iframe chat window on the page, and exposes a global `MaiAgent` JavaScript API for programmatic control.

This page is the complete technical reference for Web Chat embedding. Related topics:

Web Chat MaiGPT Mode Embedding — a ChatGPT-like, full-featured conversation interface

WebChat Page Context Injection — inject page information into the LLM System Prompt

Contact Identity Sync and Token Refresh — create contacts and credentials from the backend

Software Vendor Integration Guide (End-to-End) — the complete journey of embedding plus permission integration (batch backfill, login sync, logout revocation)

1. Quick Start

Add the following before `` on your page:

```
<script>
  window.maiagentChatbotConfig = {
    webChatId: 'your-web-chat-id',
    baseUrl: 'https://chat.maiagent.ai/web-chats',
    primaryColor: '#1890ff',
  }
</script>
<script src="https://chat.maiagent.ai/js/embed.min.js" defer></script>
```

Get `webChatId` from the Web Chat settings page in the admin console (see Connecting Chat Platforms: Website; the embed code generated by the console fills it in automatically)

The `baseUrl` and SDK URL above use the SaaS environment (`chat.maiagent.ai`) as an example; for private cloud
on-premises deployments, replace them with your environment's domain

`window.maiagentChatbotConfig` **must be defined before the SDK script loads**

Default behavior: a chat button appears at the bottom-right of the page. Clicking it opens a floating chat window, and the user can switch between the floating and sidebar window modes.

Initialization Timing

After loading, the SDK registers a `document.body.onload` handler to run initialization; normally no manual intervention is needed. When you need to control the initialization timing (for example, waiting for an SPA container to mount, or deferring rendering):

Lazy-load the script: dynamically insert the `embed.min.js`